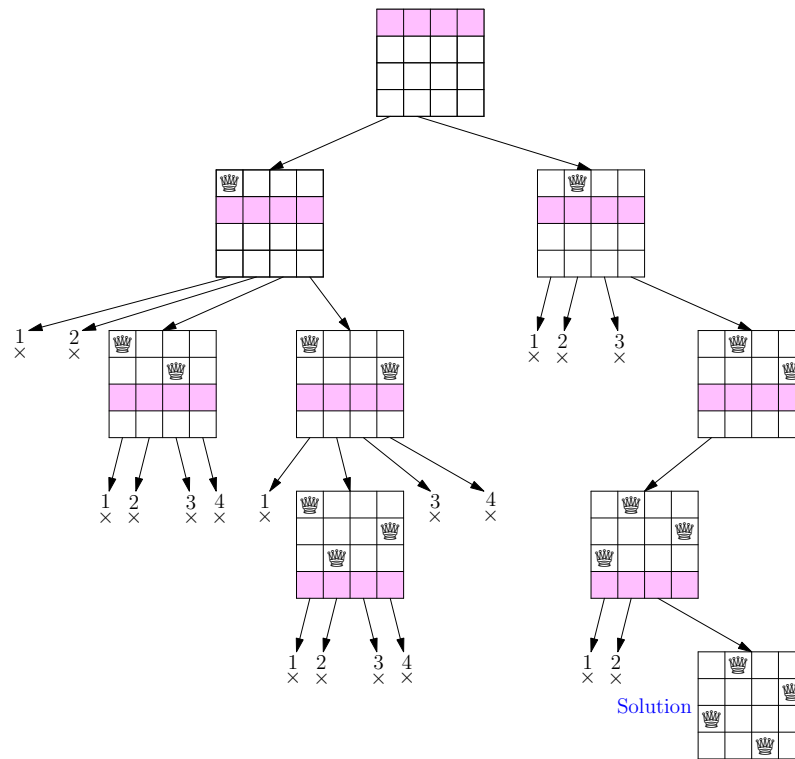


Algorithms and Analysis

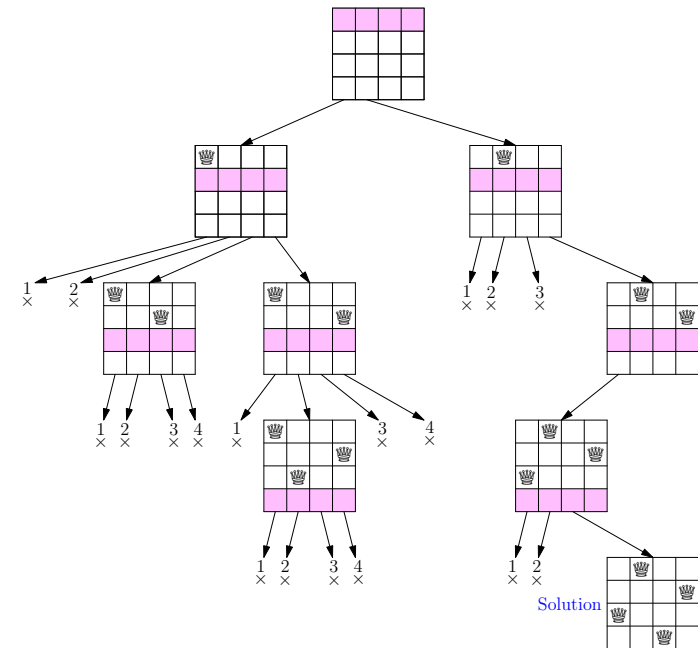
Lesson 20: *Know how to Search*



Backtracking, Branch and Bound

Outline

1. **Search Trees**
2. Backtracking
3. Branch and Bound
4. Search in AI



State Space Representation

- Many real world problems involve taking a series of actions to manipulate the state of the system
- This is the area of planning and search which sits within the domain of artificial intelligence
- One of the key props to help us develop algorithms is to think of the states as nodes of a graph which are linked if there exists an action taking us from one state to another
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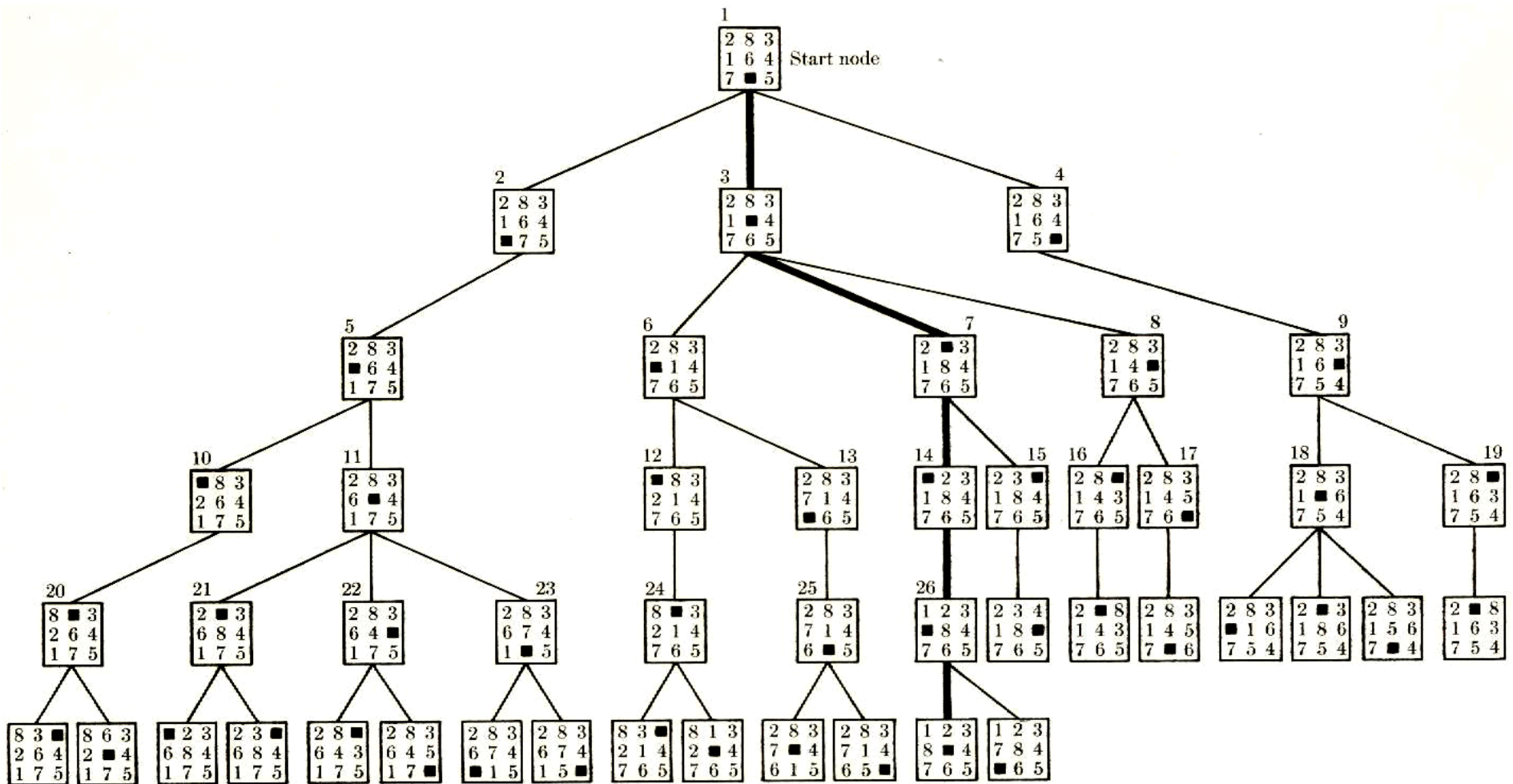
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8-Puzzle Example



Large State Spaces

- The search space typically increase exponentially with the problem size
- We can find the quickest solution to the 8-puzzle (and the 15 puzzle) using breadth first search, but larger puzzles soon become intractable
- Nevertheless, a lot of important problems involve very large state spaces and we have to find algorithms to explore them

Large State Spaces

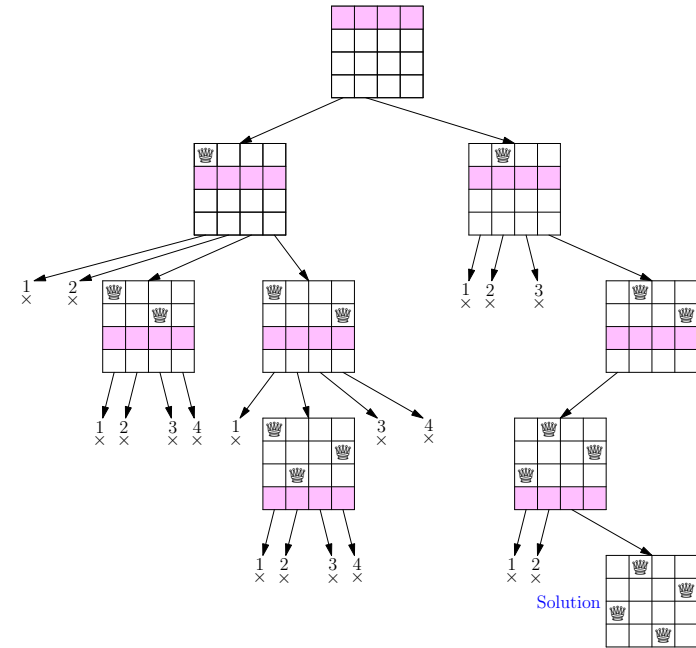
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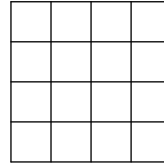
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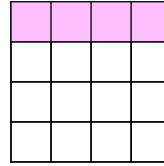
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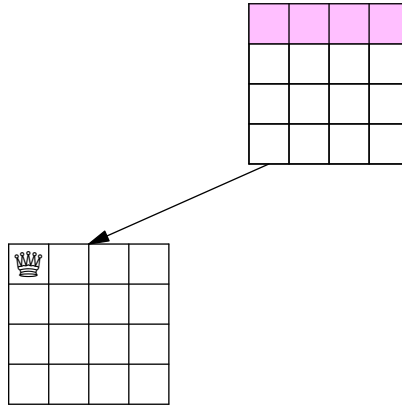
4-Queens Problem



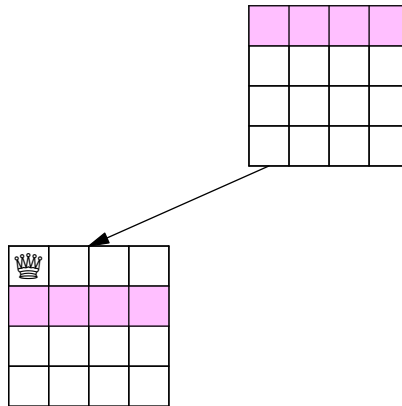
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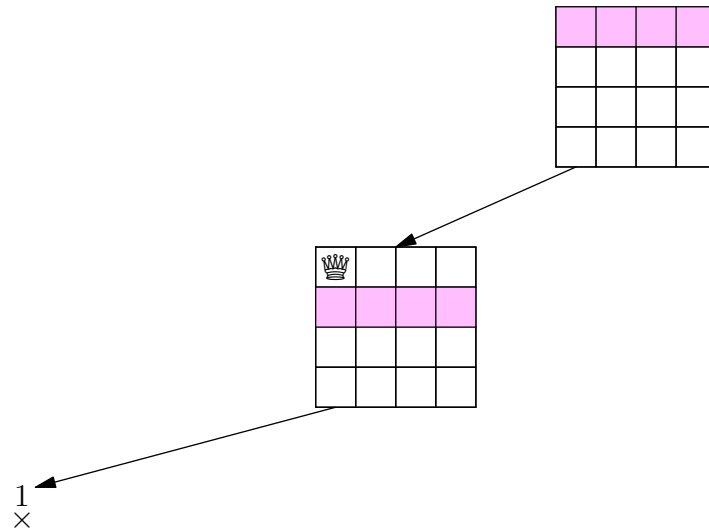
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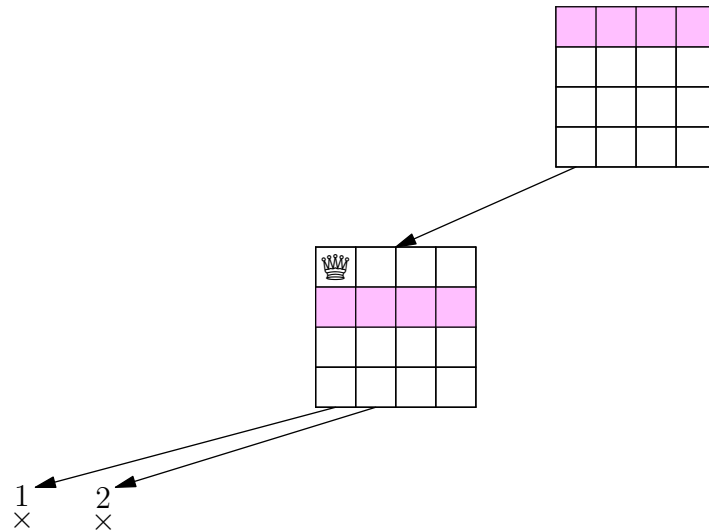
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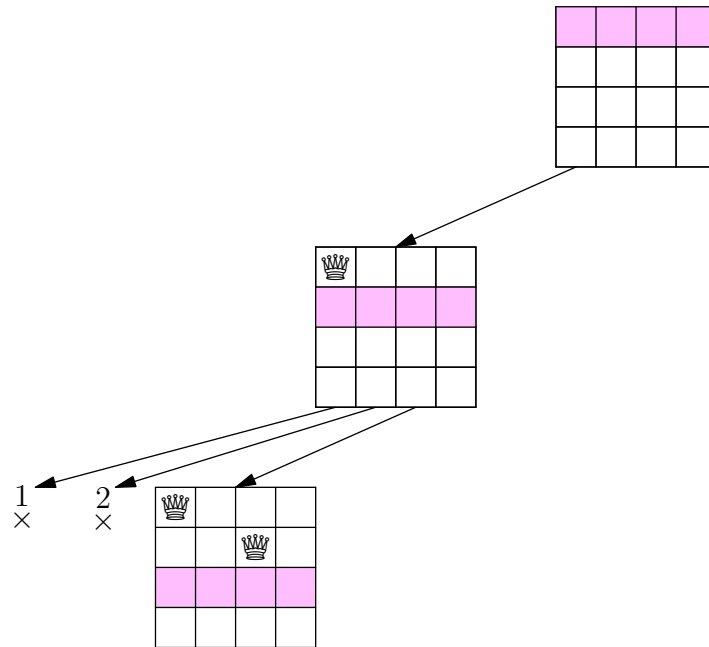
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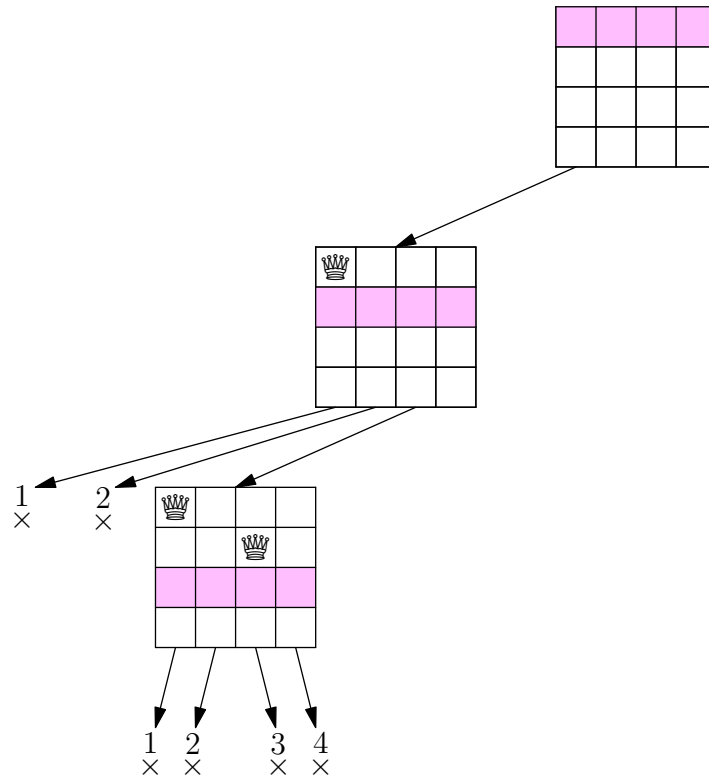
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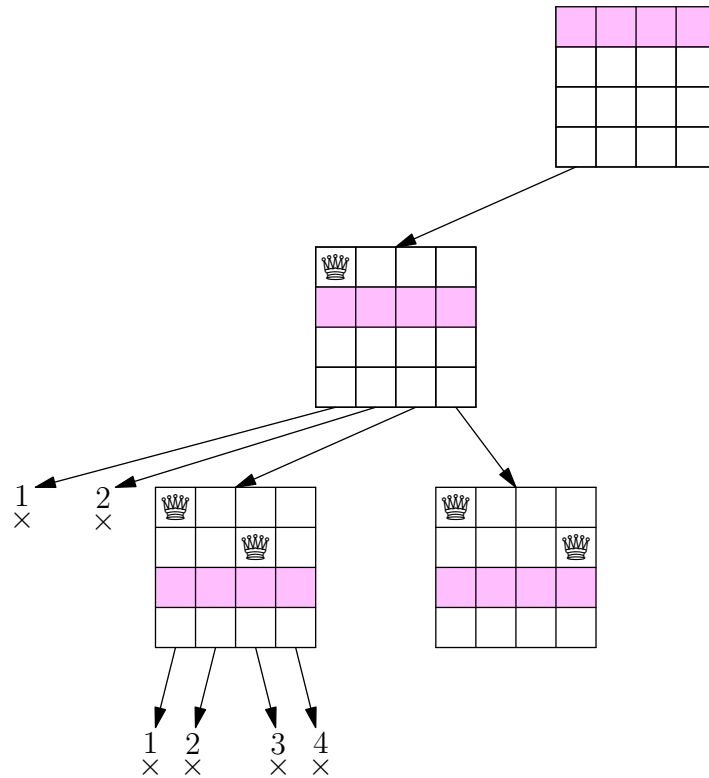
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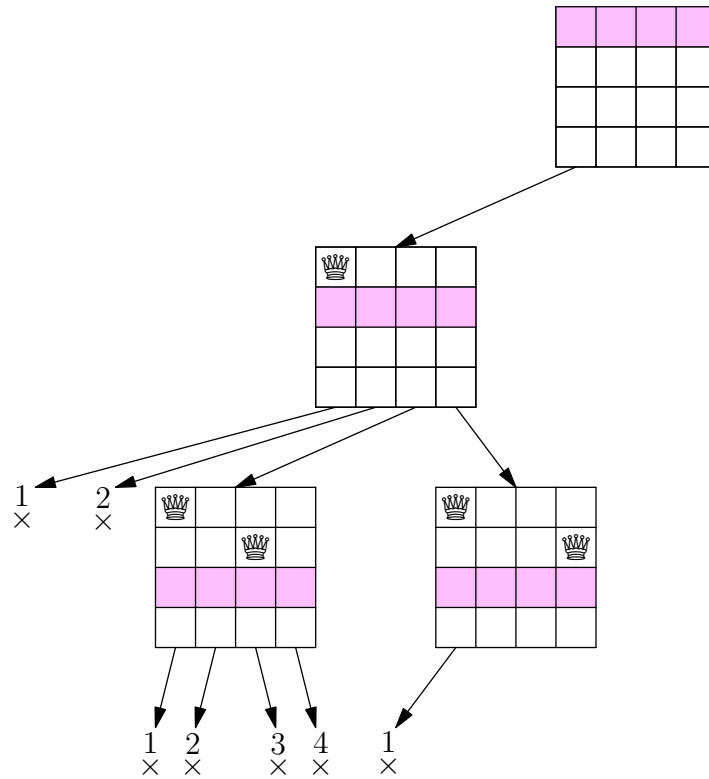
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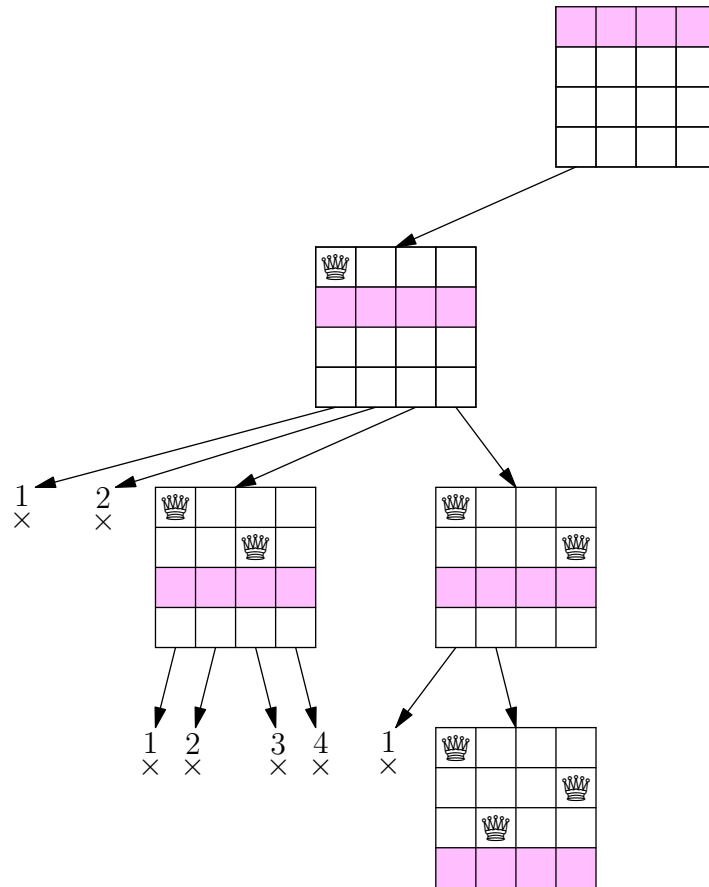
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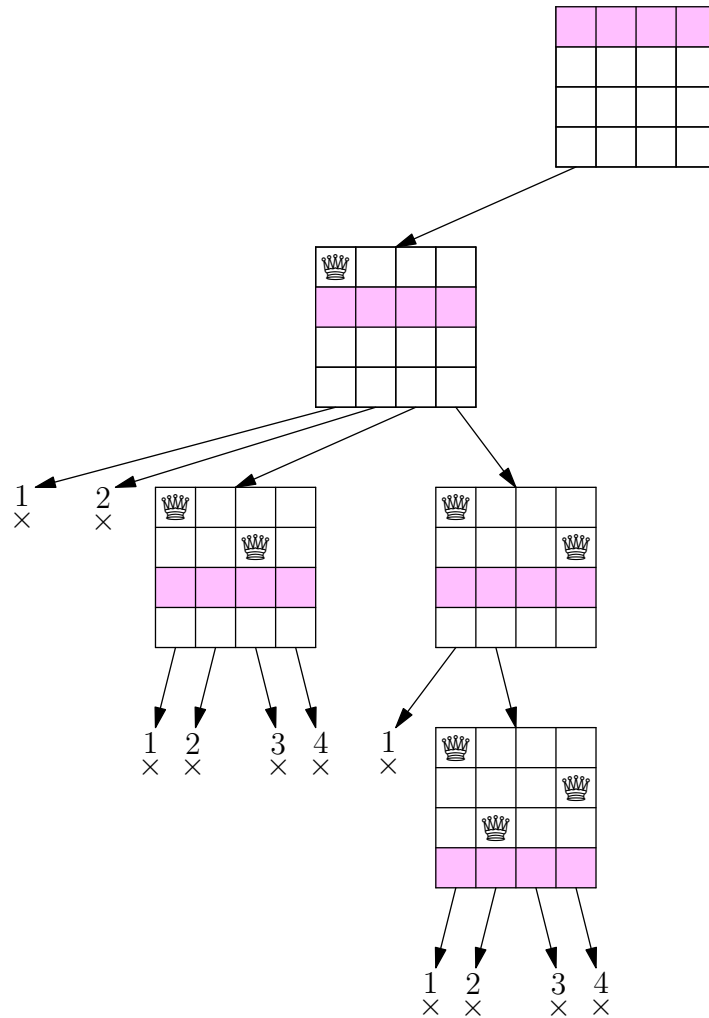
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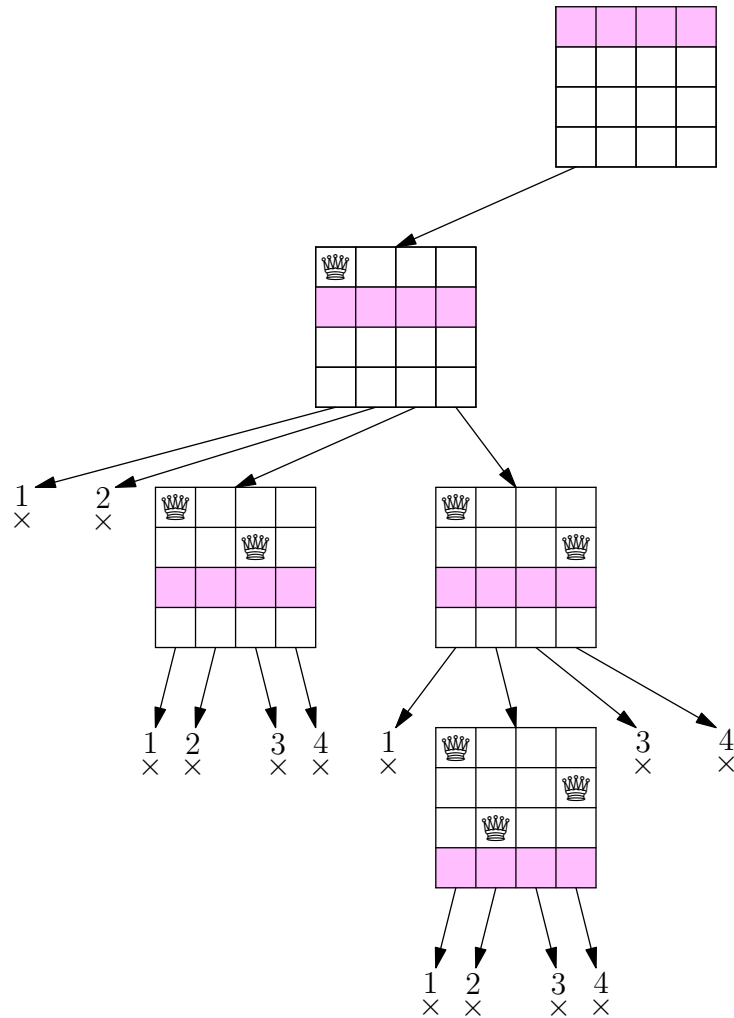
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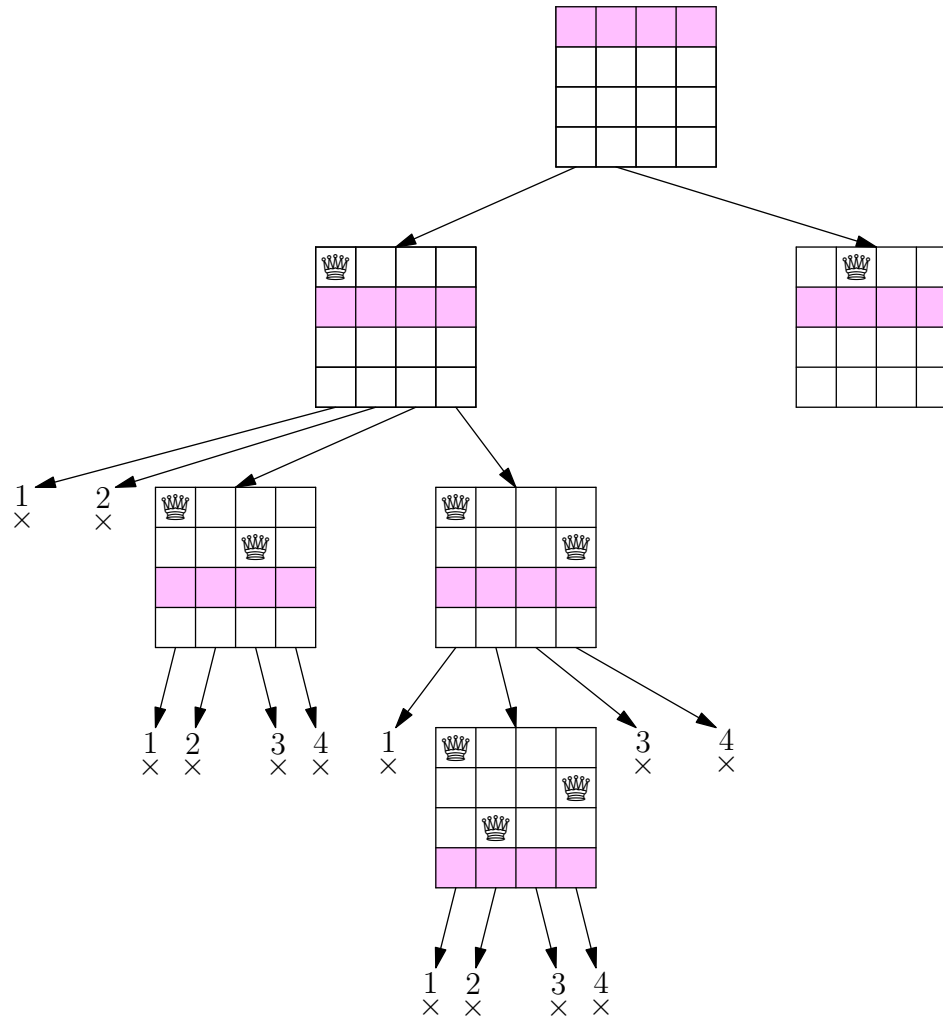
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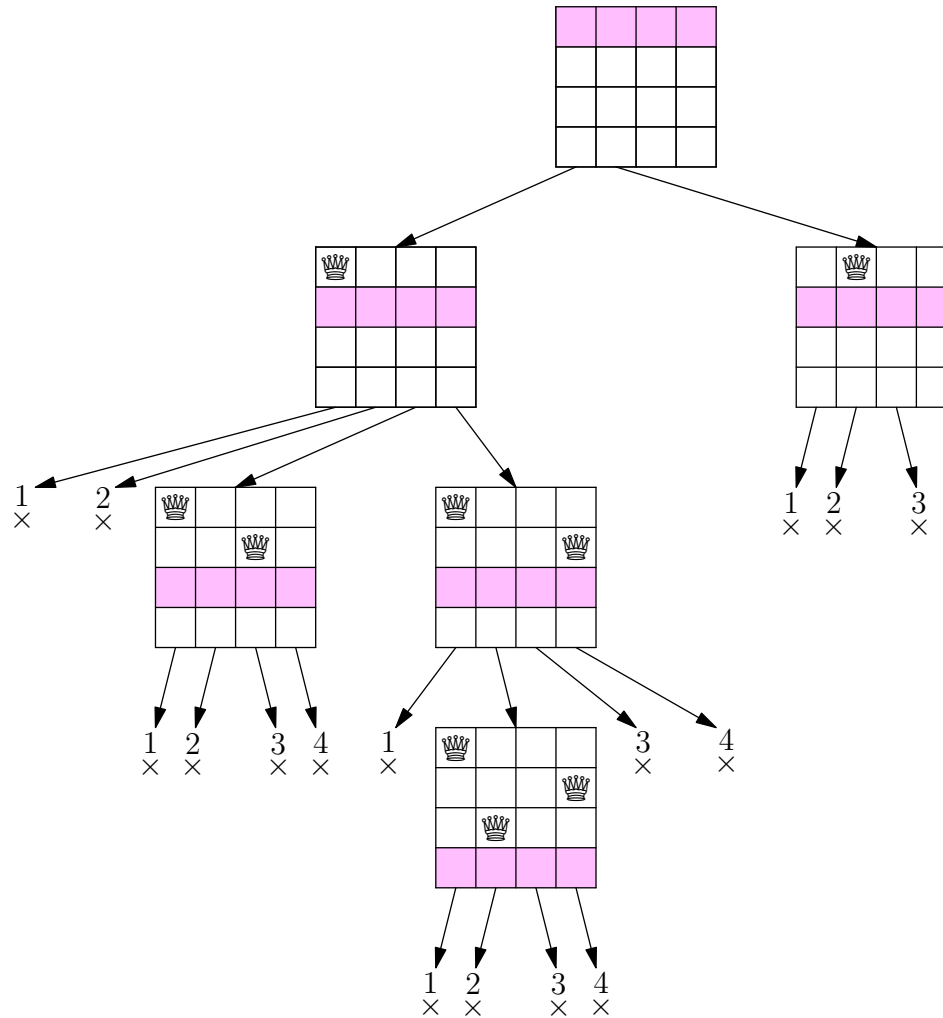
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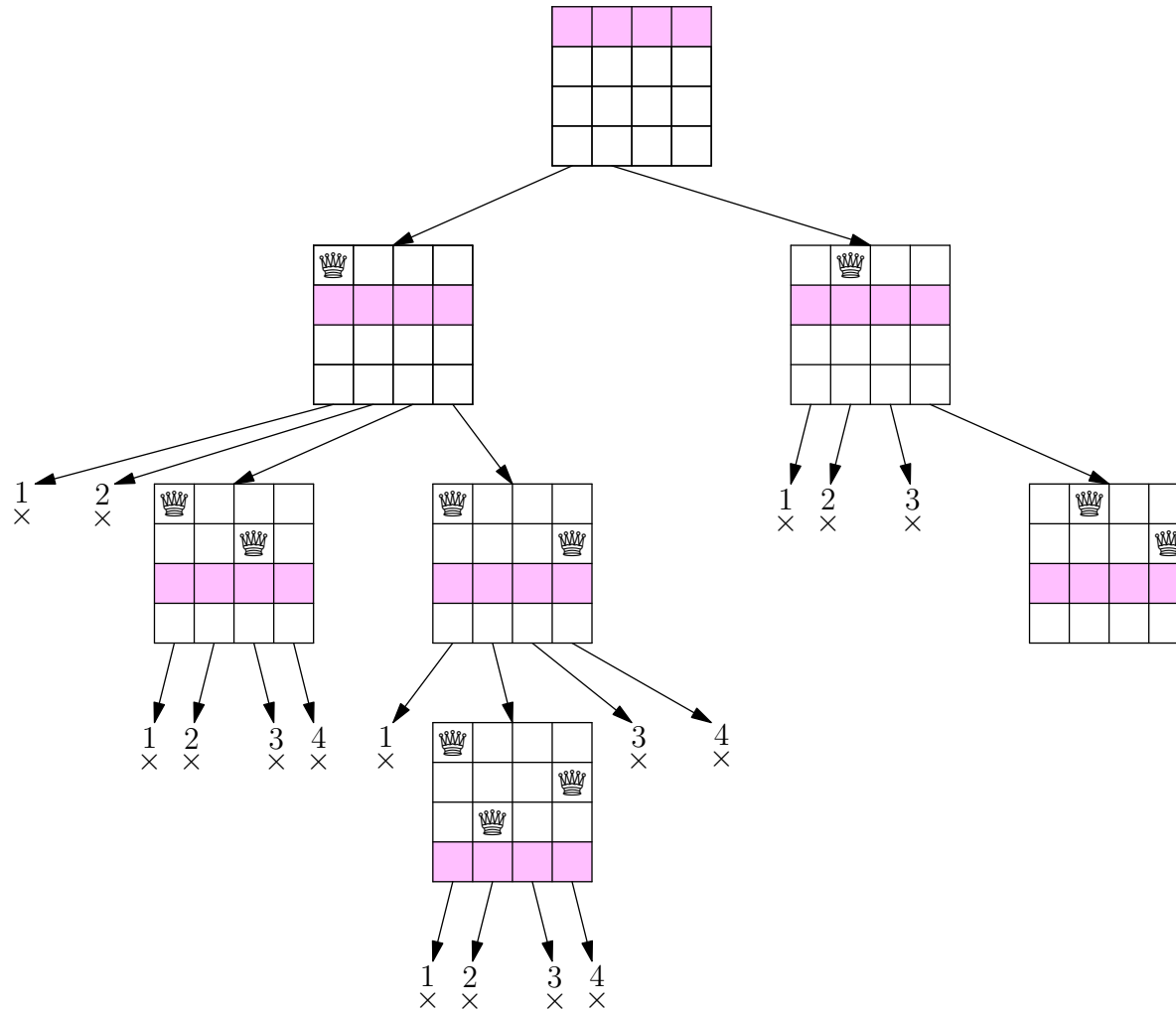
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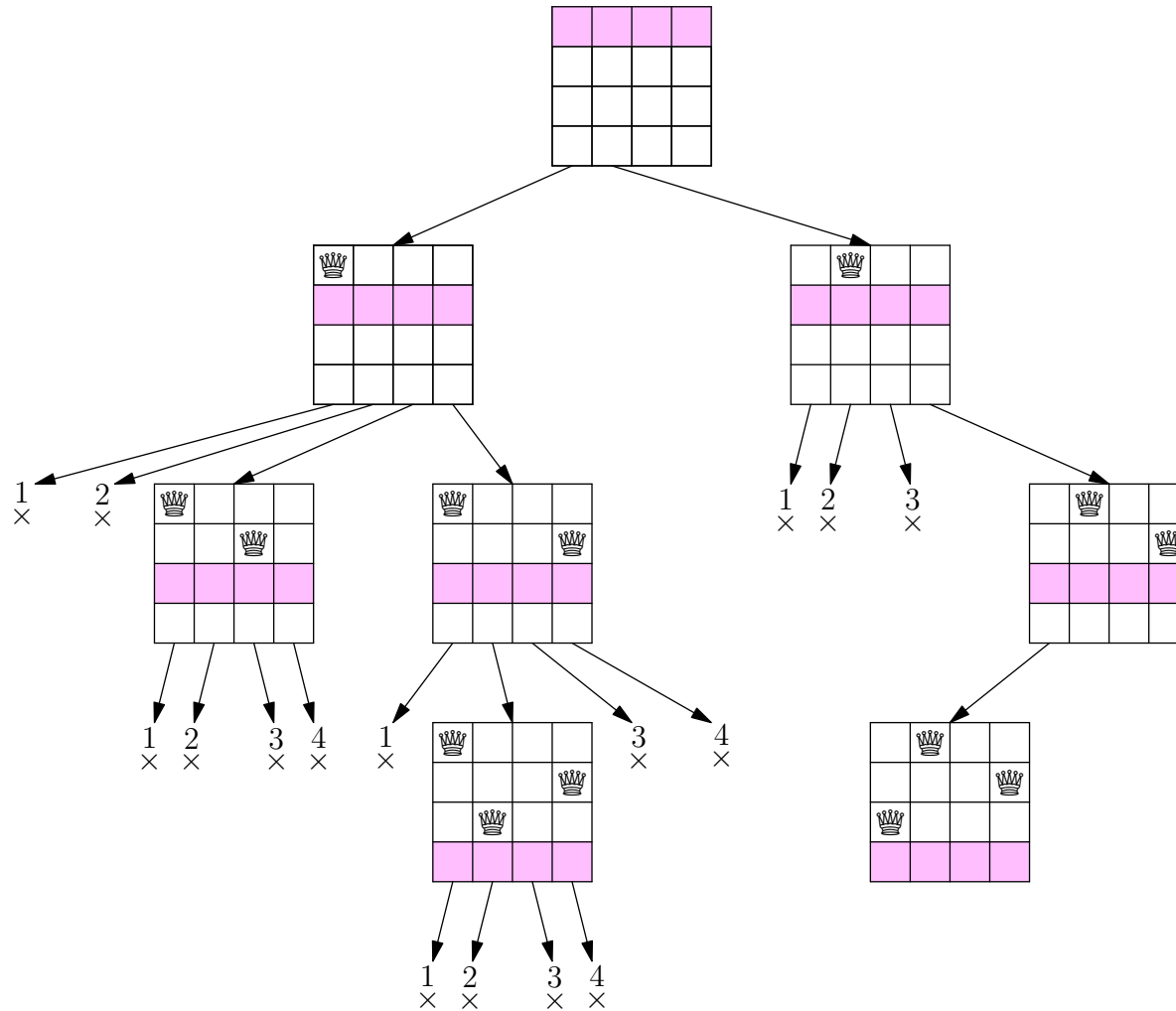
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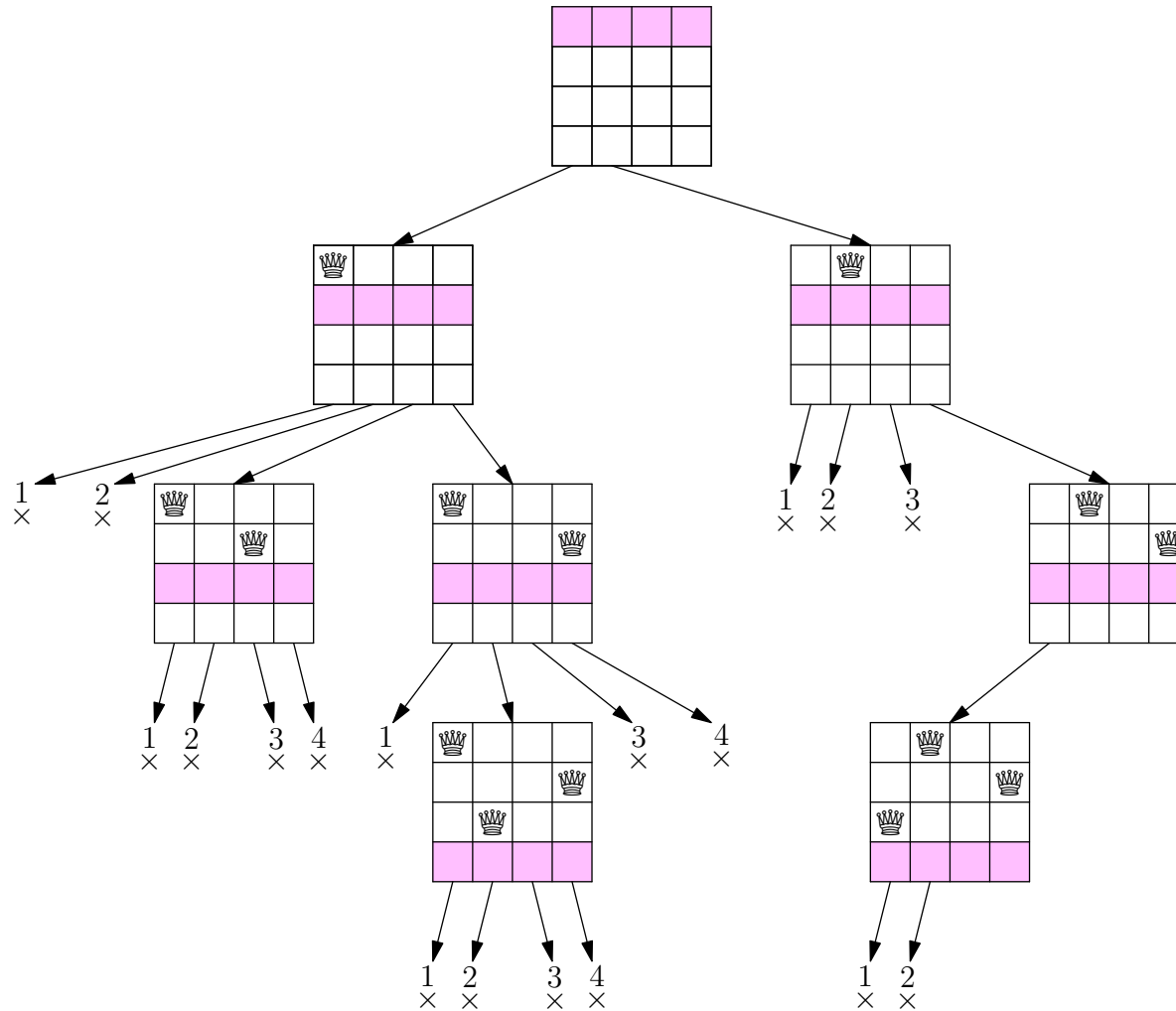
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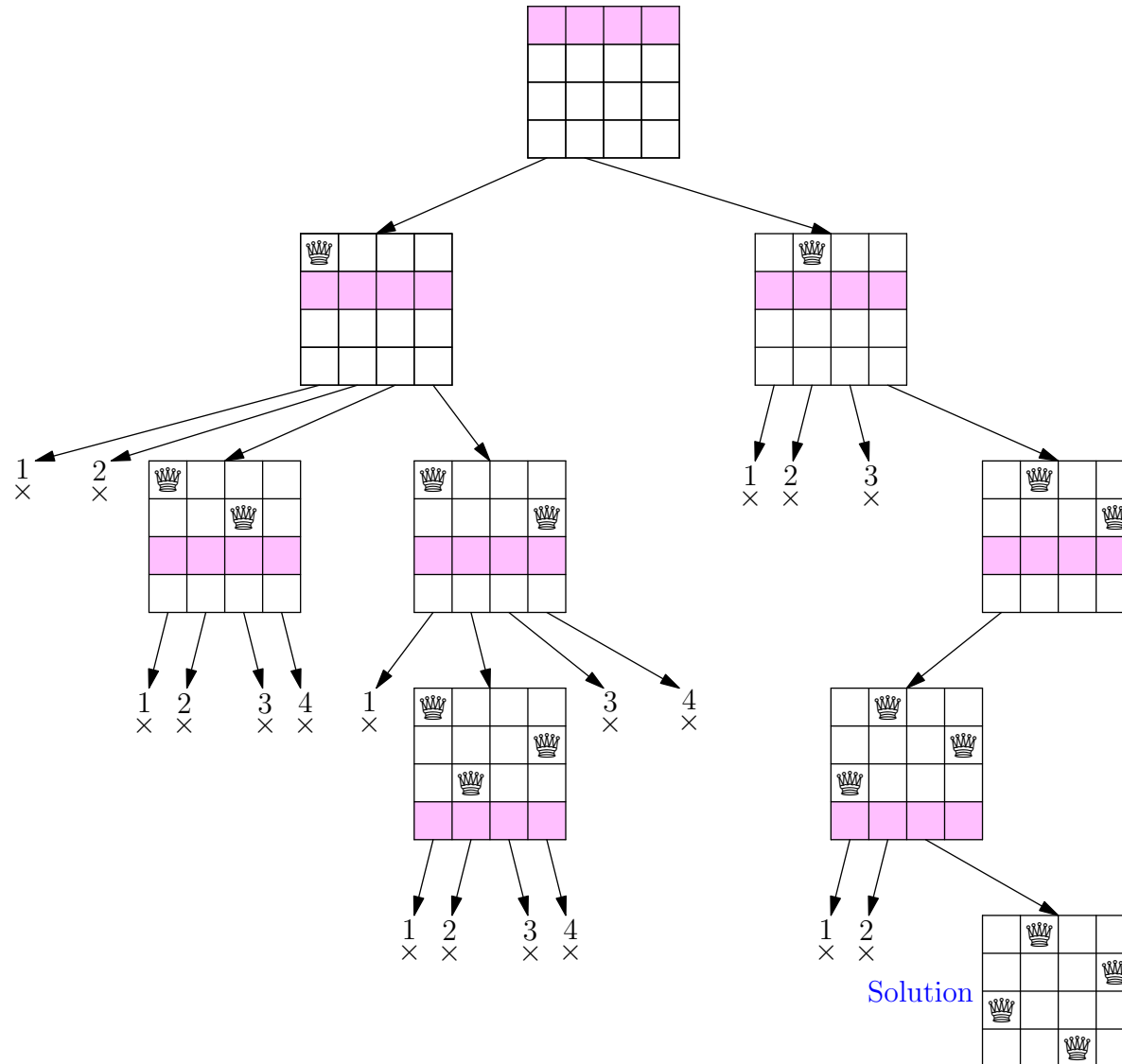
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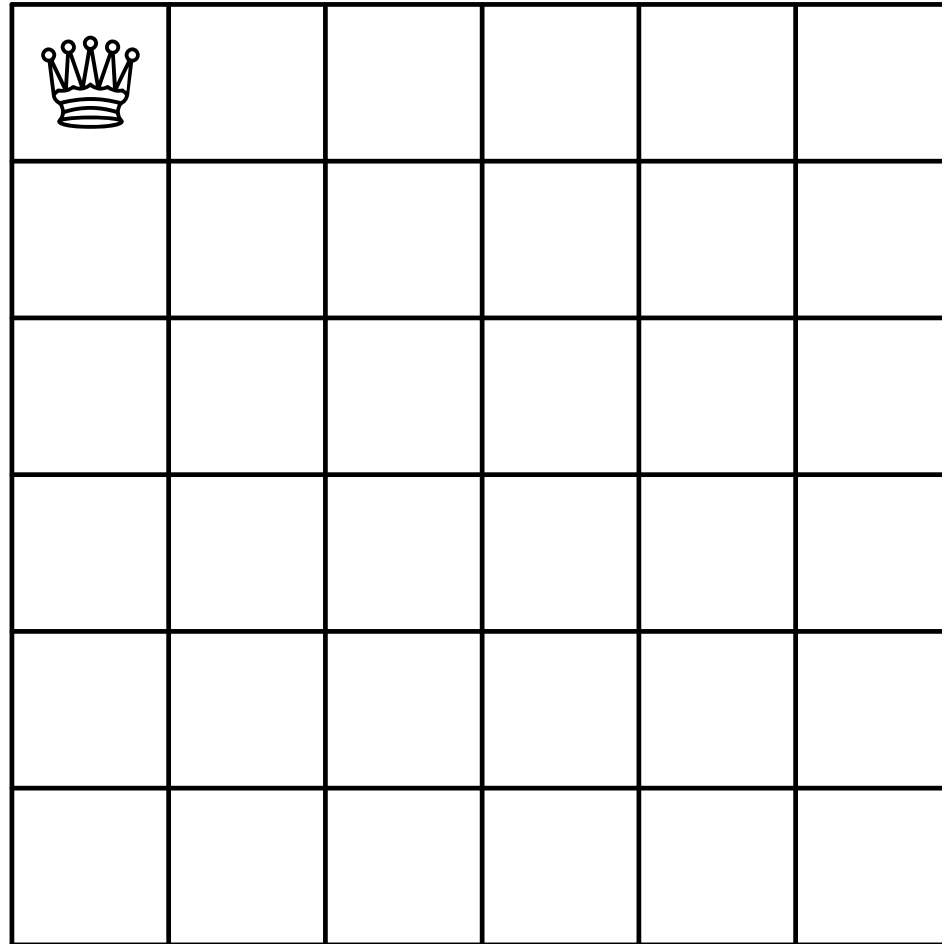


4-Queens Problem

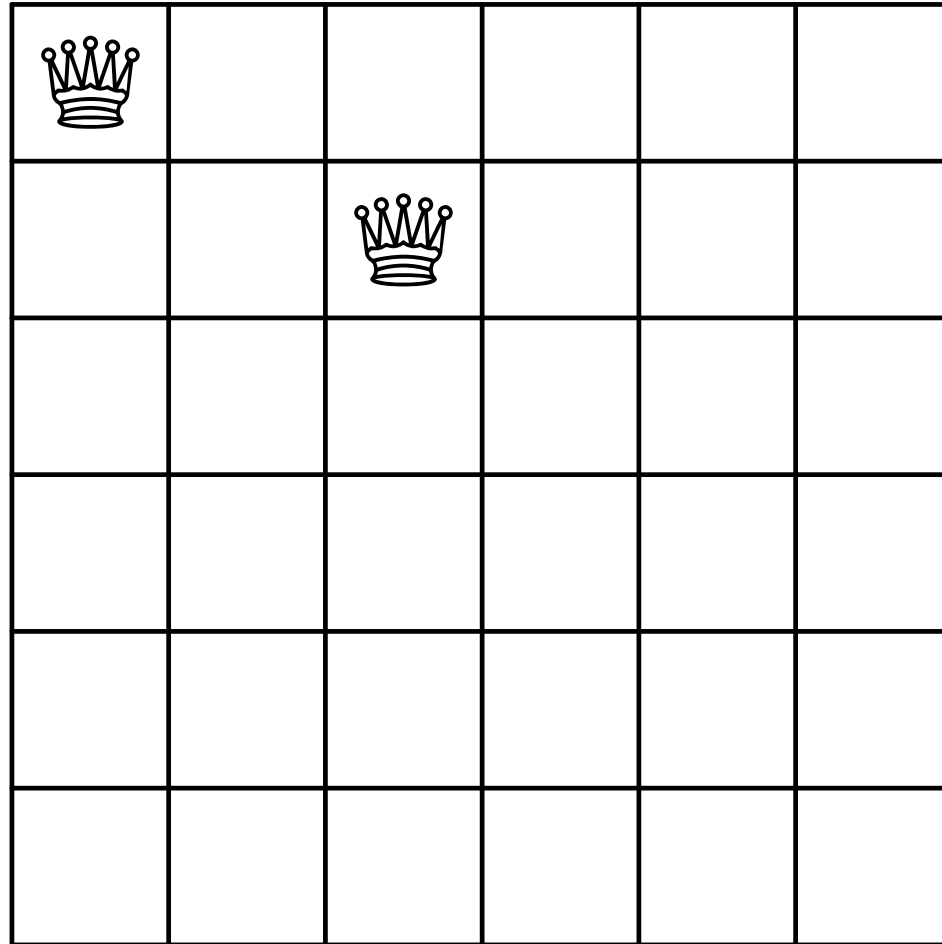


6-Queens Problem

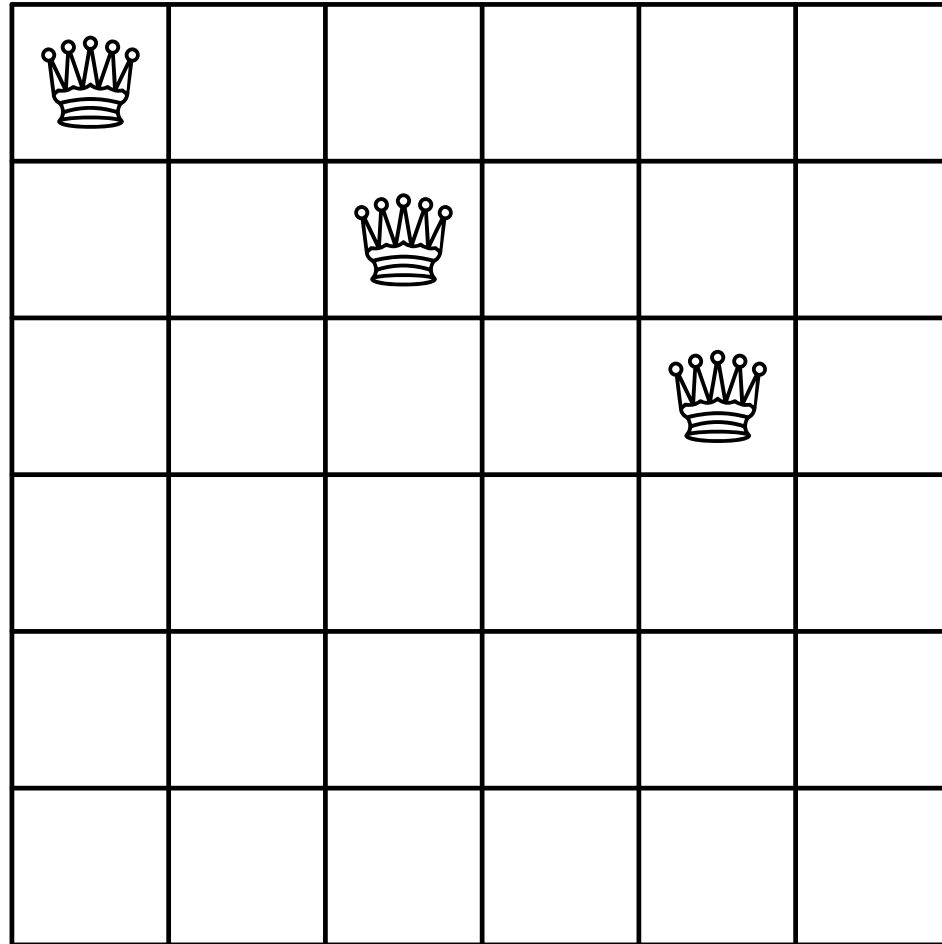
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Implementing n -Queens

- Implementing backtracking is easily done using recursion
- Recall depth-first search is easily implemented using recursion
- We just need a recursive function `next(n, row, sol)` which for a n -Queens problem searches new solutions in `row` given queens in previous rows given in `sol`
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Code

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List nextRow(int noRows, int row, List queenPositions) {
    if (row==noRows) {return queenPositions;}
    for (int col=0; col<noRows; ++col) {
        if (legalQueen(col, row, queenPositions)) {
            queenPositions.add(col);
            List solution = nextRow(noRows, row+1, queenPositions);
            if (solution!=null)
                return solution;
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    return null;
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bool LegalQueen(int col, int row, List sol) {
    for(int r=0; r<row: ++r) {
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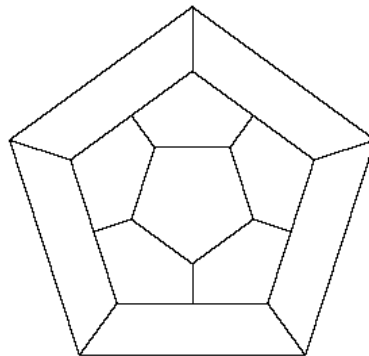
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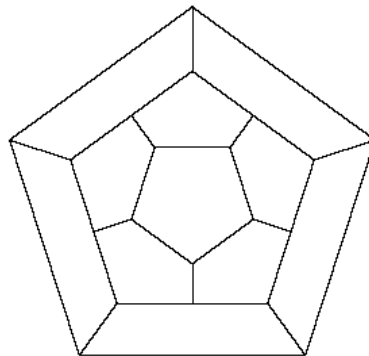
Hamiltonian Circuit

- A Hamiltonian cycle is a tour through a graph which visits every vertex once only and returns to the start
- It is a hard problem in that there are no known algorithms that are guaranteed to find a Hamiltonian cycle in polynomial time
- For many graphs it is not too hard



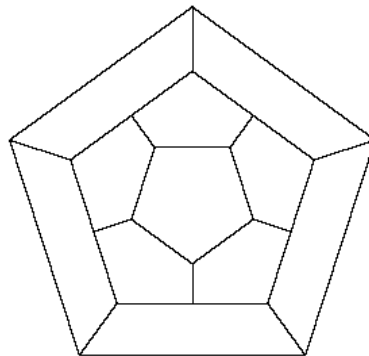
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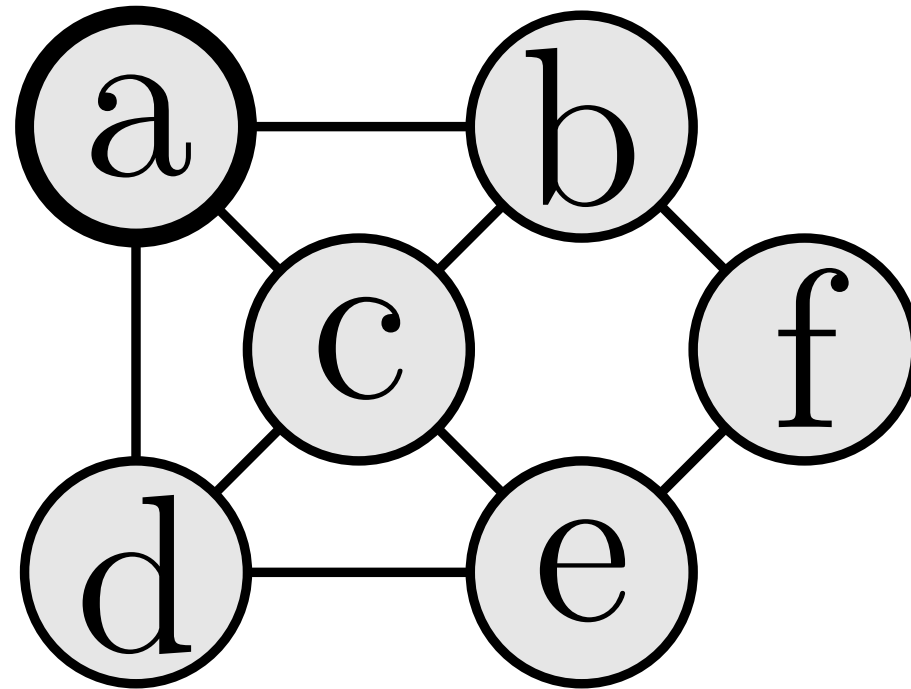


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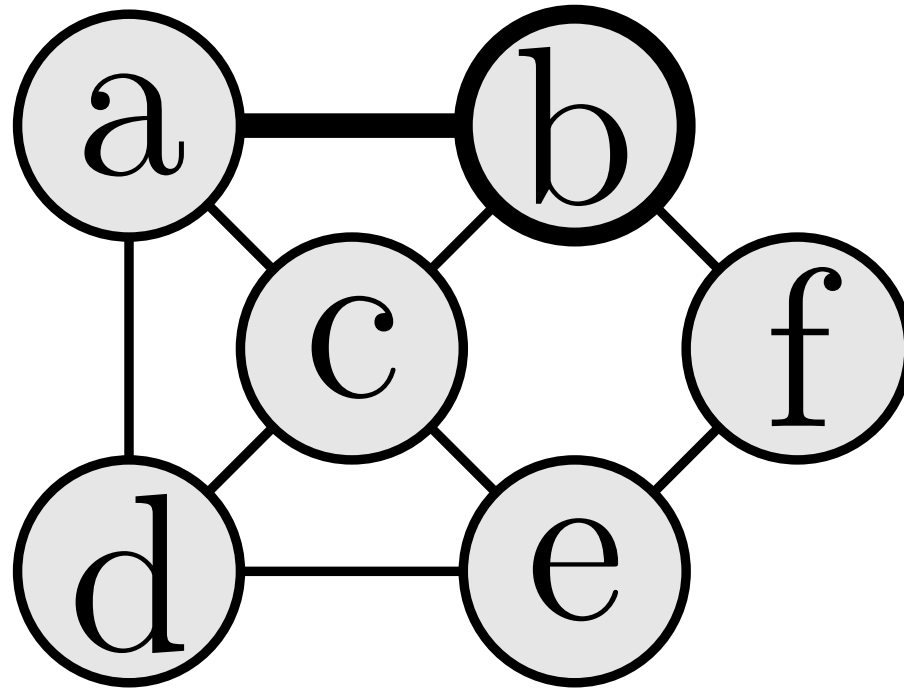
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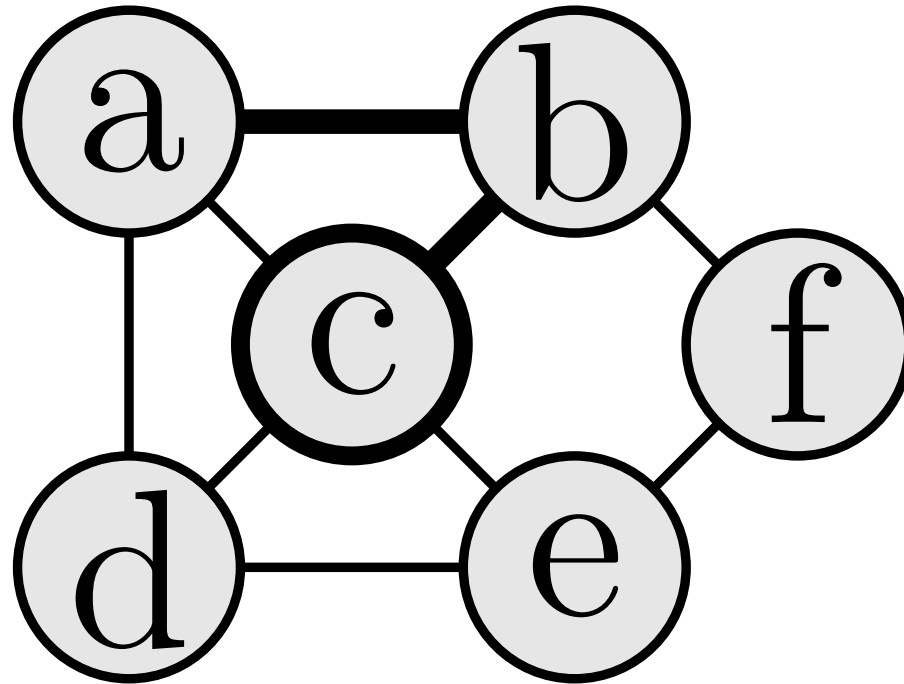
Hamiltonian Circuit Example



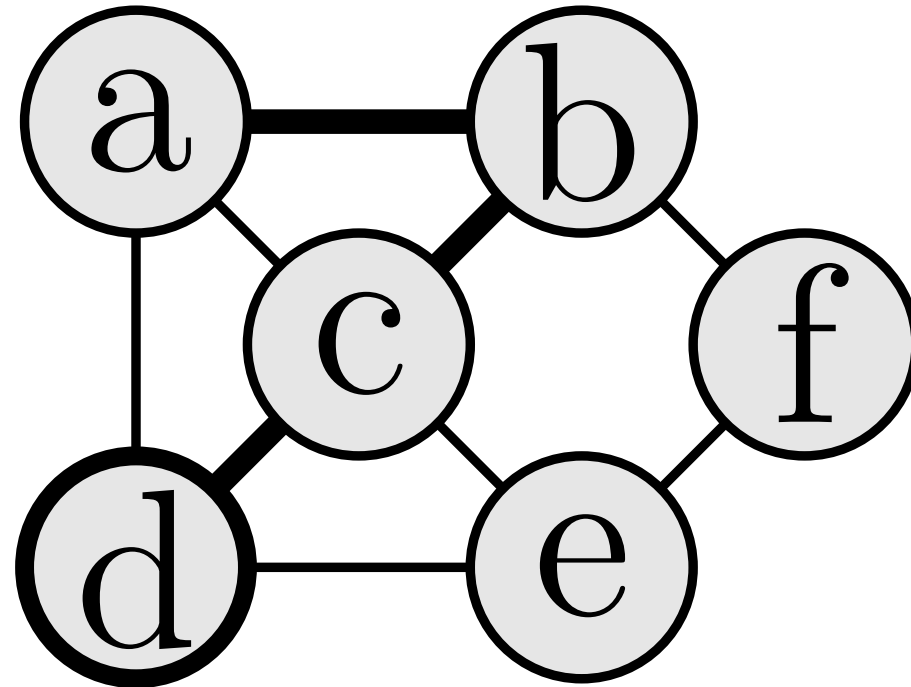
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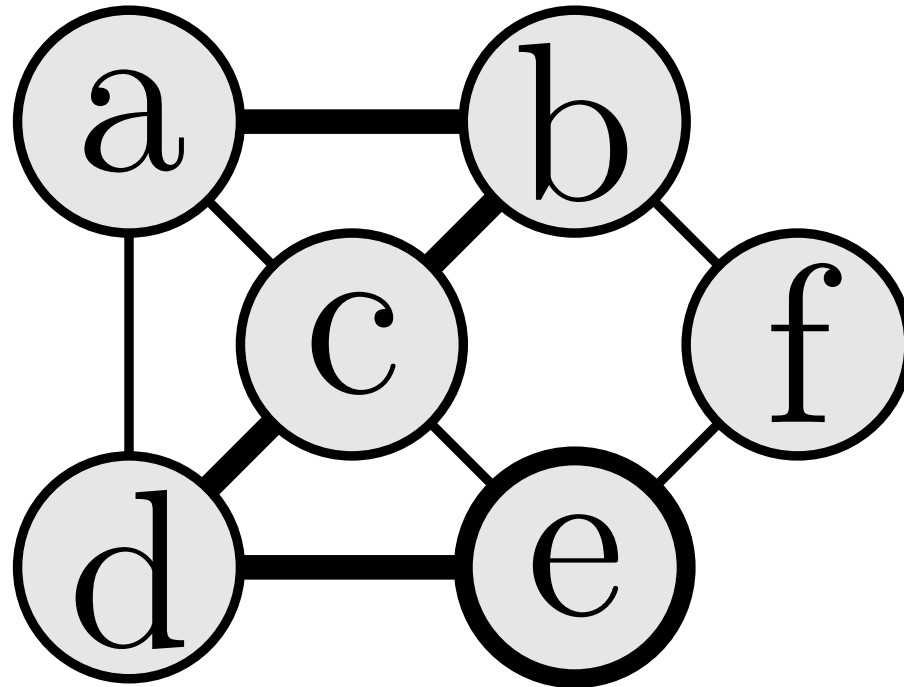
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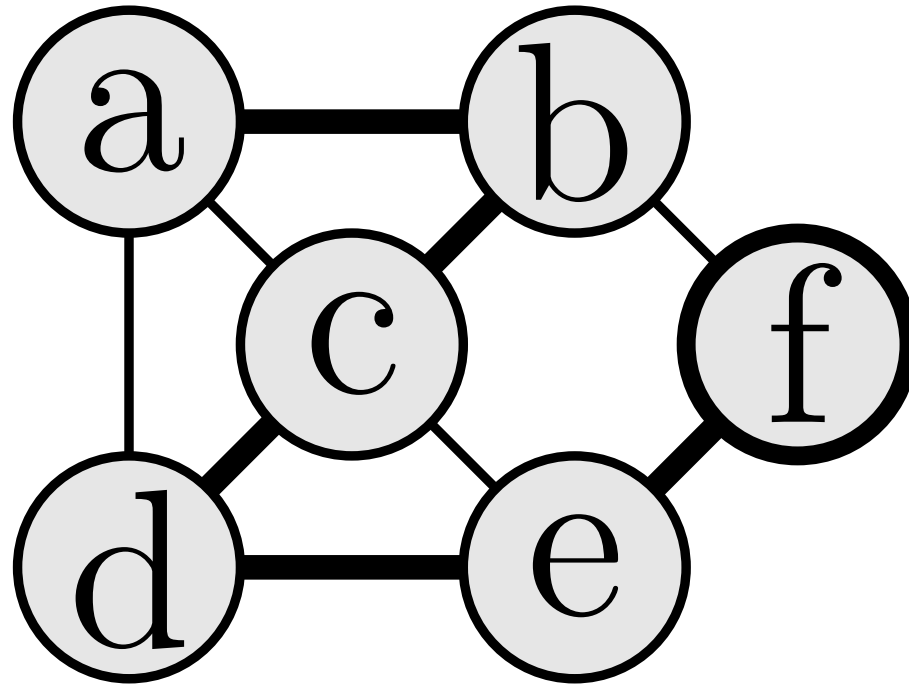
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Backtracking

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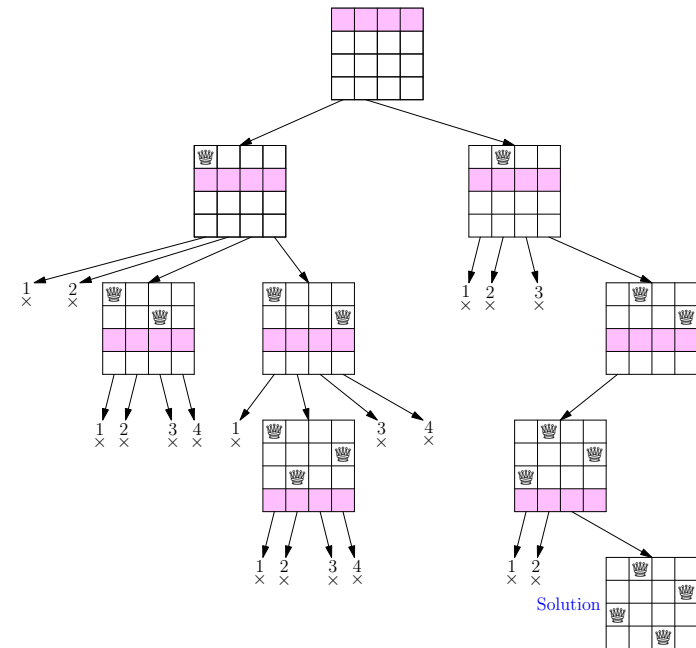
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2. Backtracking
3. **Branch and Bound**
4. Search in AI



Optimisation Problems

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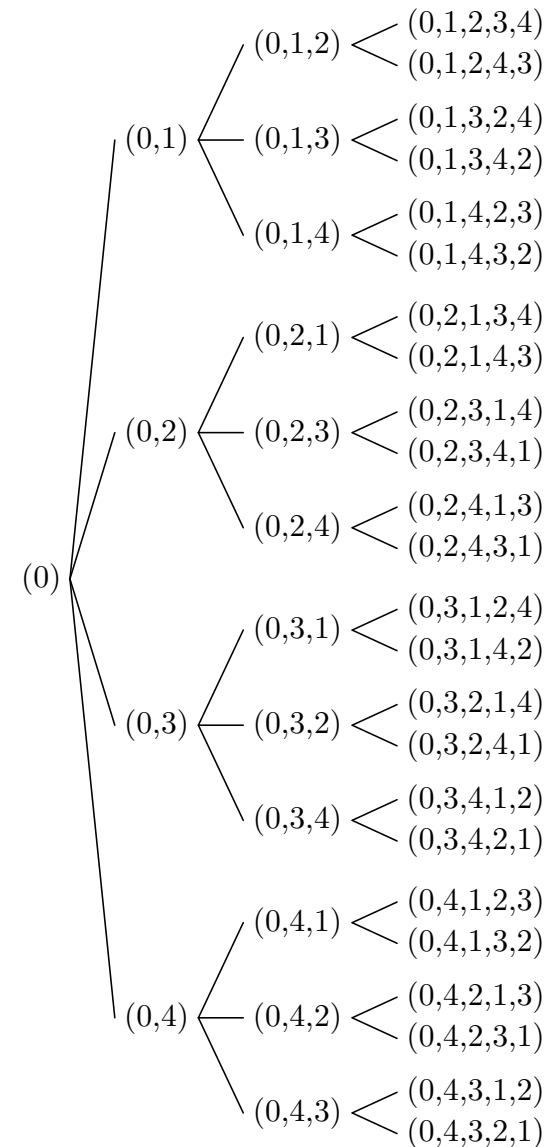
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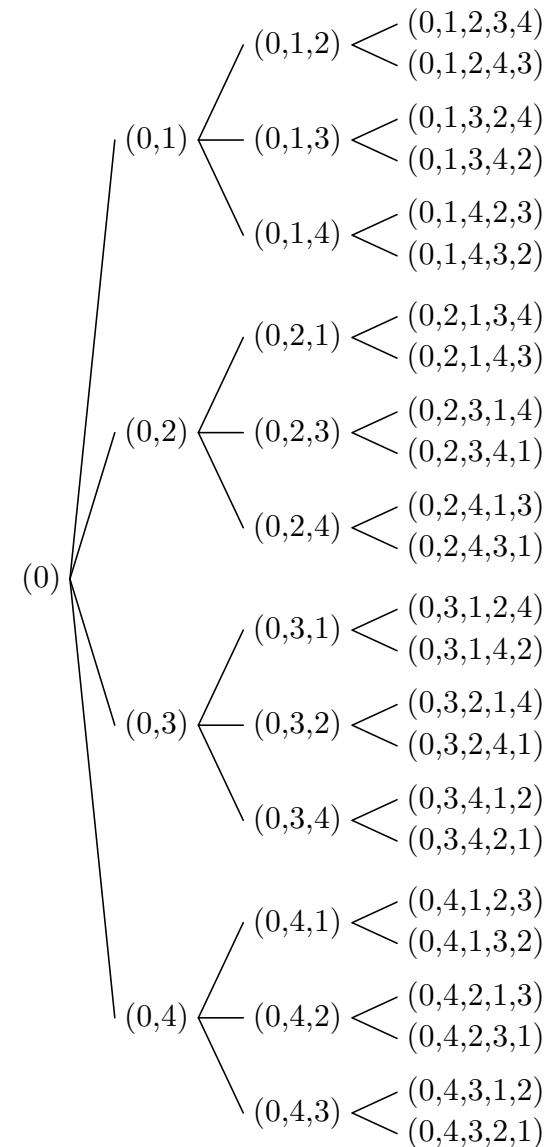
Cutting the Search Tree

- We can think of exact enumeration as exploring a giant search tree
- If we know a partial solution is worse than our bound we cut the search tree
- The earlier we cut the tree the more we can save



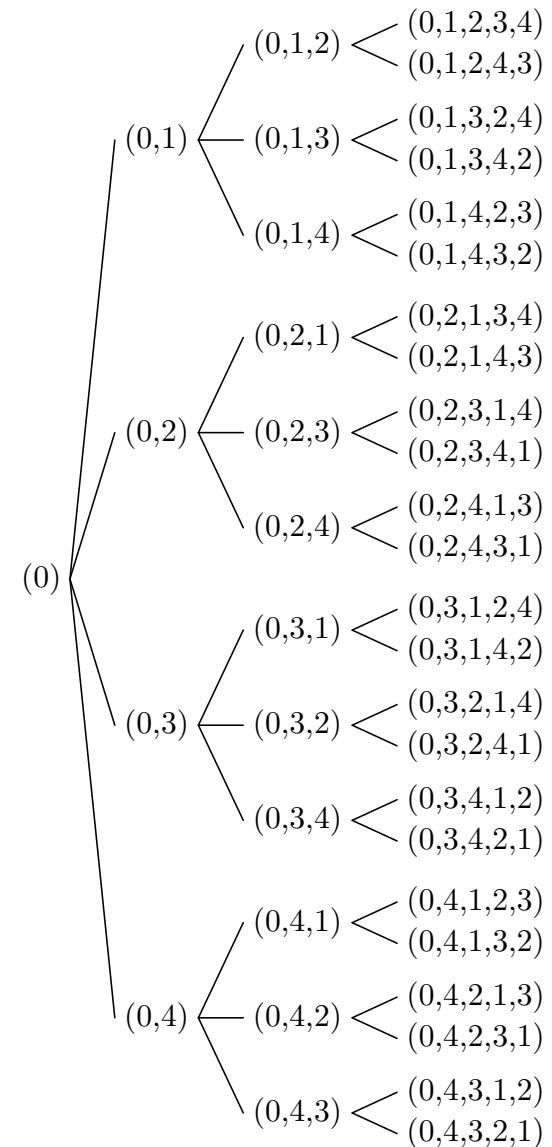
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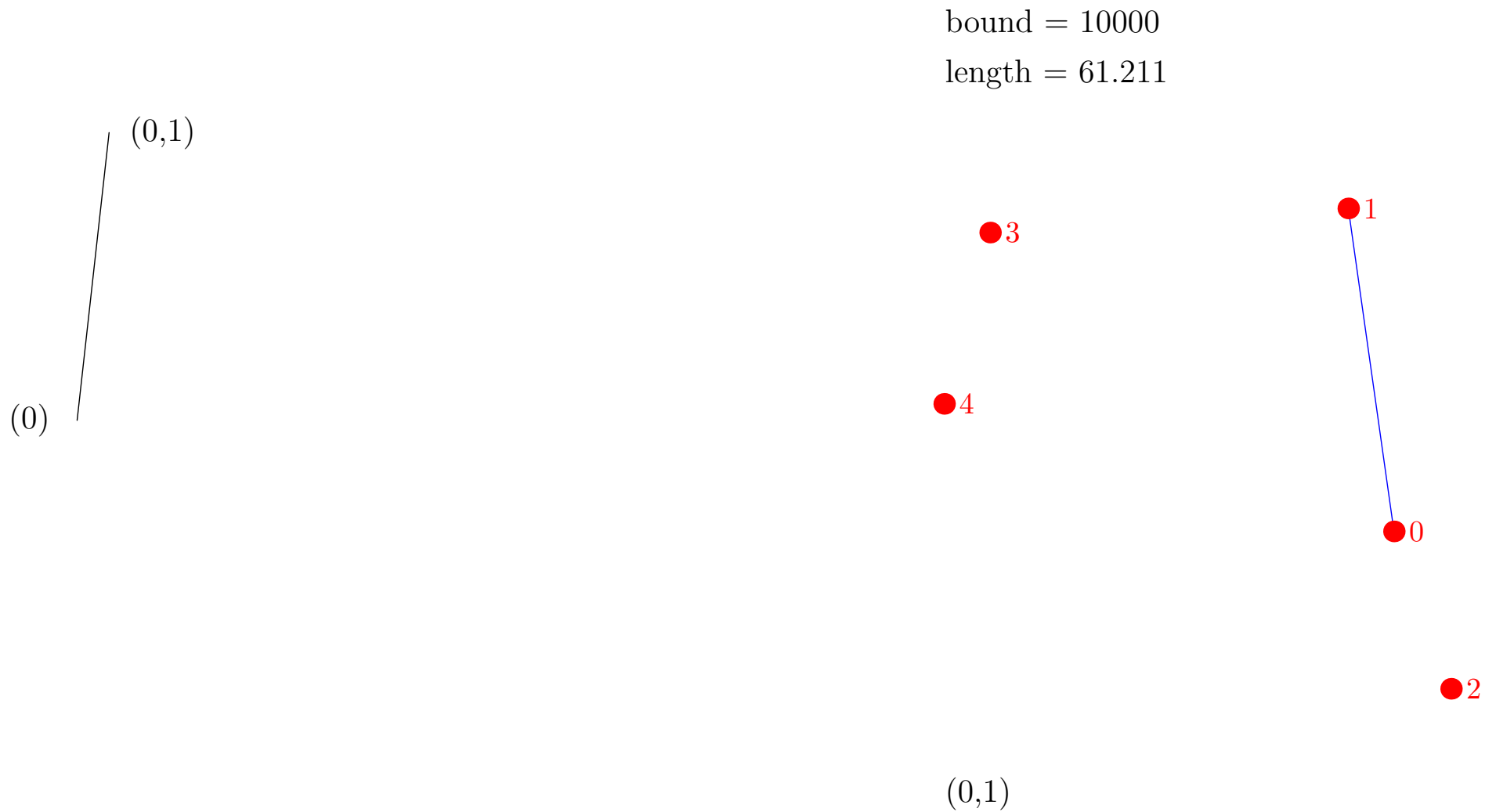


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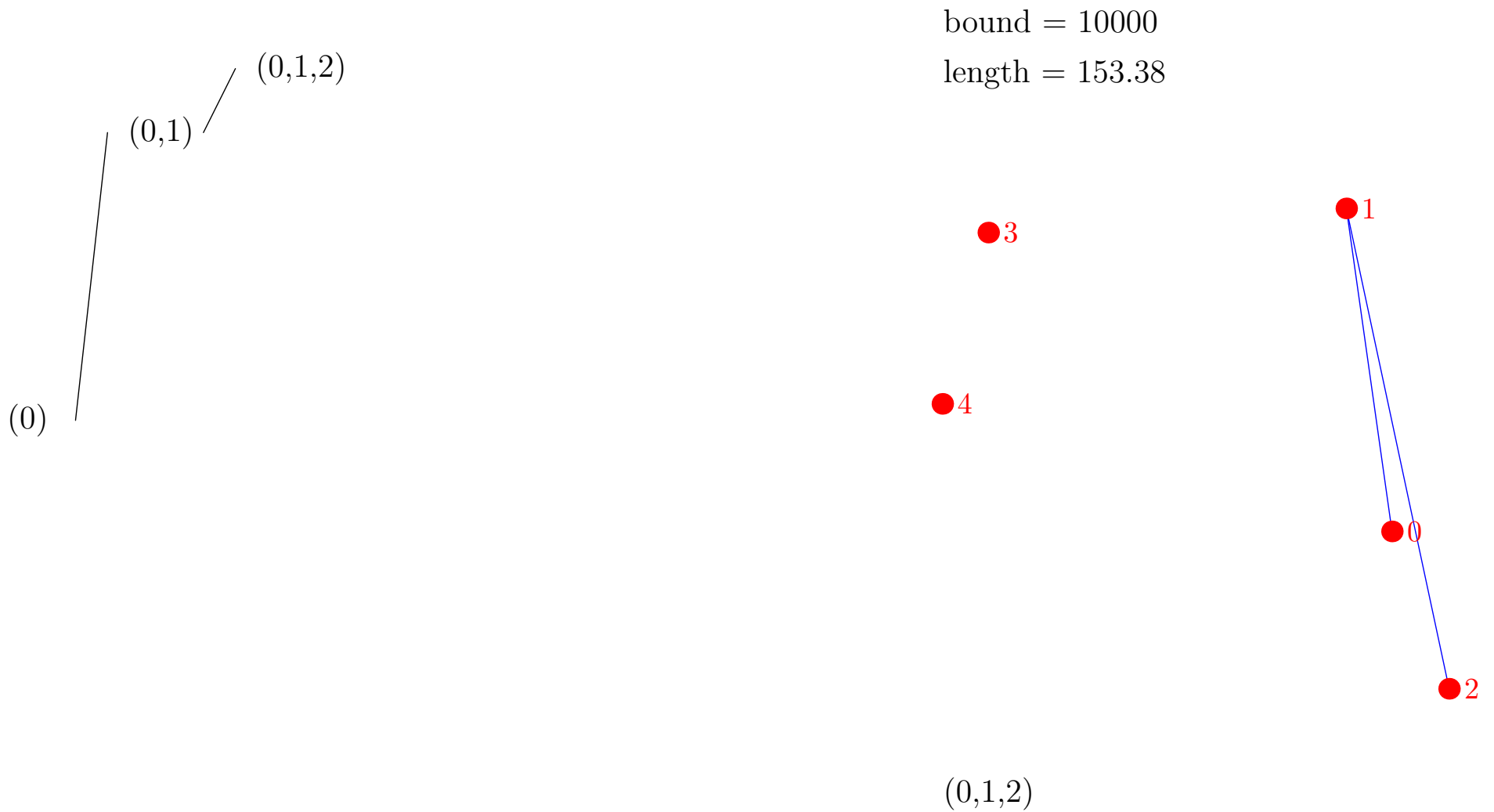
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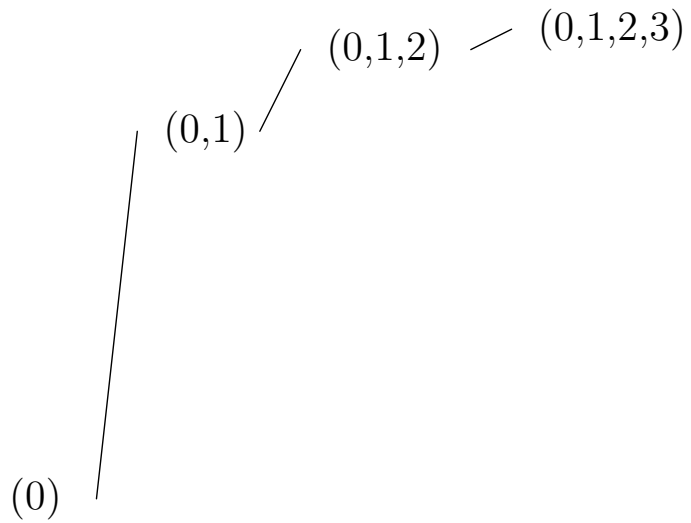
Branch and Bound in Action



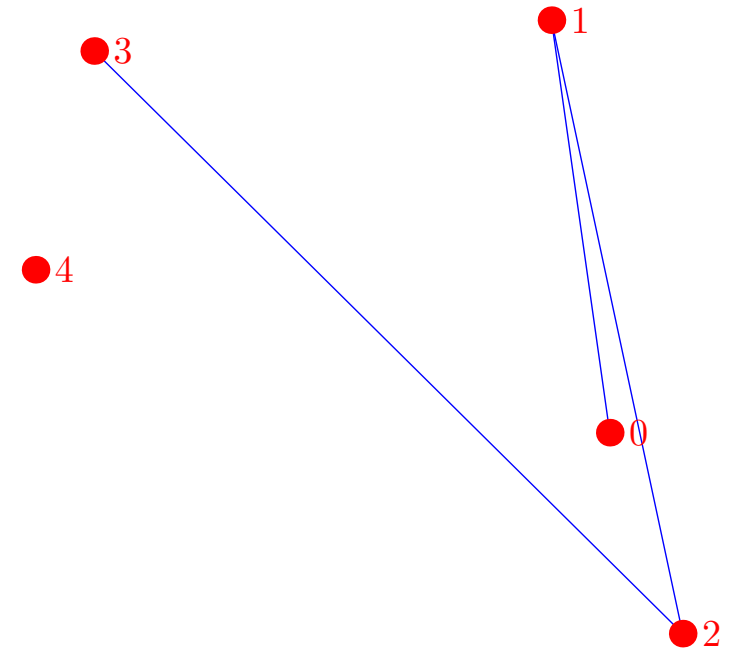
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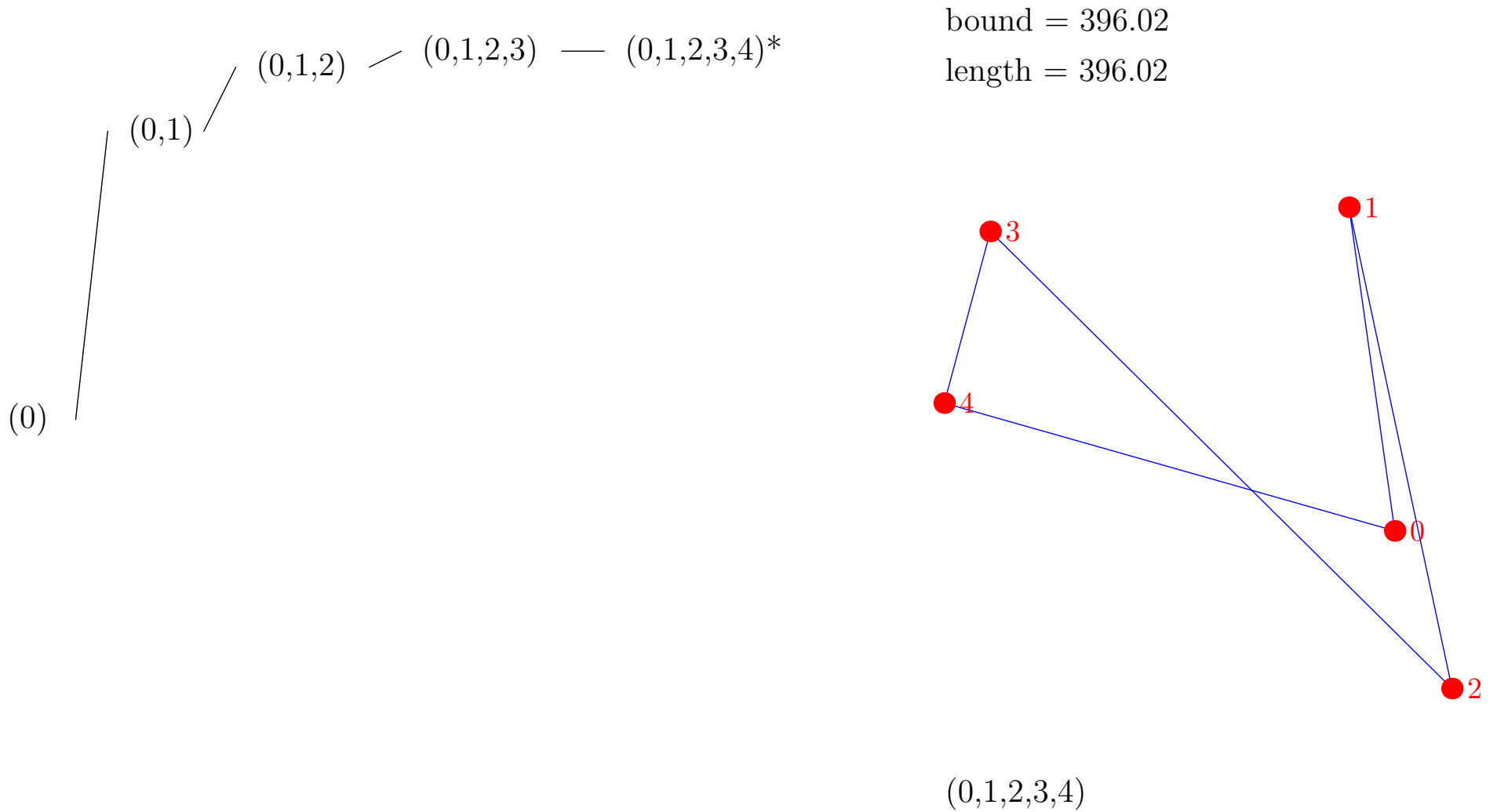


bound = 10000
length = 275.05

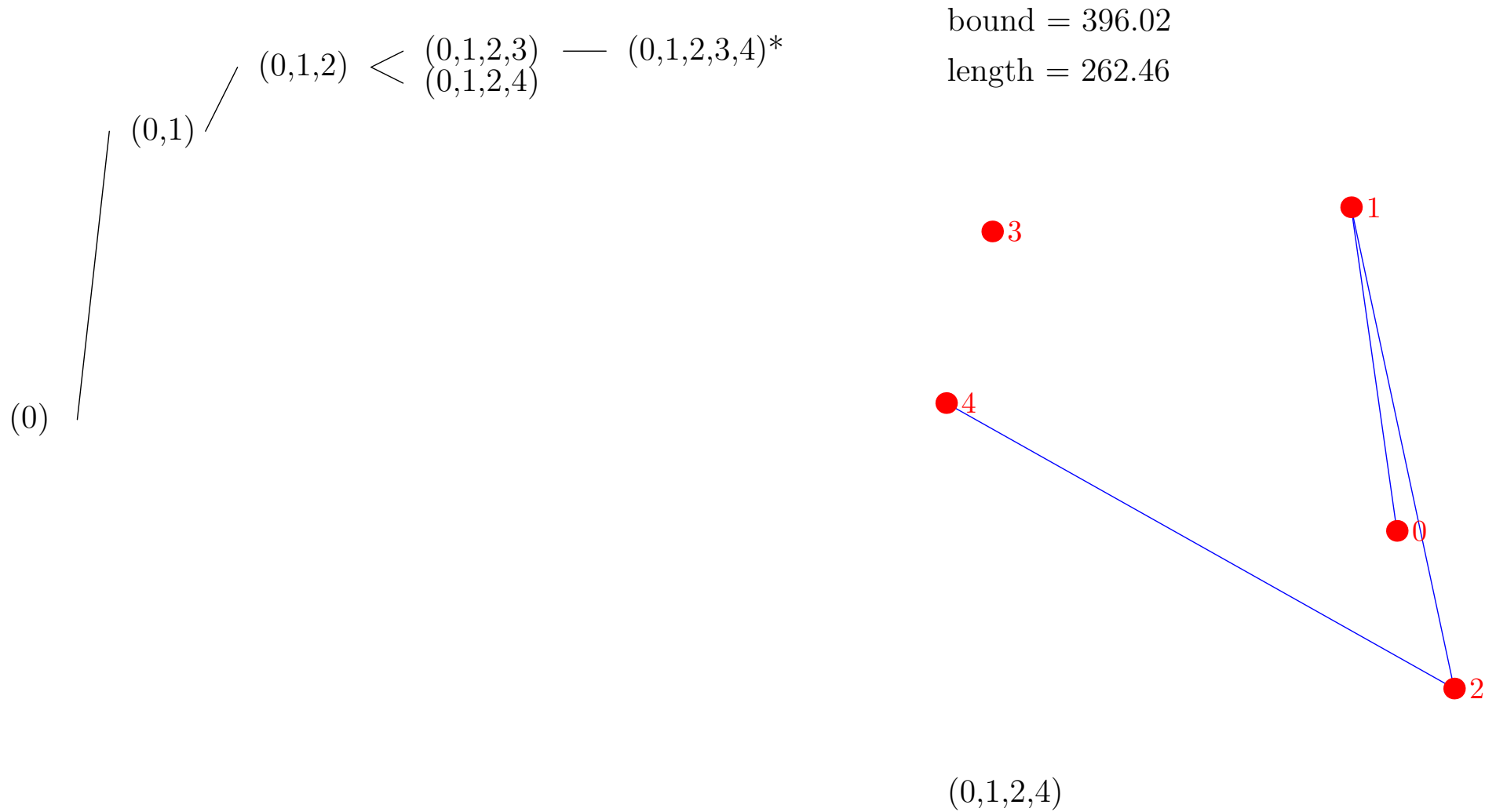


$(0,1,2,3)$

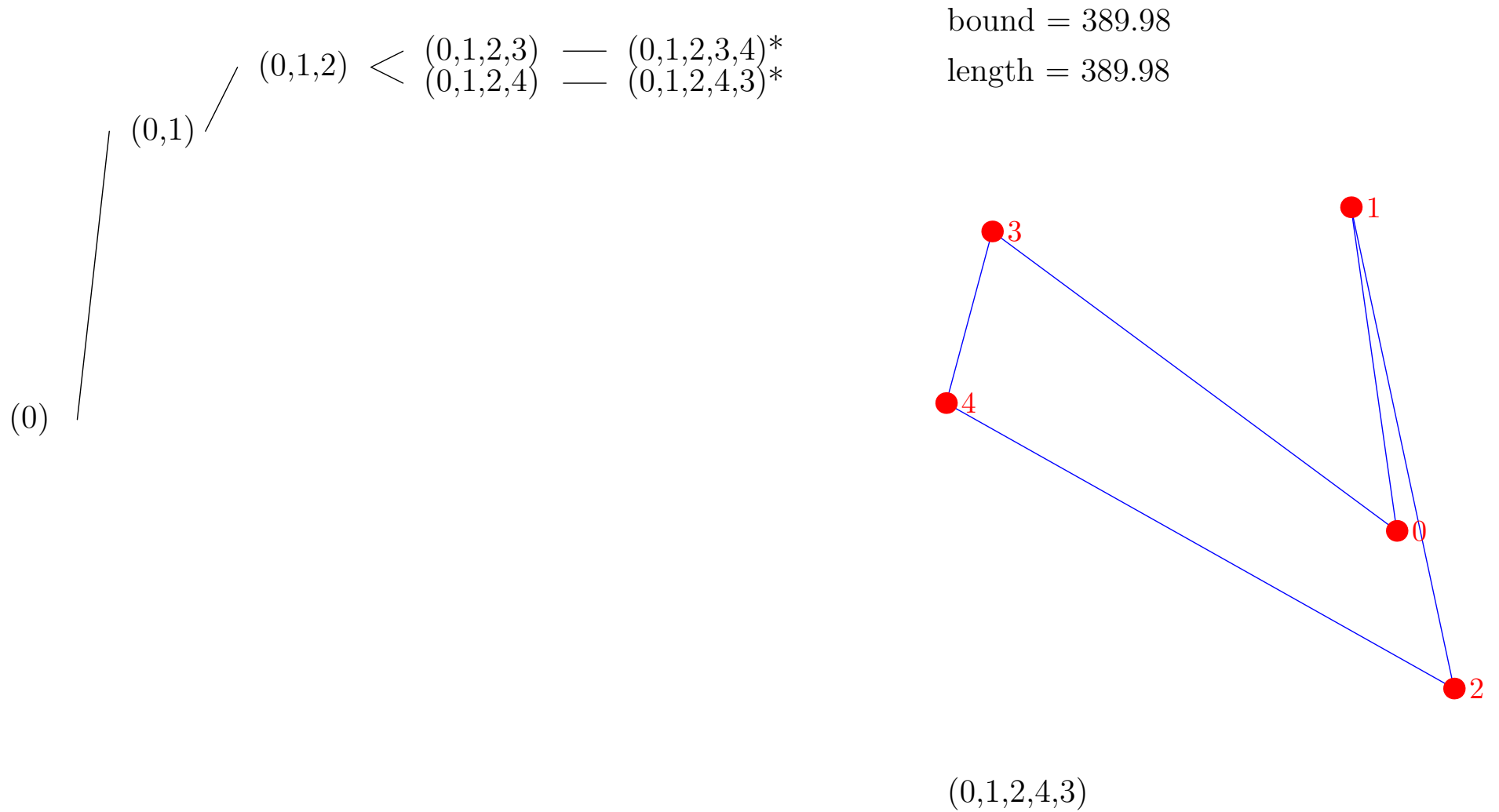
Branch and Bound in Action



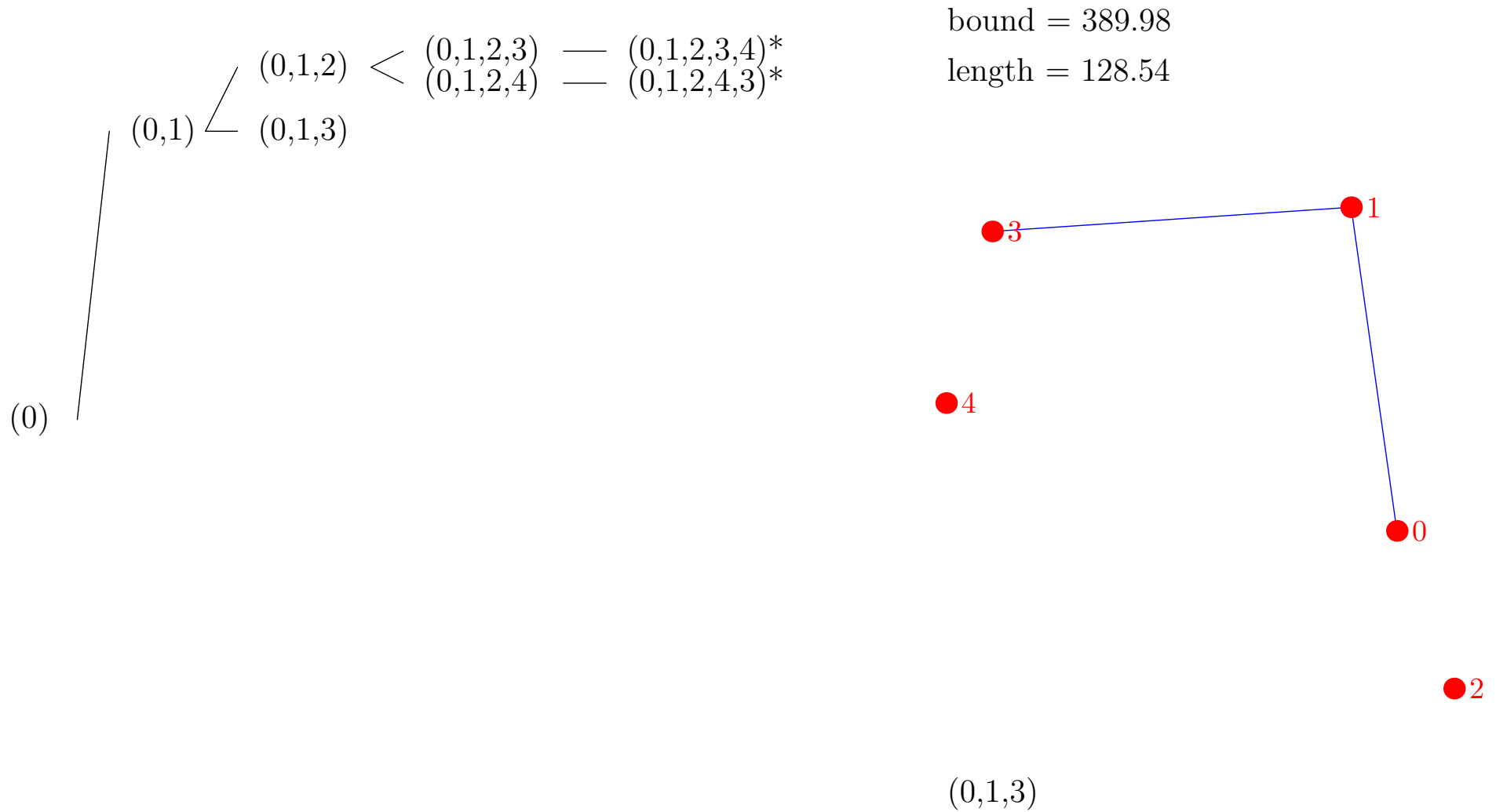
Branch and Bound in Action



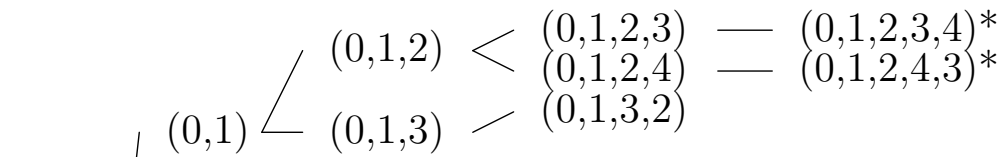
Branch and Bound in Action



Branch and Bound in Action

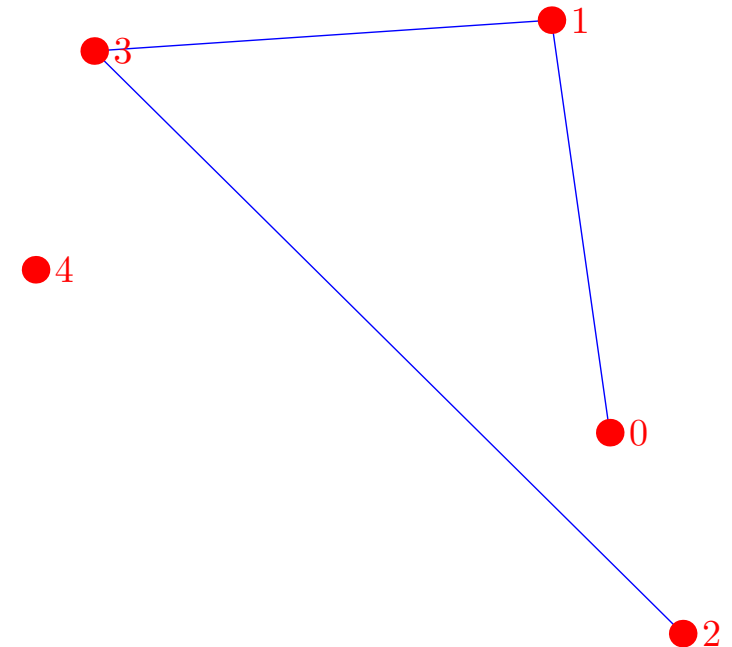


Branch and Bound in Action



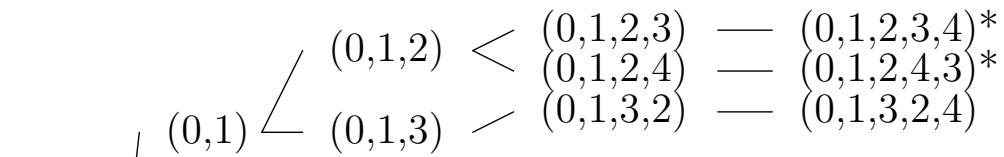
bound = 389.98

length = 250.21



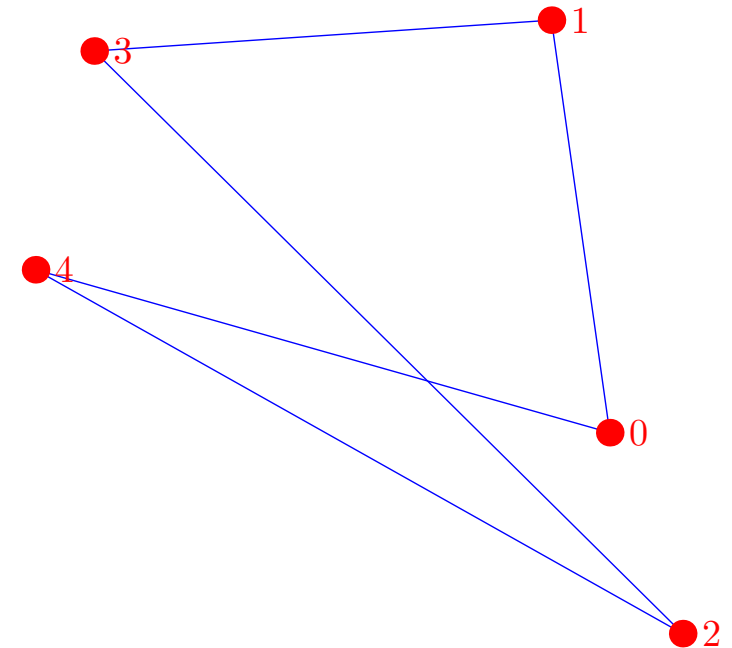
$(0,1,3,2)$

Branch and Bound in Action



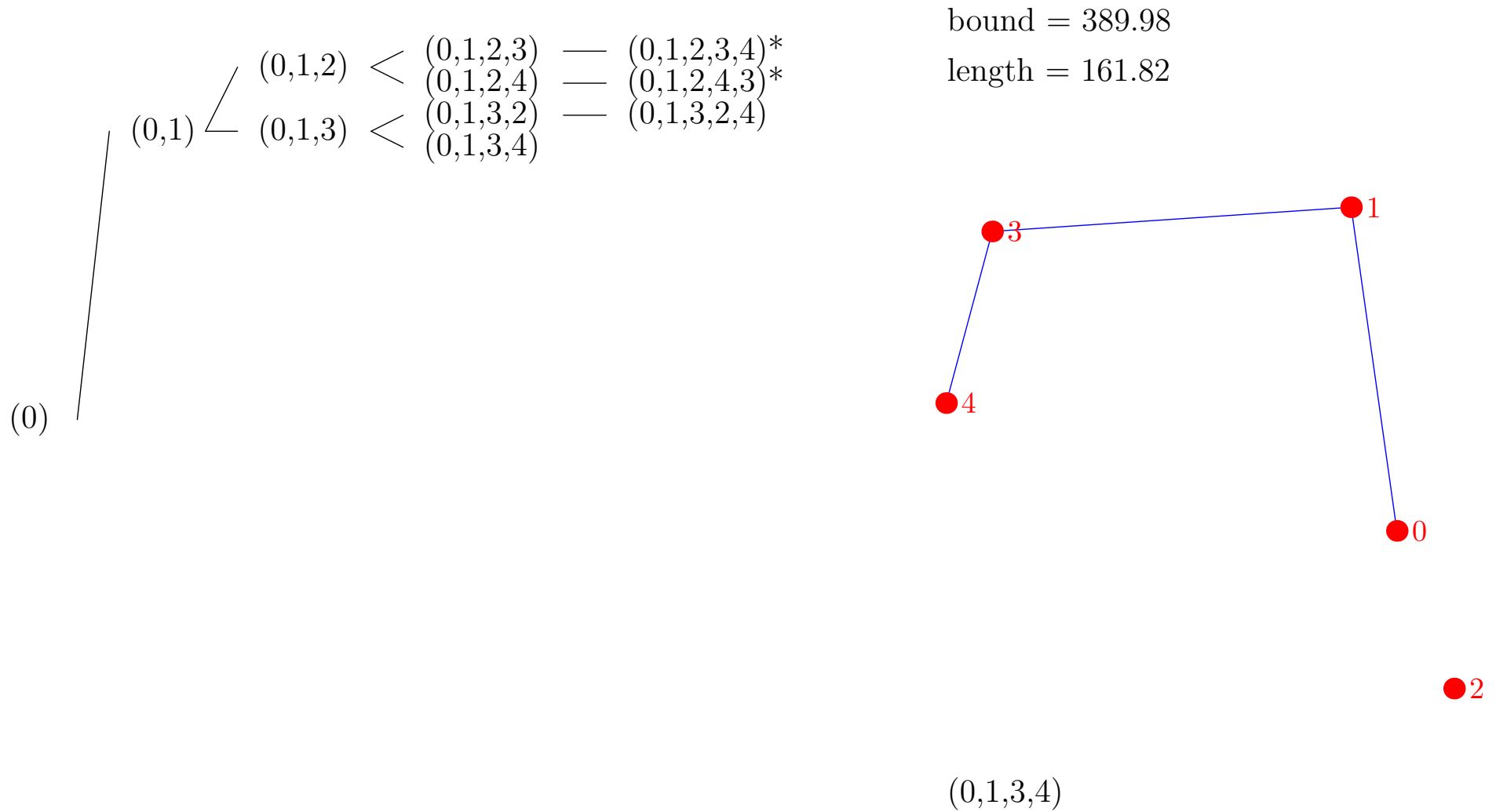
bound = 389.98

length = 446.99

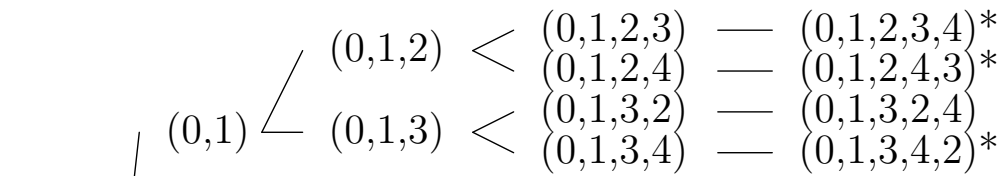


$(0,1,3,2,4)$

Branch and Bound in Action

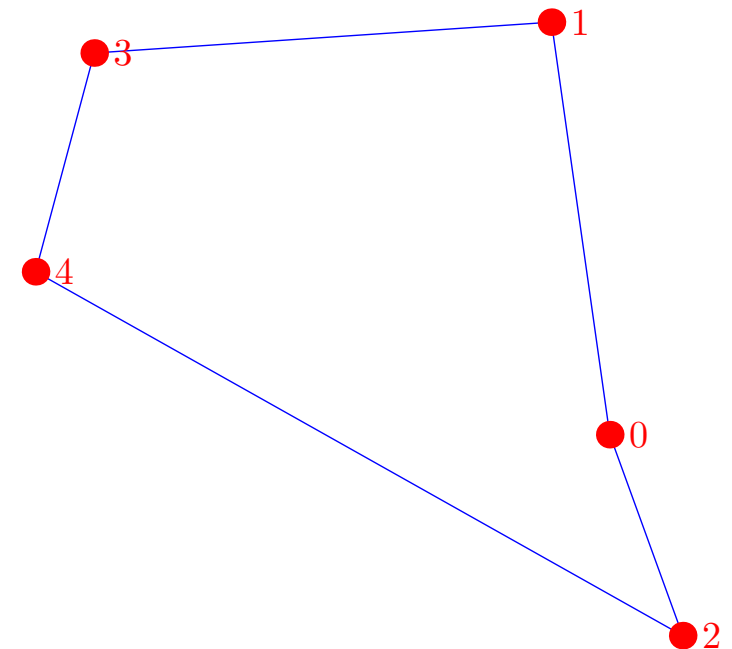


Branch and Bound in Action



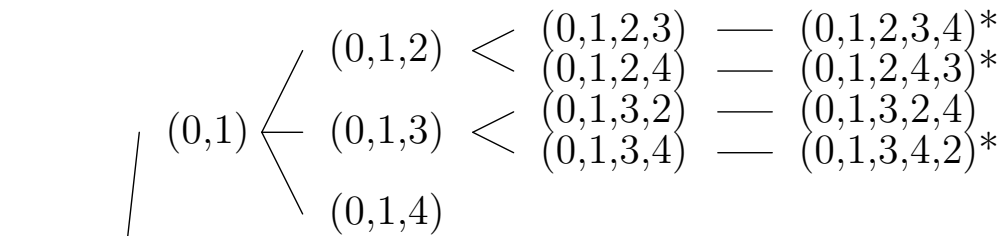
bound = 302.31

length = 302.31



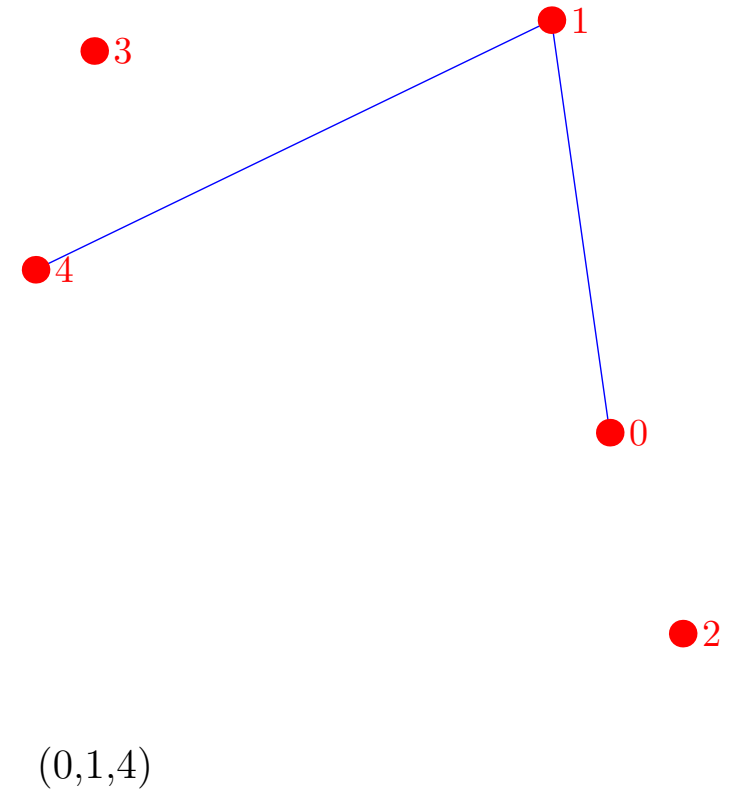
$(0,1,3,4,2)$

Branch and Bound in Action

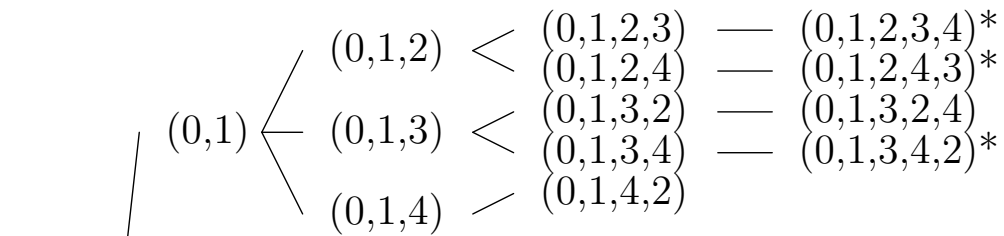


bound = 302.31

length = 145.41

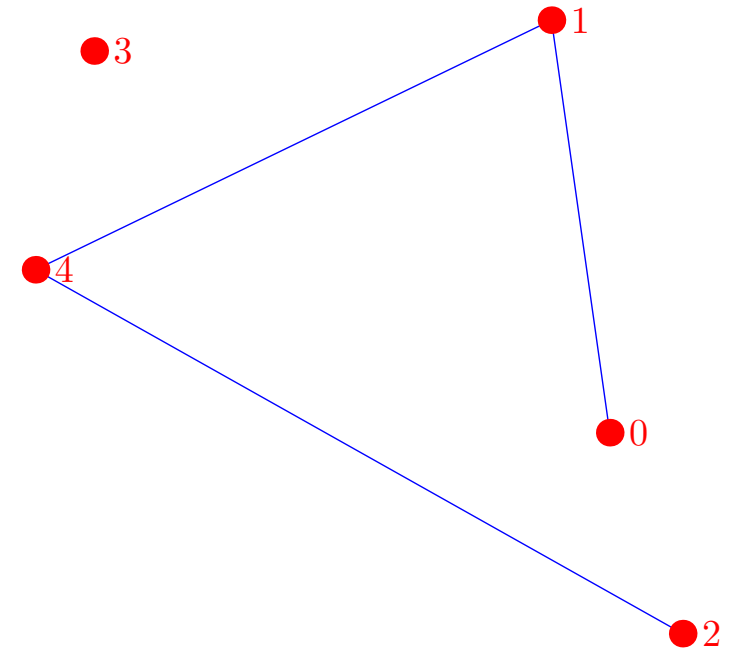


Branch and Bound in Action



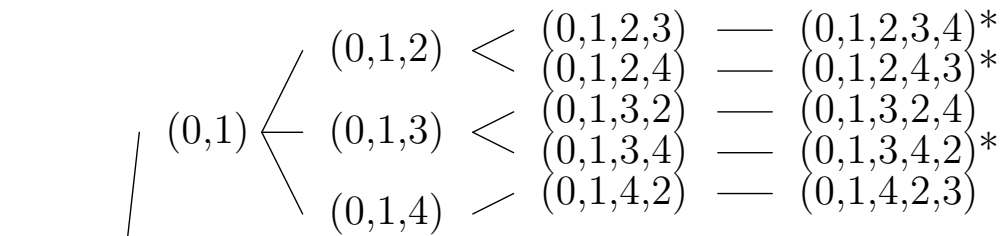
bound = 302.31

length = 254.49



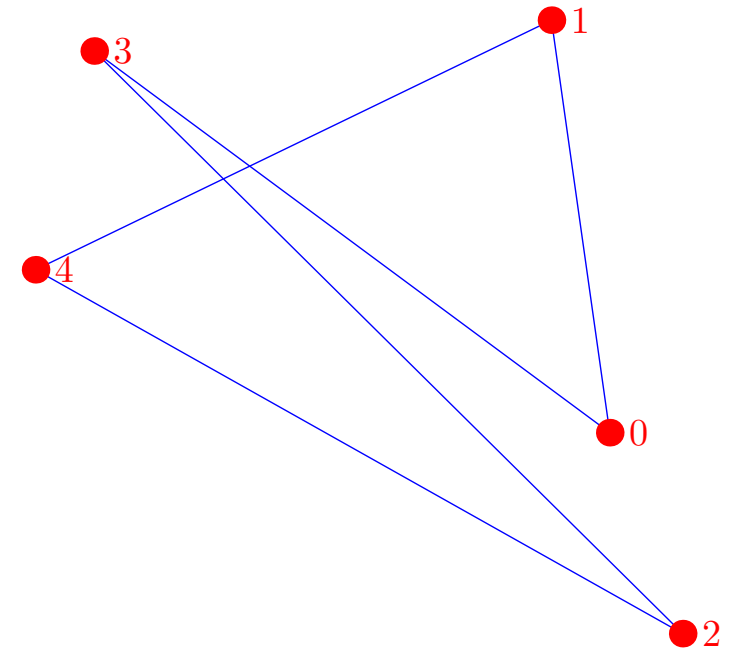
(0,1,4,2)

Branch and Bound in Action



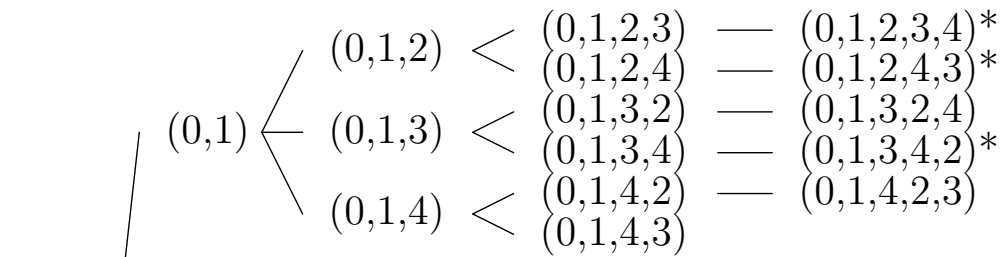
bound = 302.31

length = 470.41



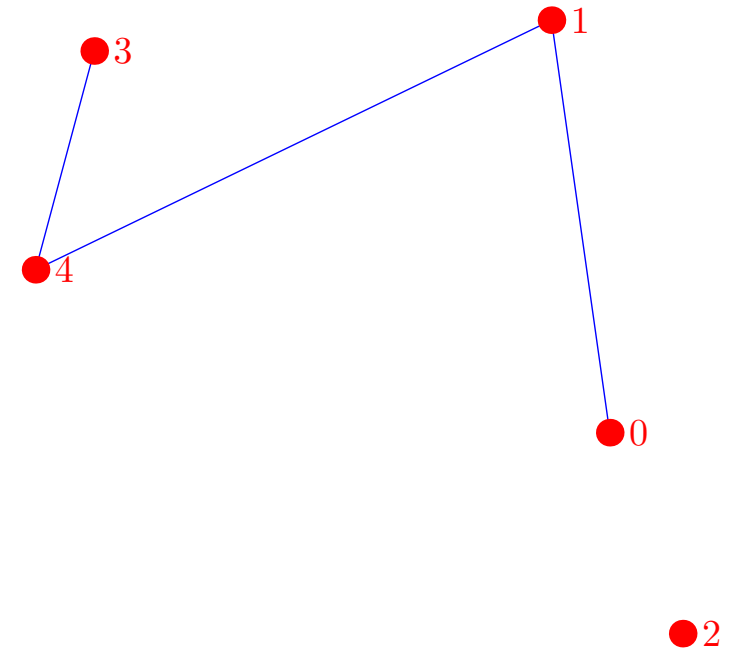
$(0,1,4,2,3)$

Branch and Bound in Action



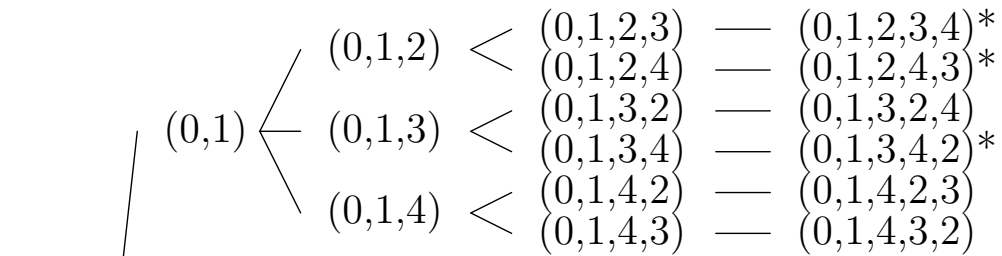
bound = 302.31

length = 178.69



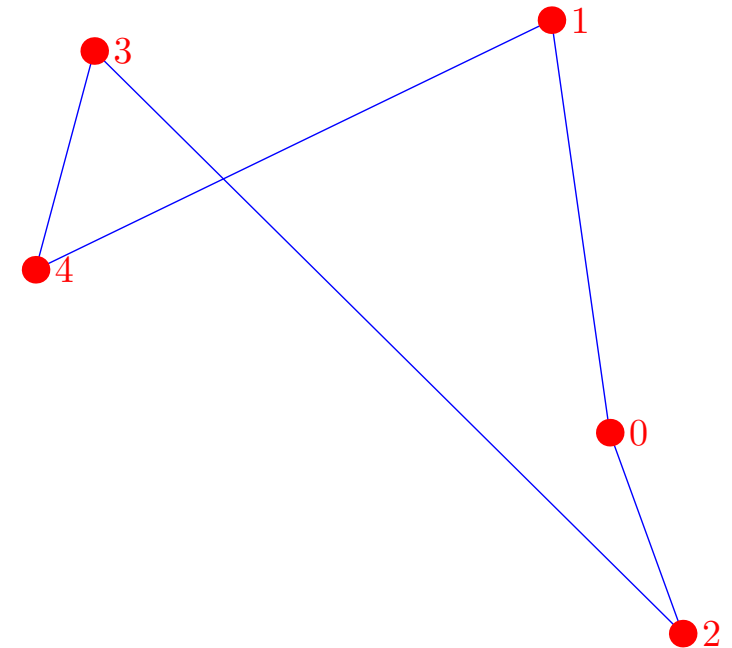
$(0,1,4,3)$

Branch and Bound in Action



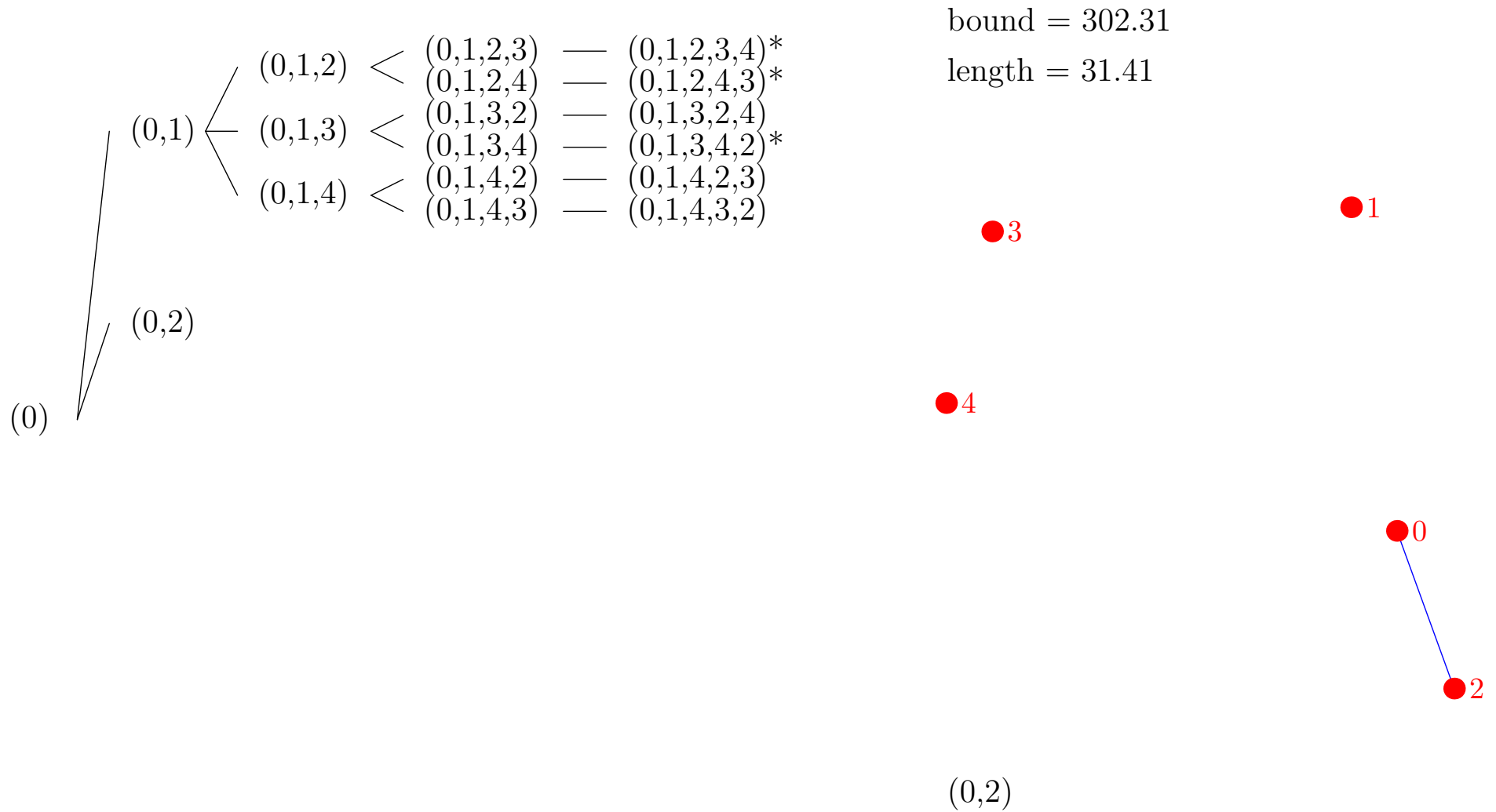
bound = 302.31

length = 331.77

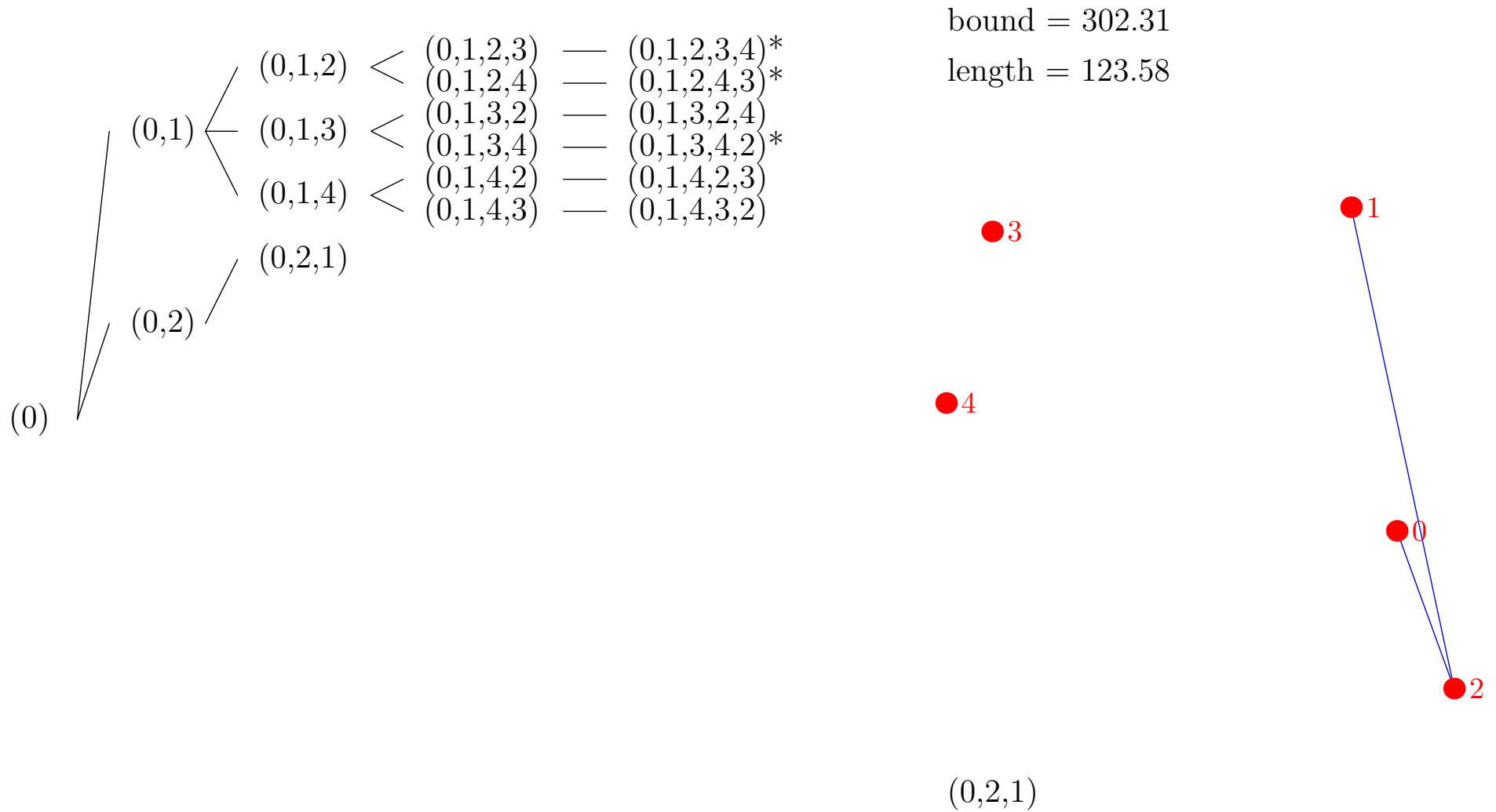


(0,1,4,3,2)

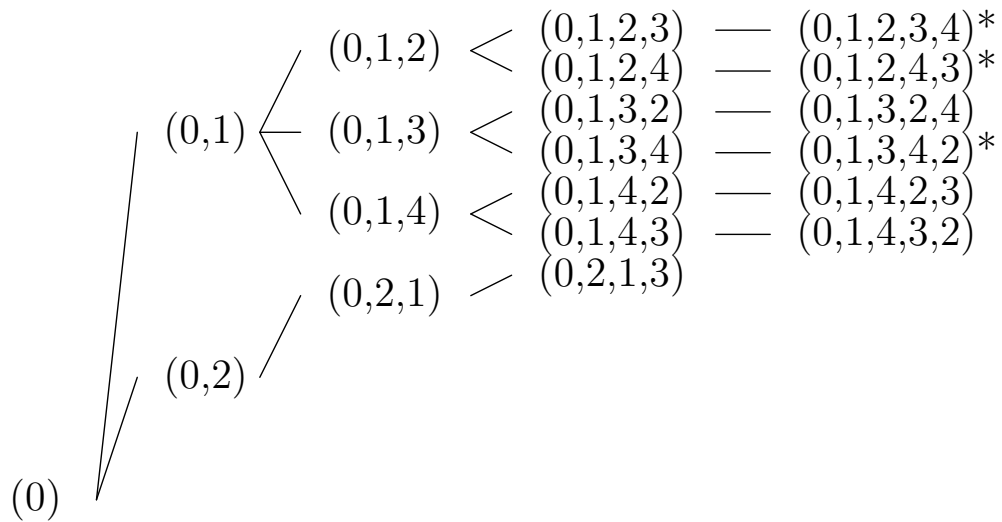
Branch and Bound in Action



Branch and Bound in Action

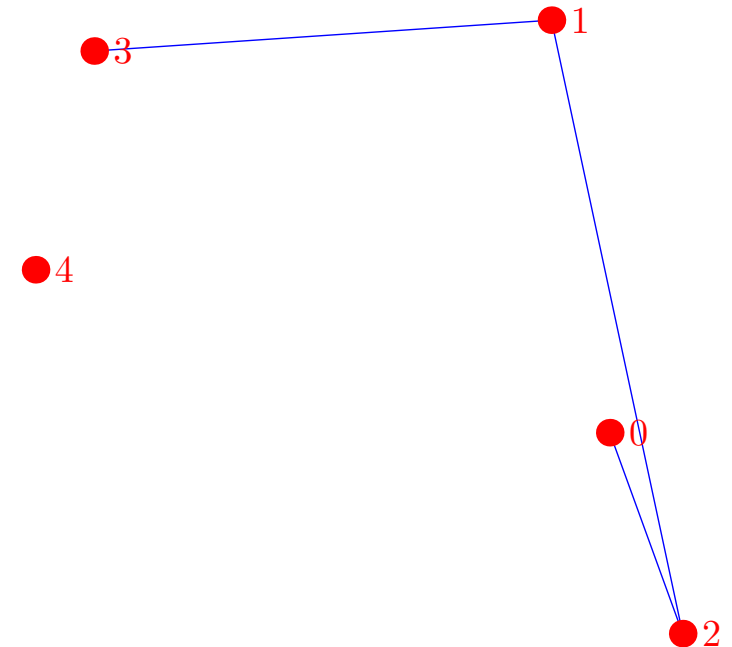


Branch and Bound in Action



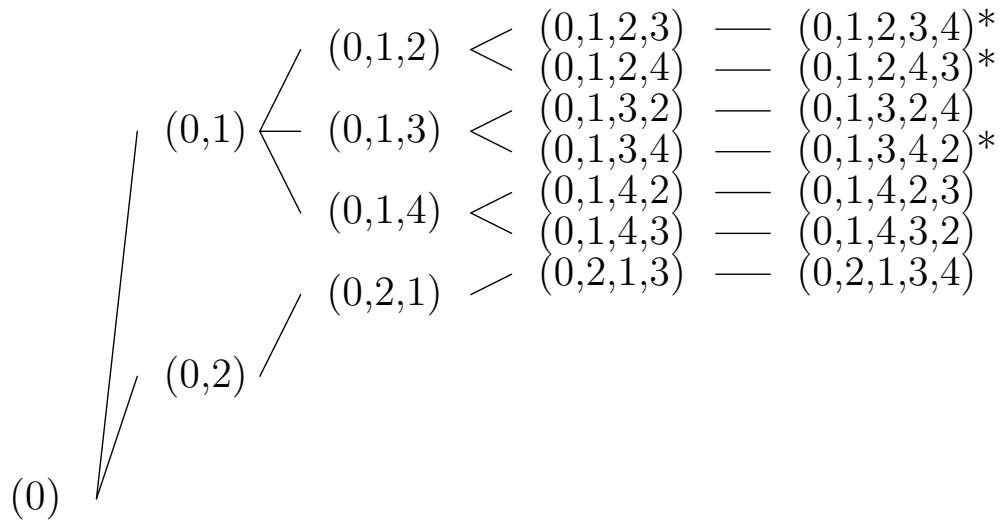
bound = 302.31

length = 190.92



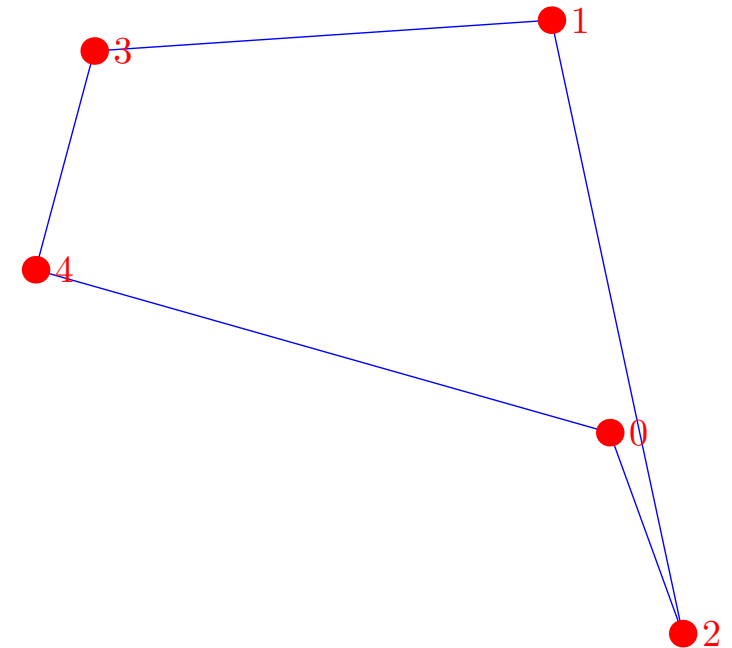
$(0,2,1,3)$

Branch and Bound in Action



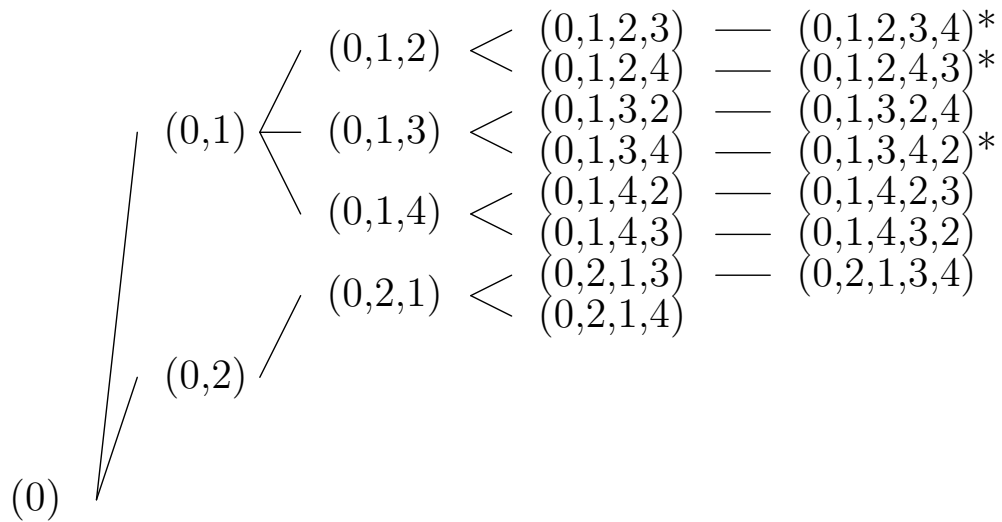
bound = 302.31

length = 311.88



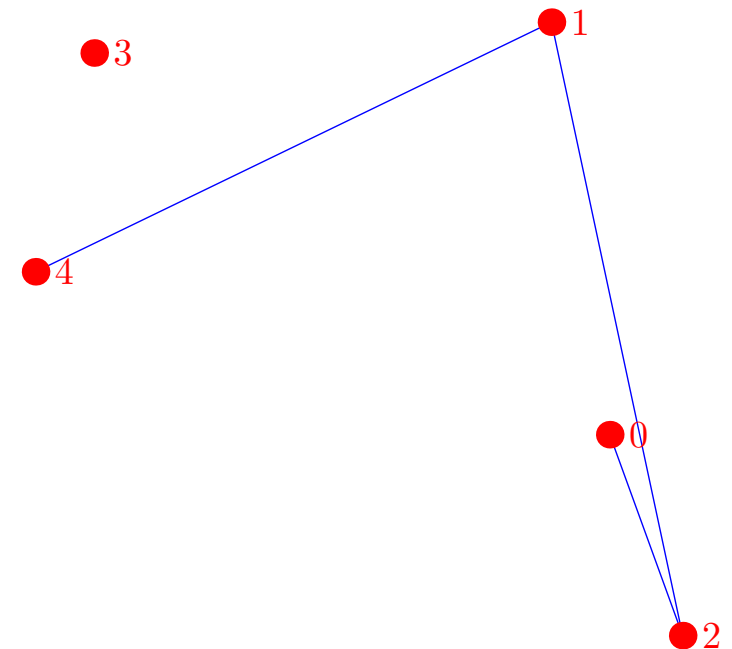
$(0,2,1,3,4)$

Branch and Bound in Action



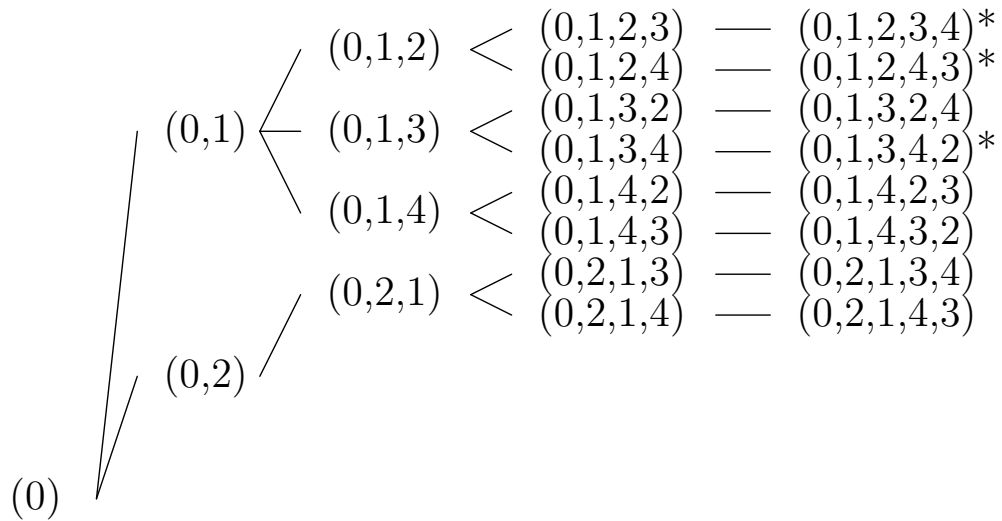
bound = 302.31

length = 207.79



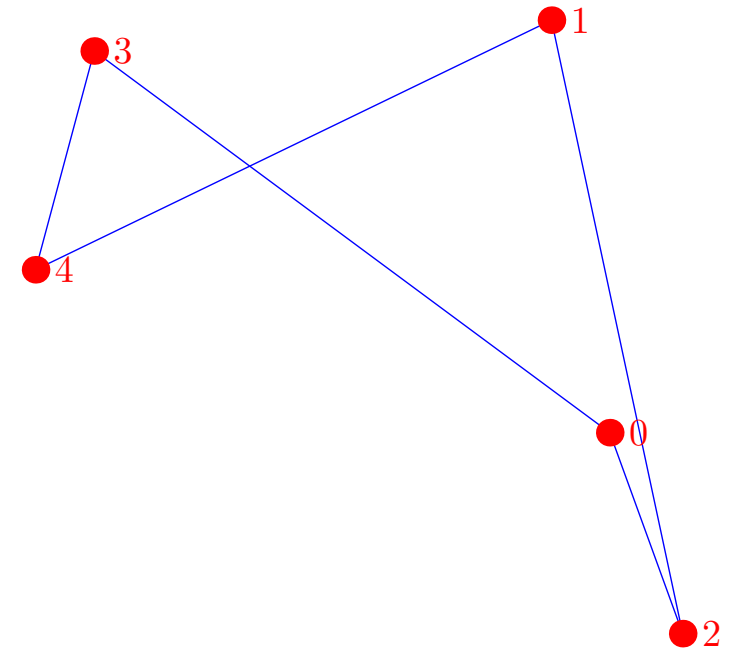
$(0,2,1,4)$

Branch and Bound in Action



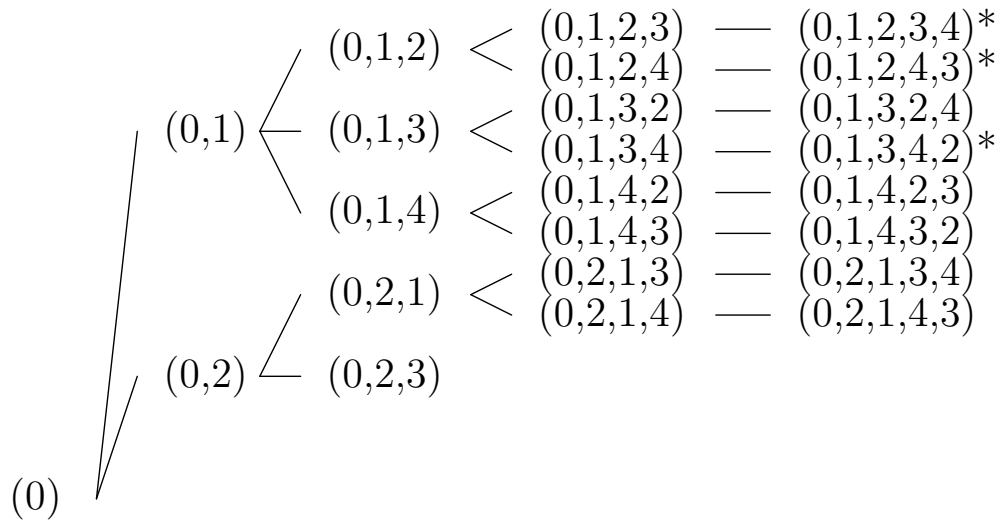
bound = 302.31

length = 335.3



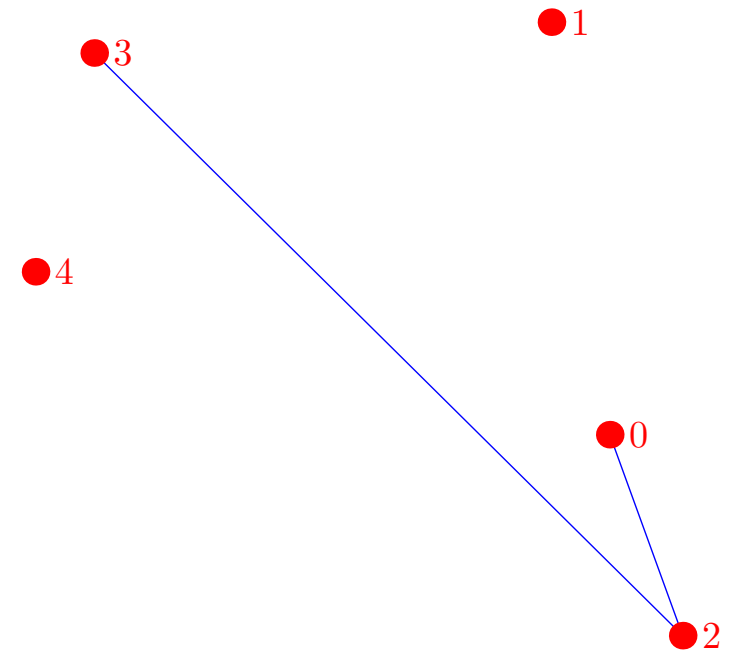
$(0,2,1,4,3)$

Branch and Bound in Action

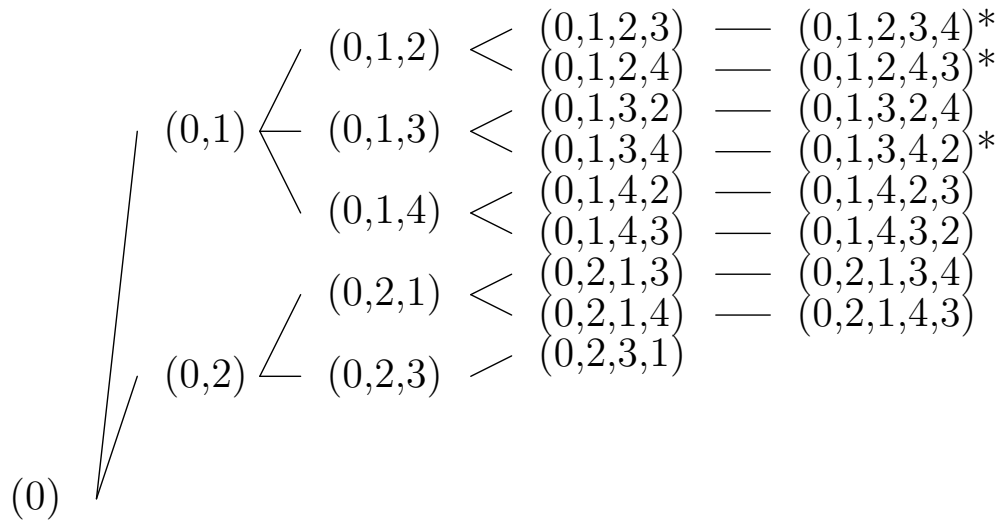


bound = 302.31

length = 153.08

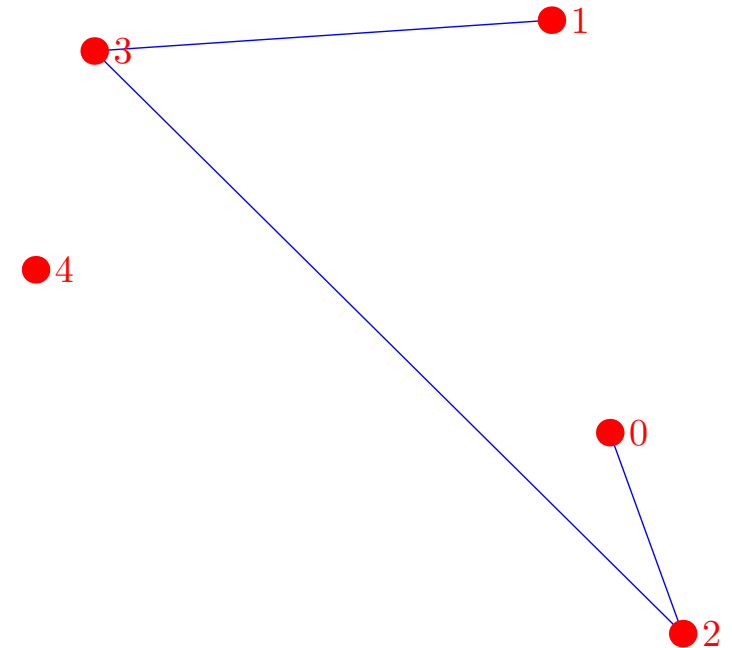


Branch and Bound in Action



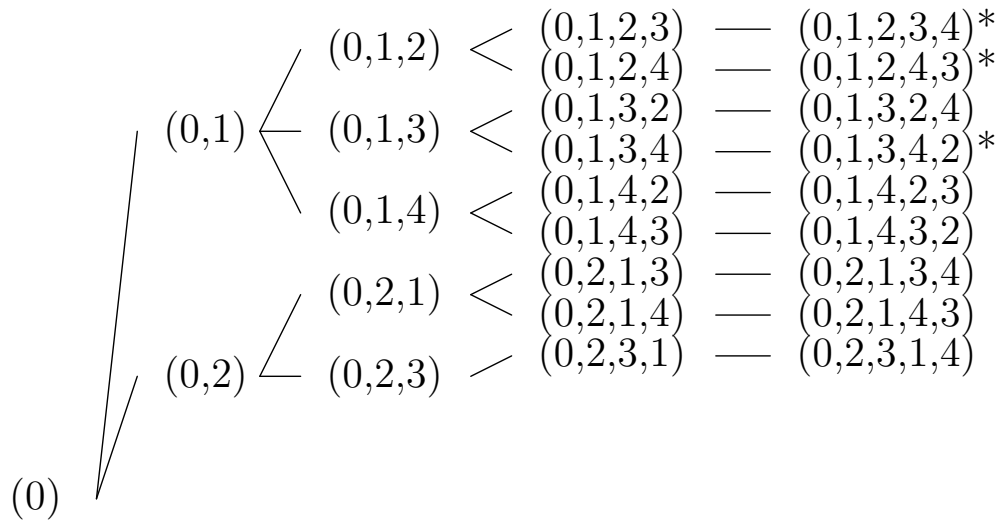
bound = 302.31

length = 220.41



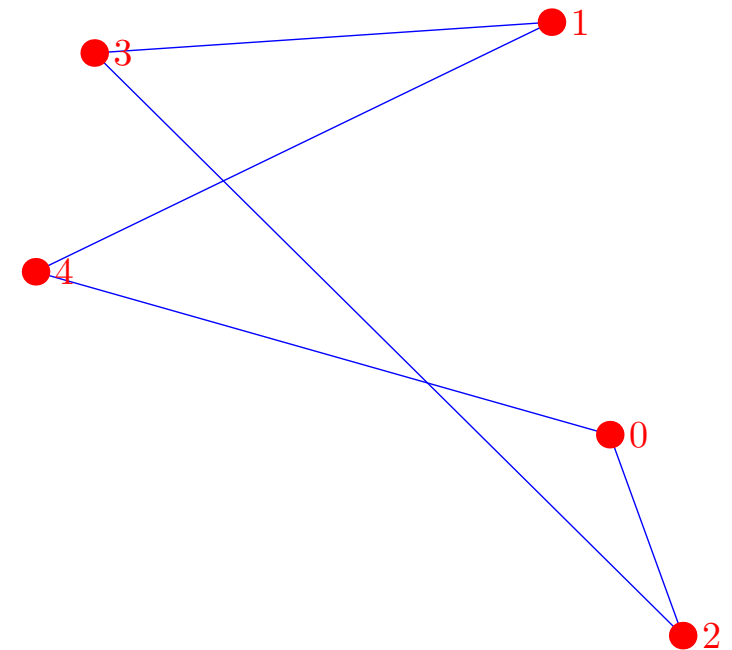
$(0,2,3,1)$

Branch and Bound in Action



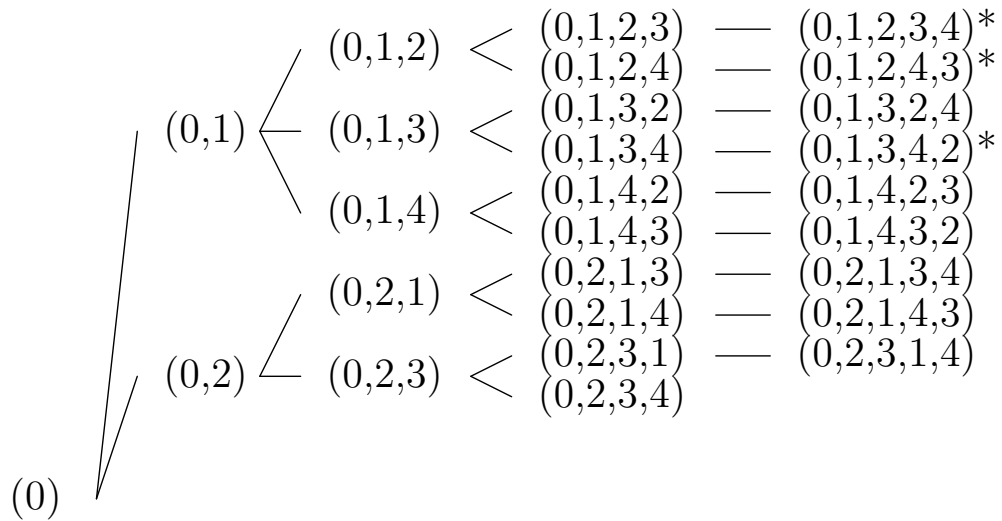
bound = 302.31

length = 392.31



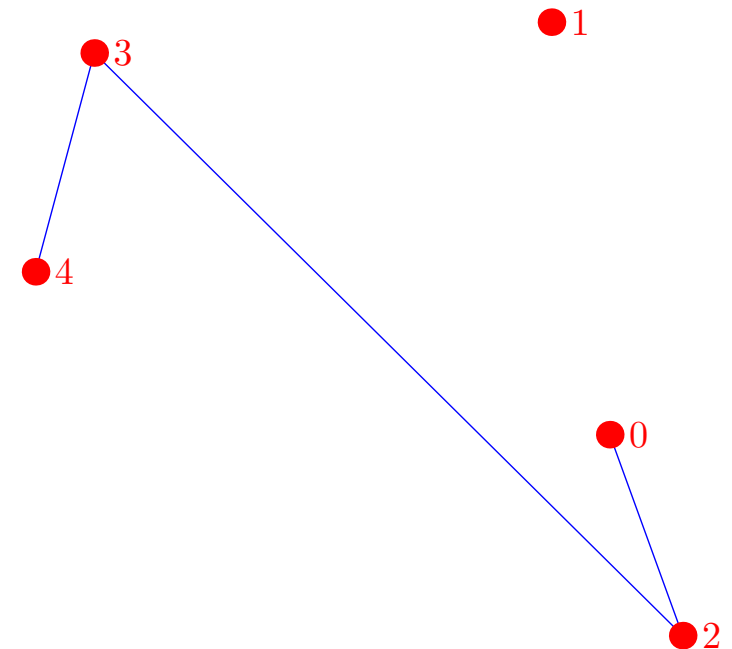
(0,2,3,1,4)

Branch and Bound in Action



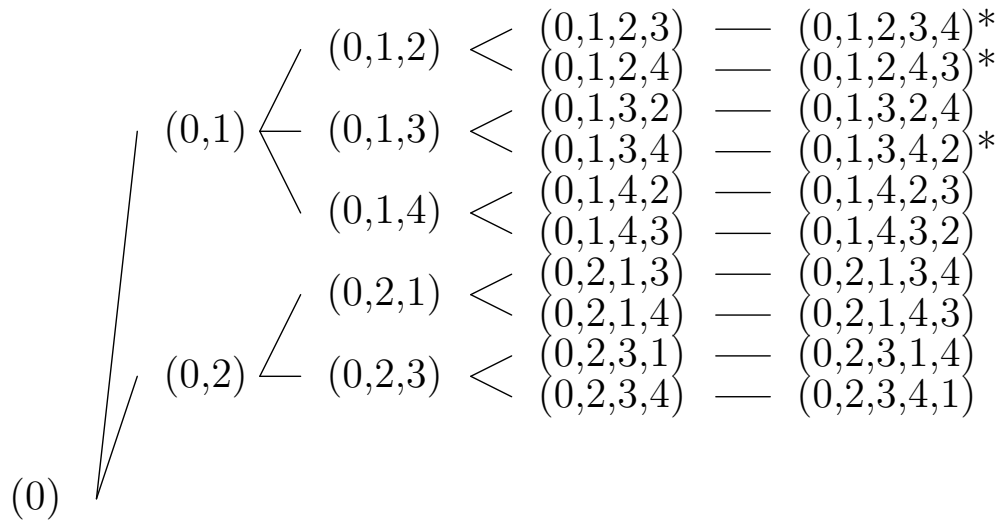
bound = 302.31

length = 186.35



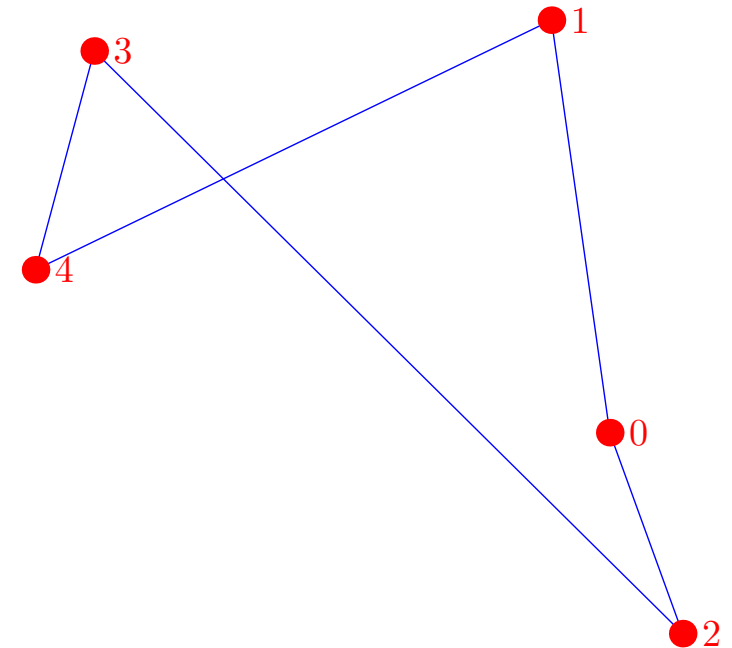
$(0,2,3,4)$

Branch and Bound in Action



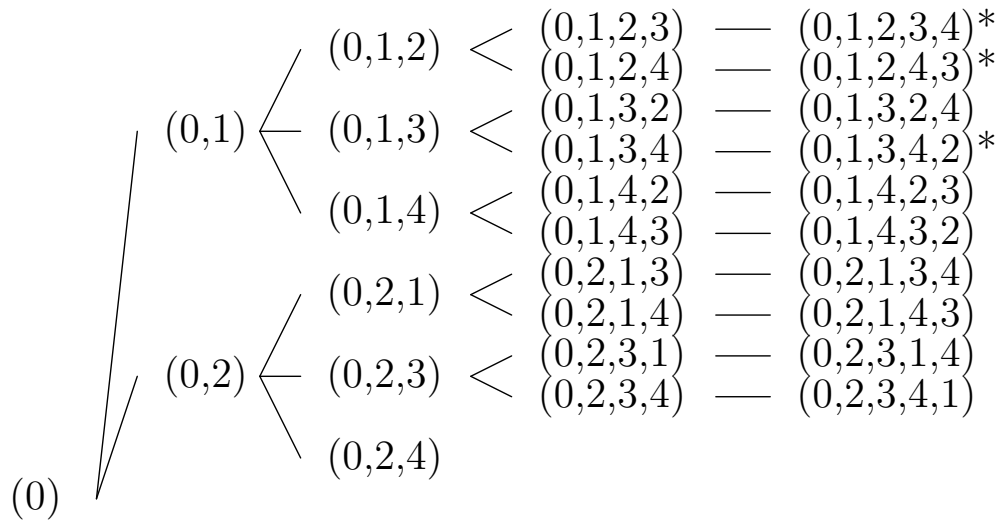
bound = 302.31

length = 331.77



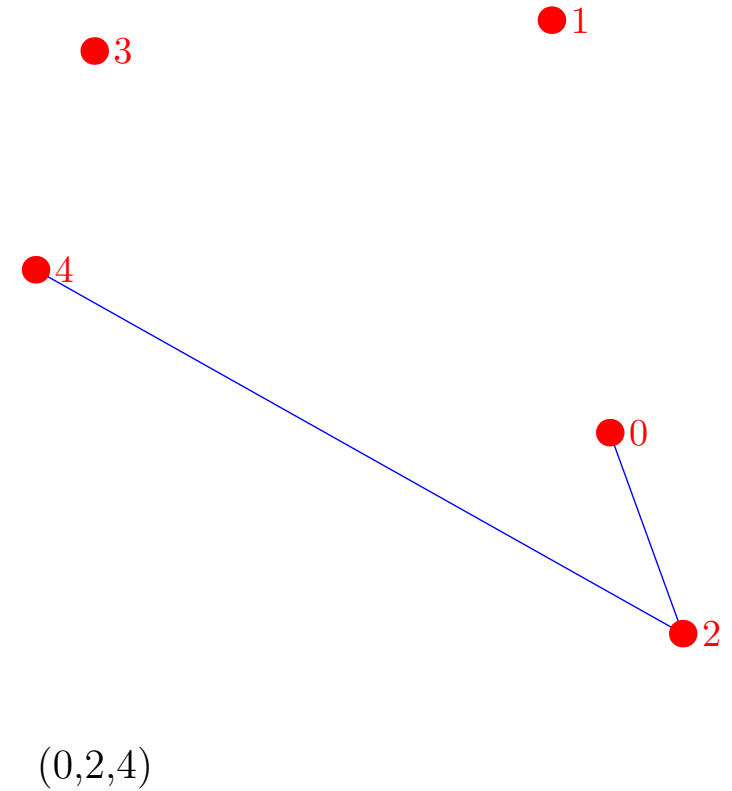
(0,2,3,4,1)

Branch and Bound in Action

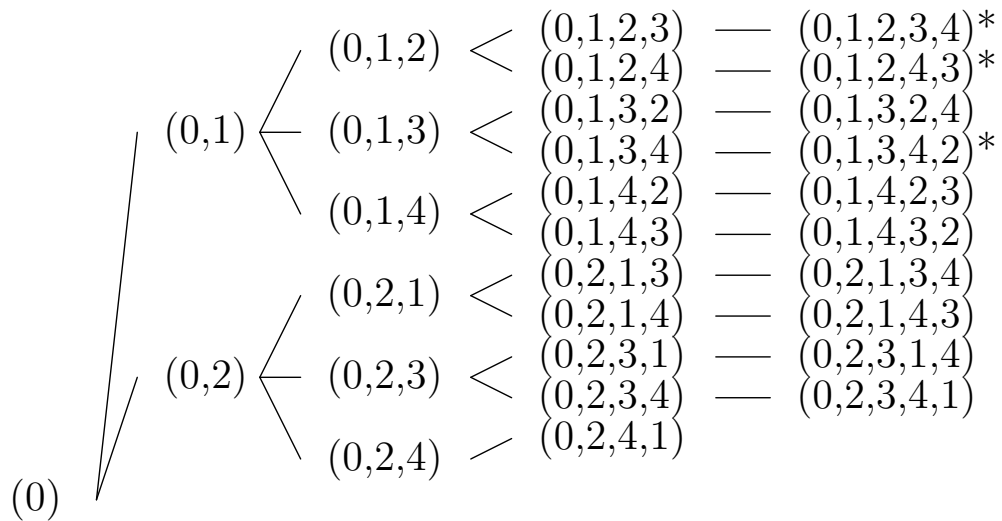


bound = 302.31

length = 140.49

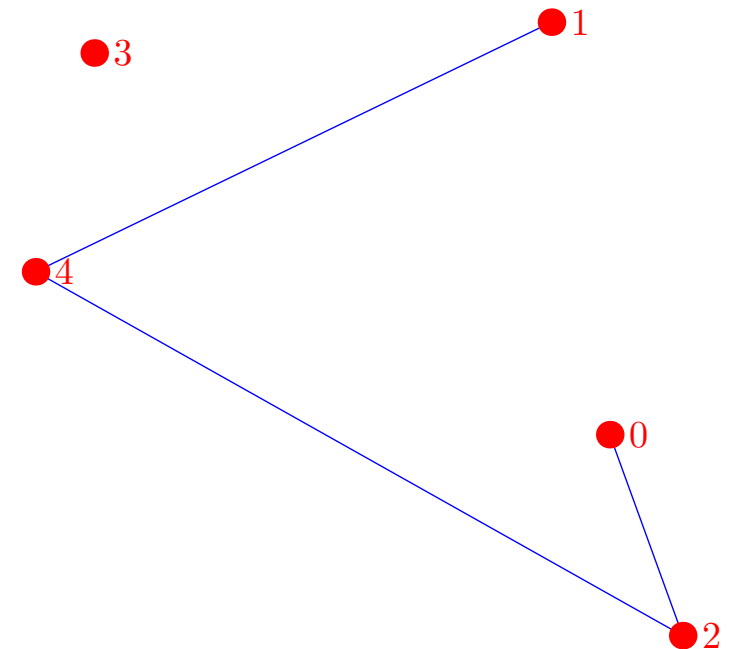


Branch and Bound in Action



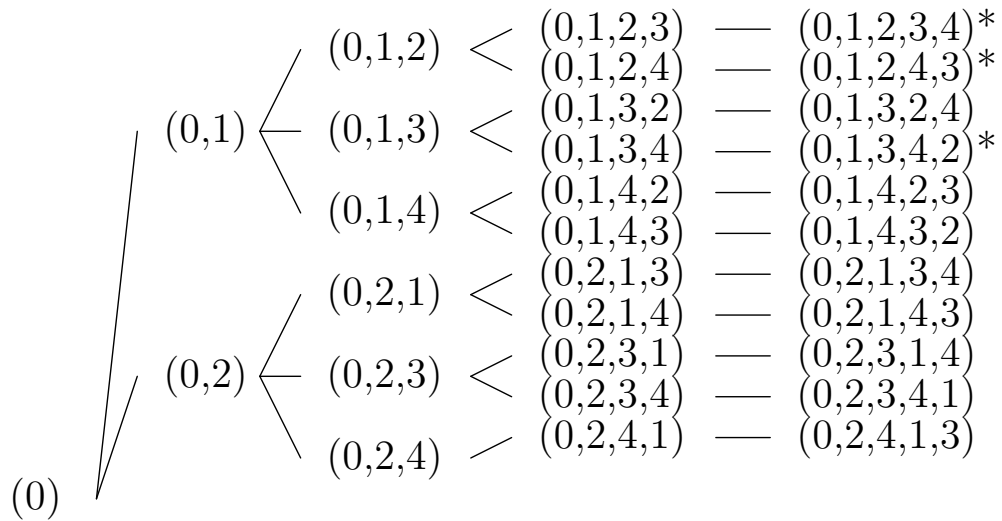
bound = 302.31

length = 224.69



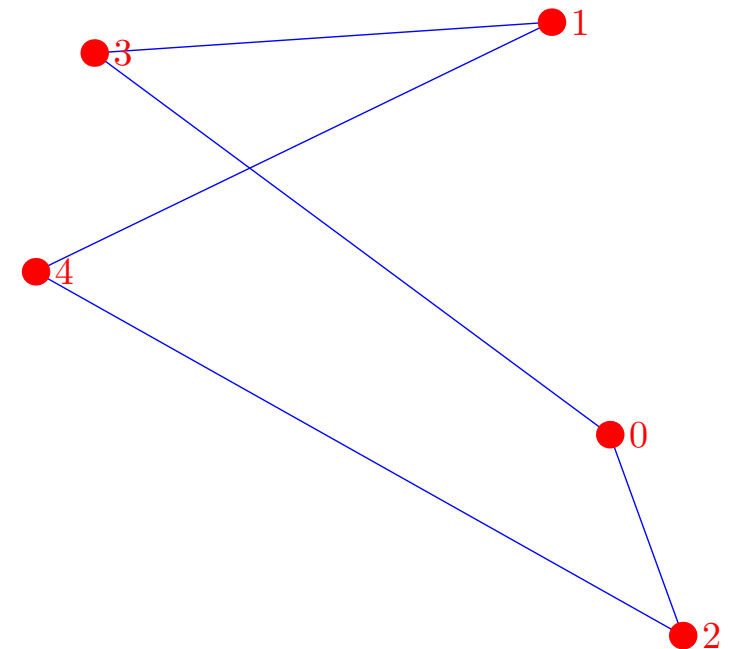
(0,2,4,1)

Branch and Bound in Action



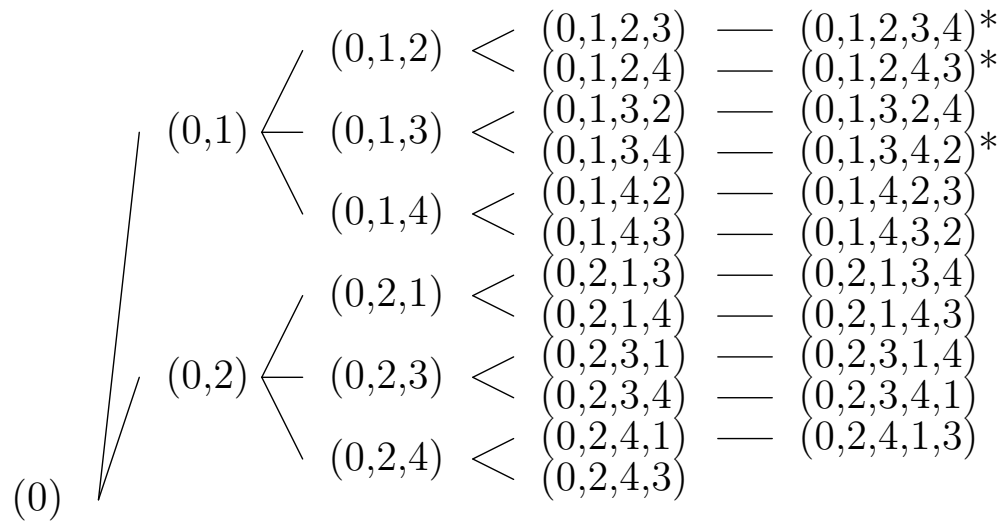
bound = 302.31

length = 386.27



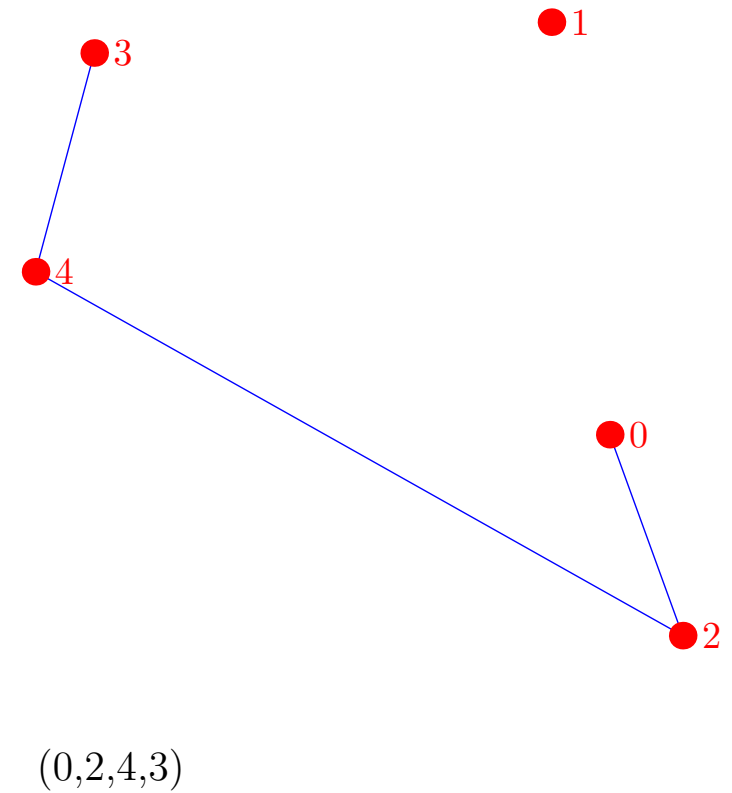
(0,2,4,1,3)

Branch and Bound in Action

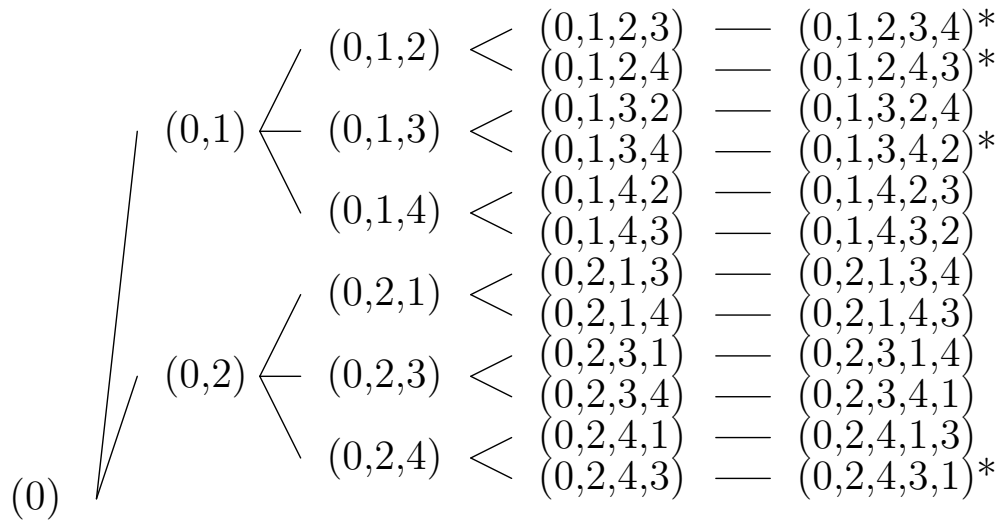


bound = 302.31

length = 173.77

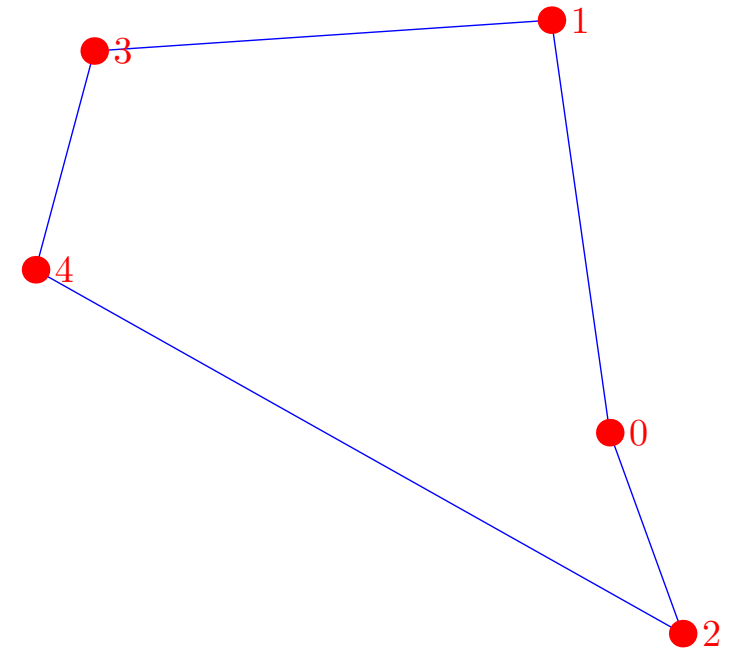


Branch and Bound in Action



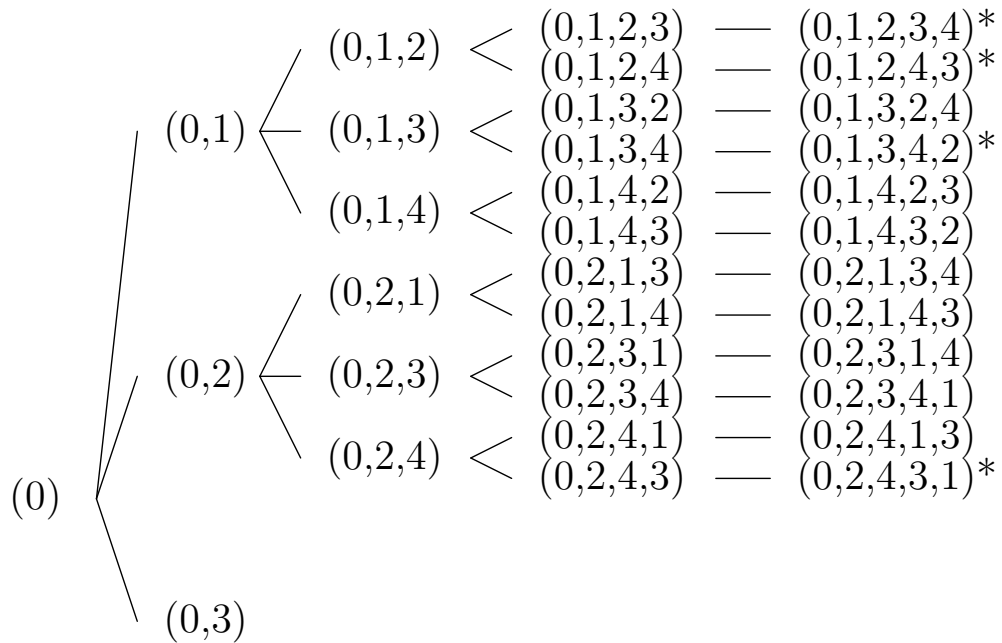
bound = 302.31

length = 302.31



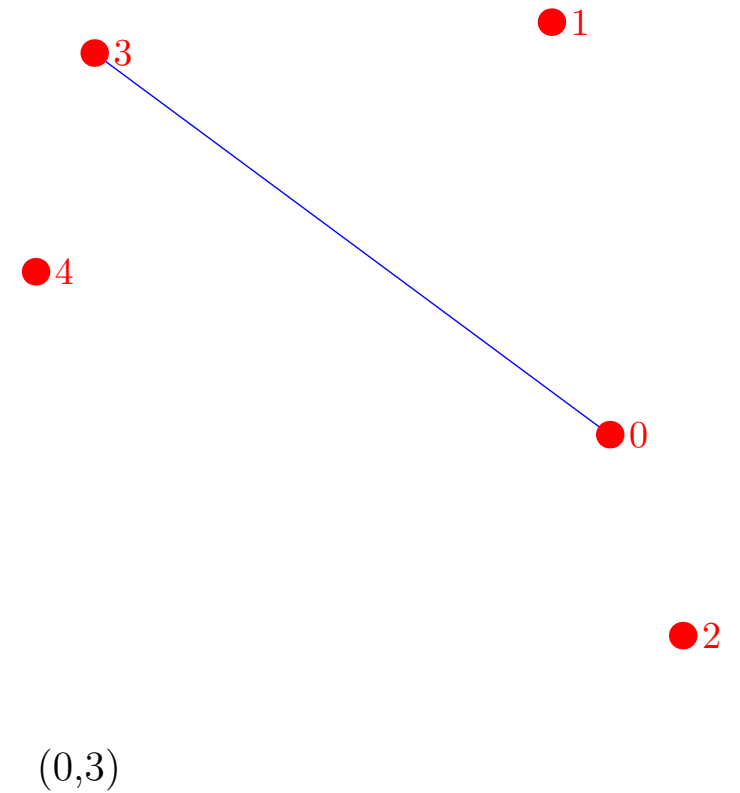
(0,2,4,3,1)

Branch and Bound in Action

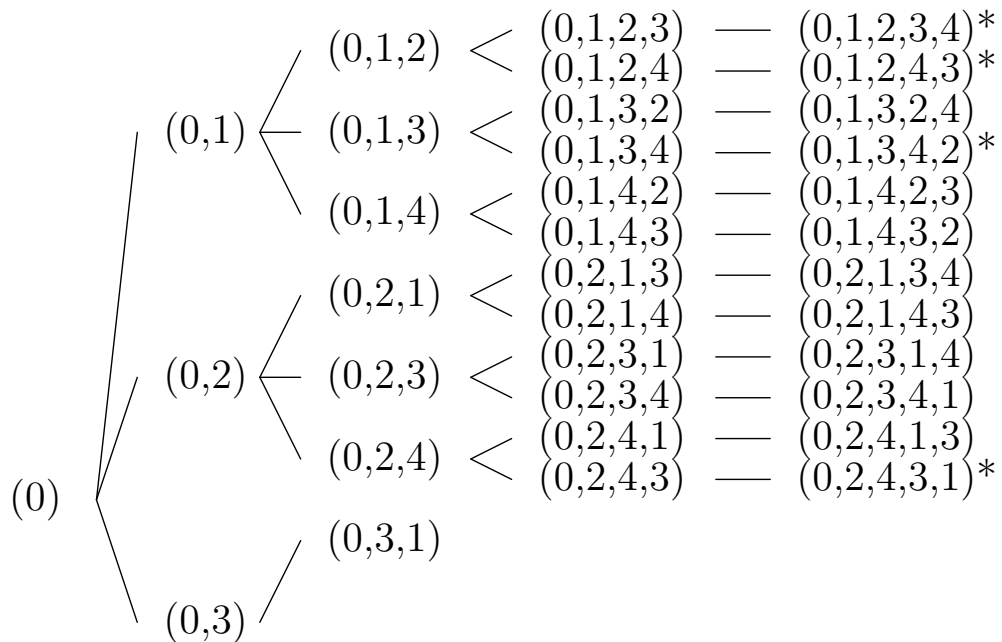


bound = 302.31

length = 94.244

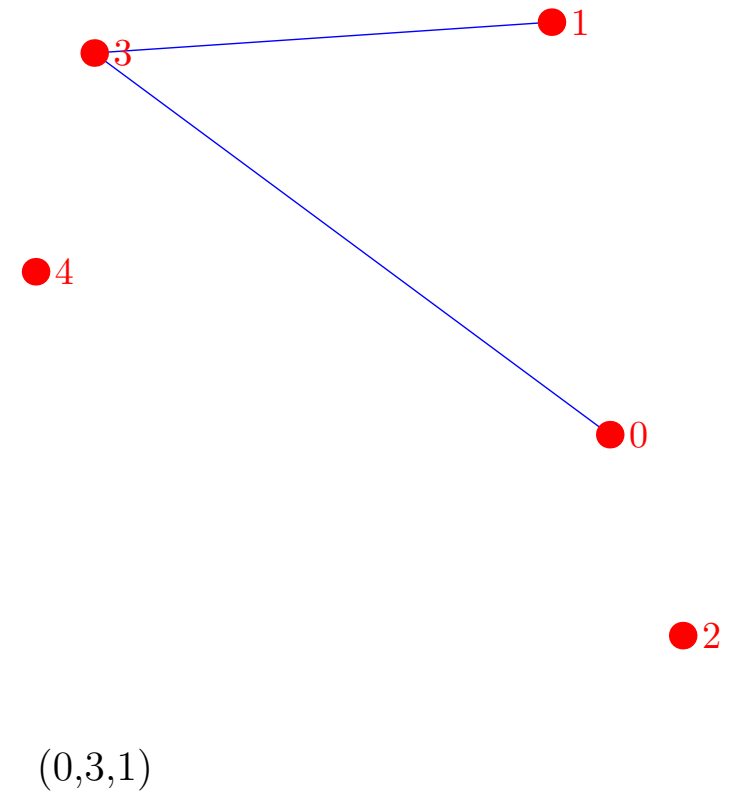


Branch and Bound in Action

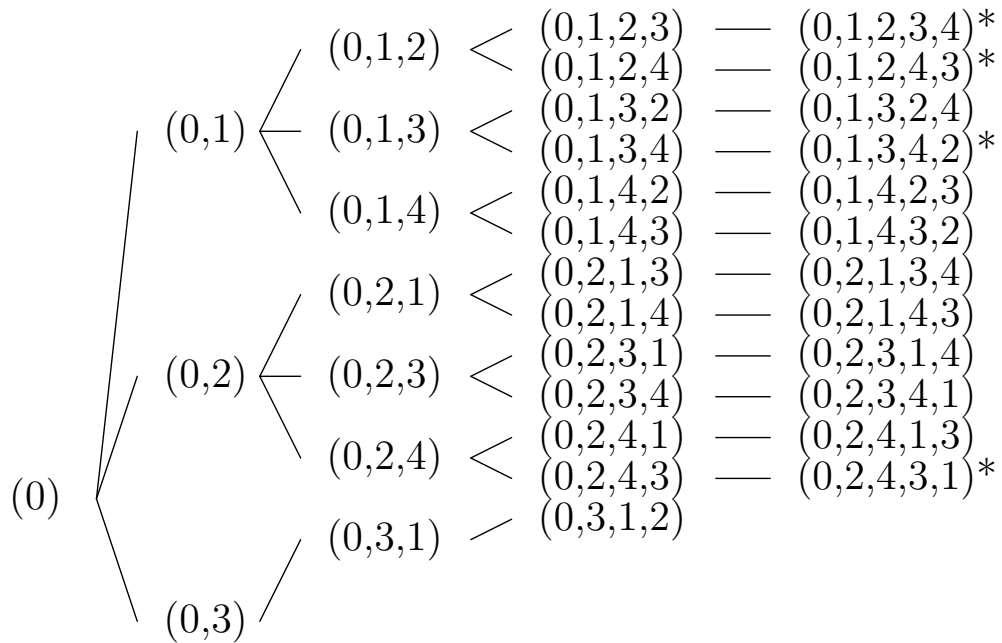


bound = 302.31

length = 161.58

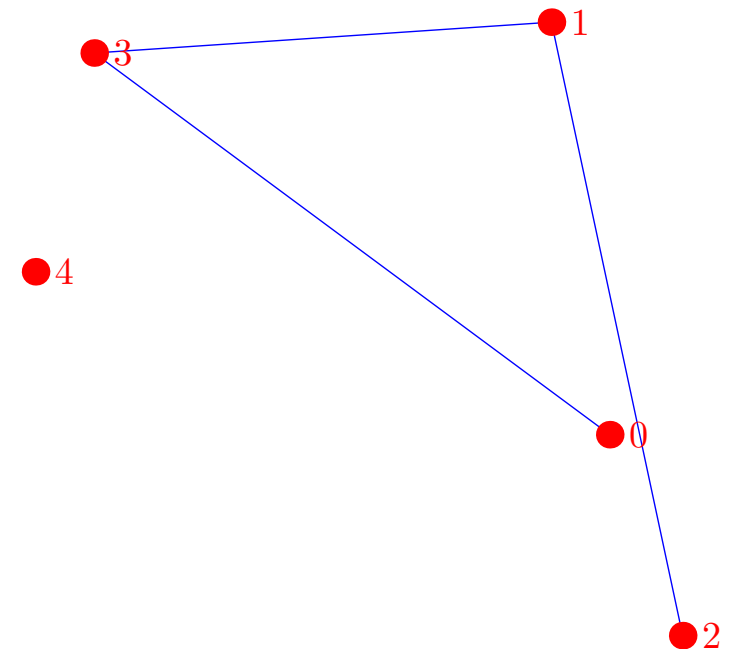


Branch and Bound in Action



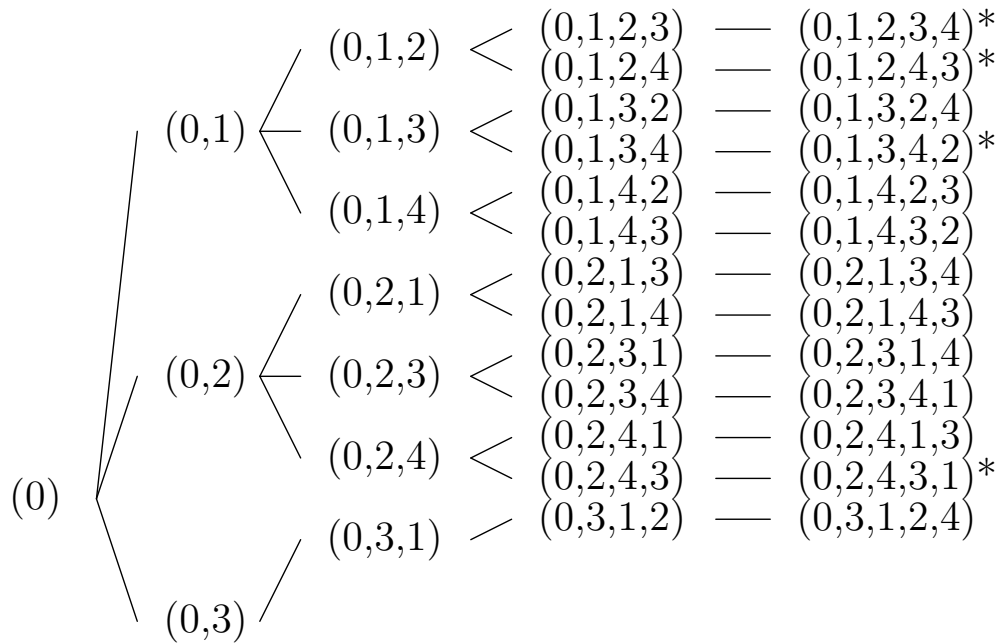
bound = 302.31

length = 253.75



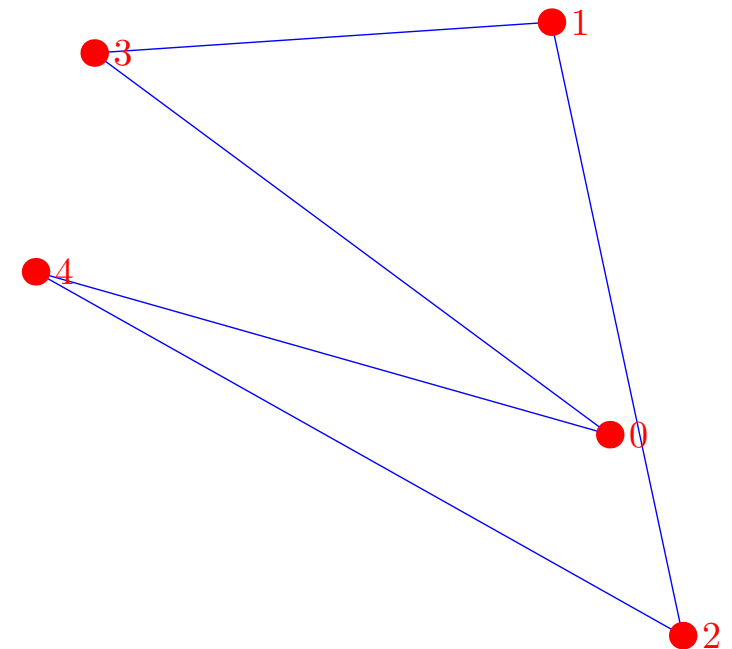
(0,3,1,2)

Branch and Bound in Action



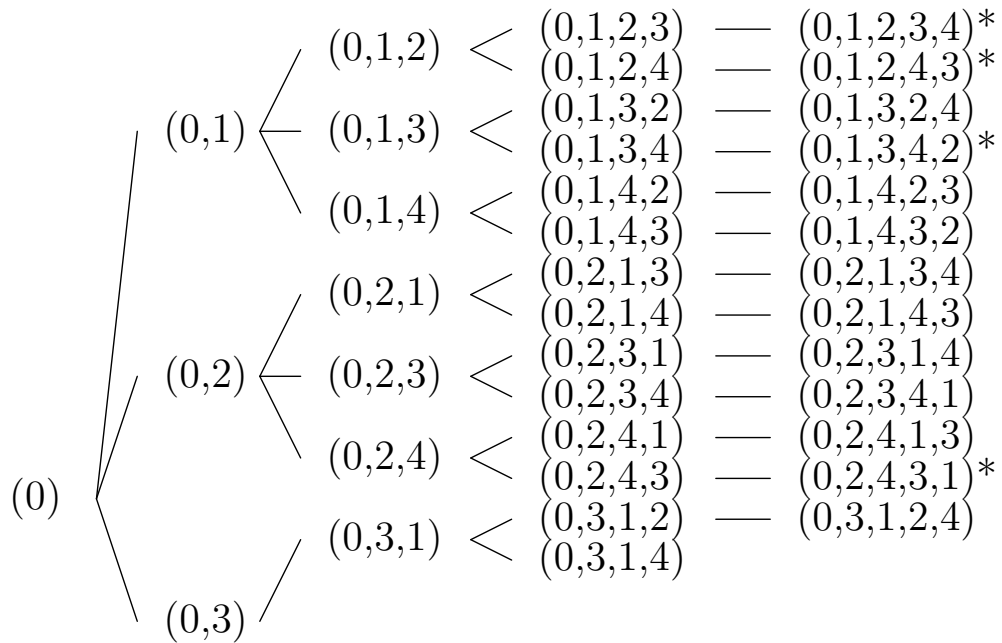
bound = 302.31

length = 450.52



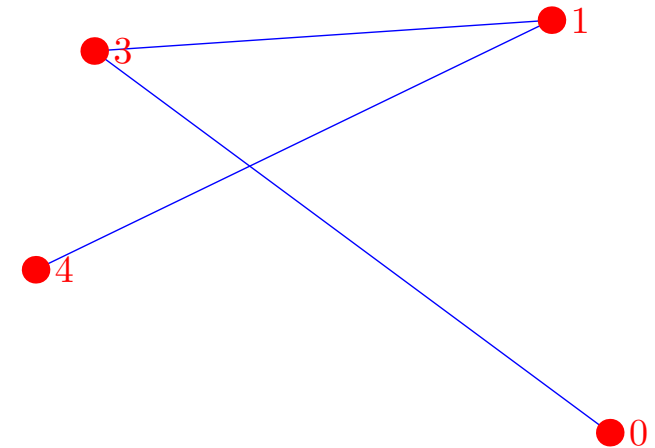
$(0,3,1,2,4)$

Branch and Bound in Action



bound = 302.31

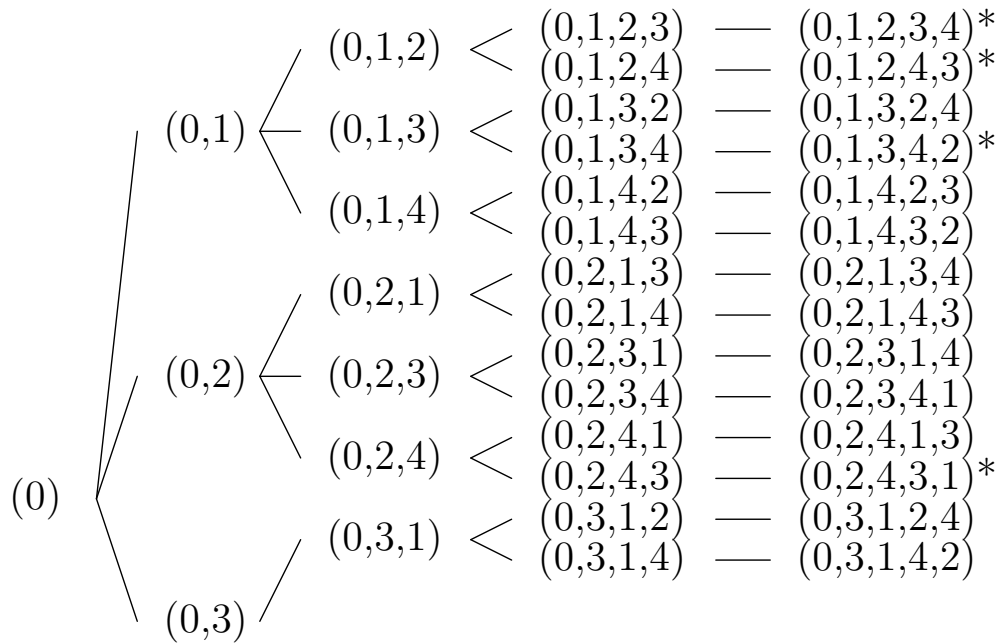
length = 245.78



● 2

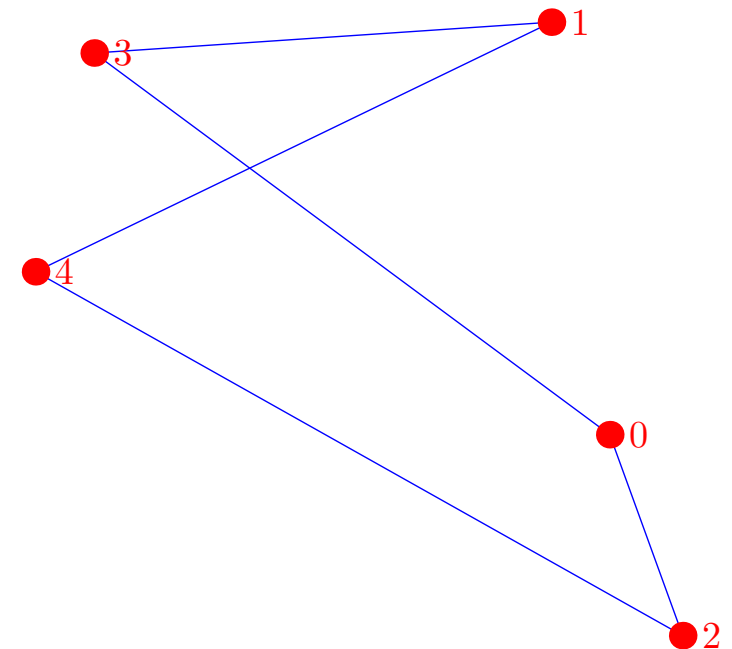
$(0,3,1,4)$

Branch and Bound in Action



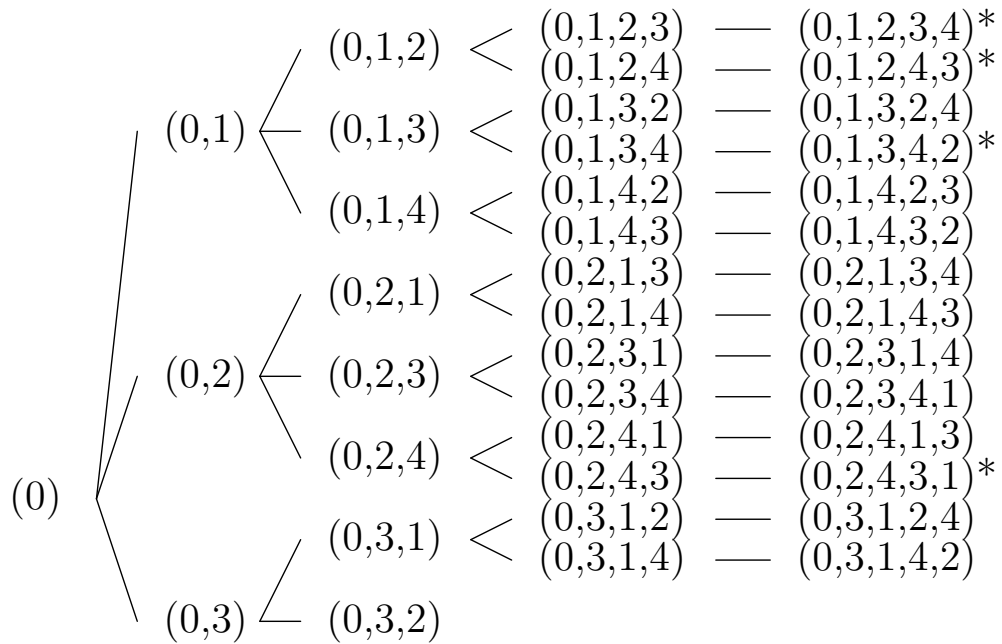
bound = 302.31

length = 386.27



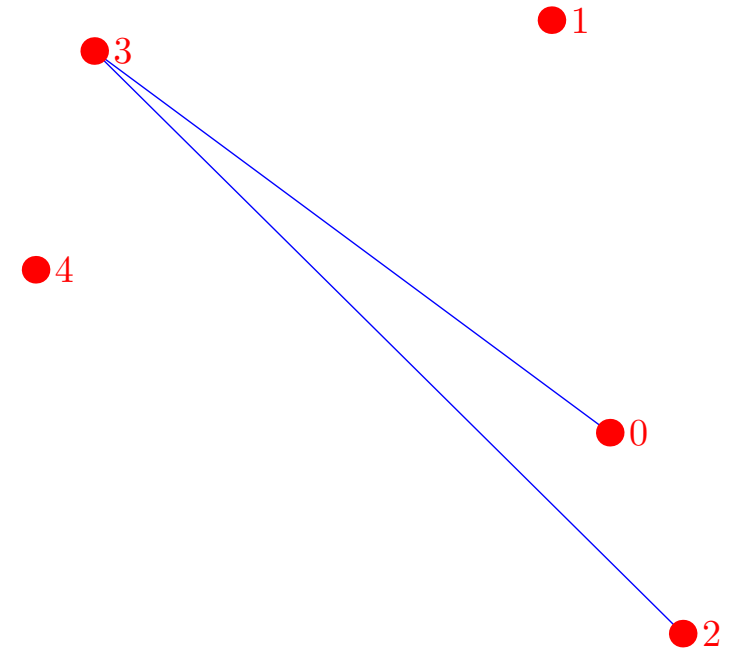
(0,3,1,4,2)

Branch and Bound in Action



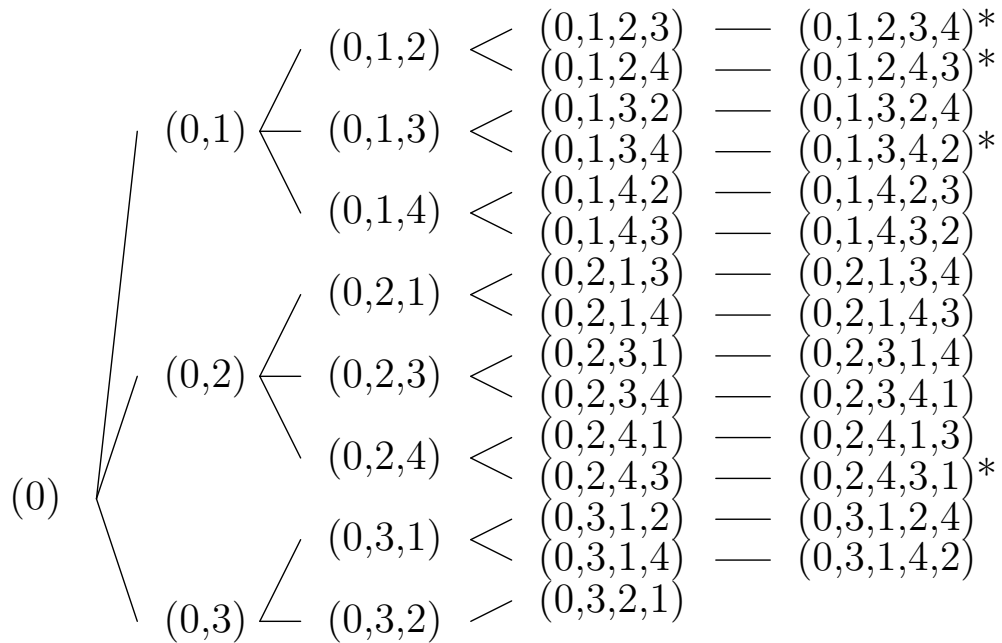
bound = 302.31

length = 215.91



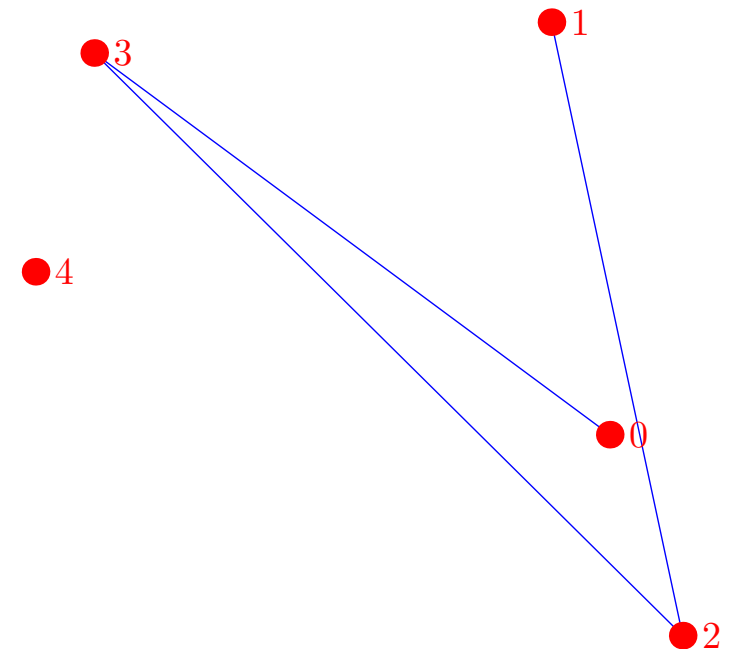
(0,3,2)

Branch and Bound in Action



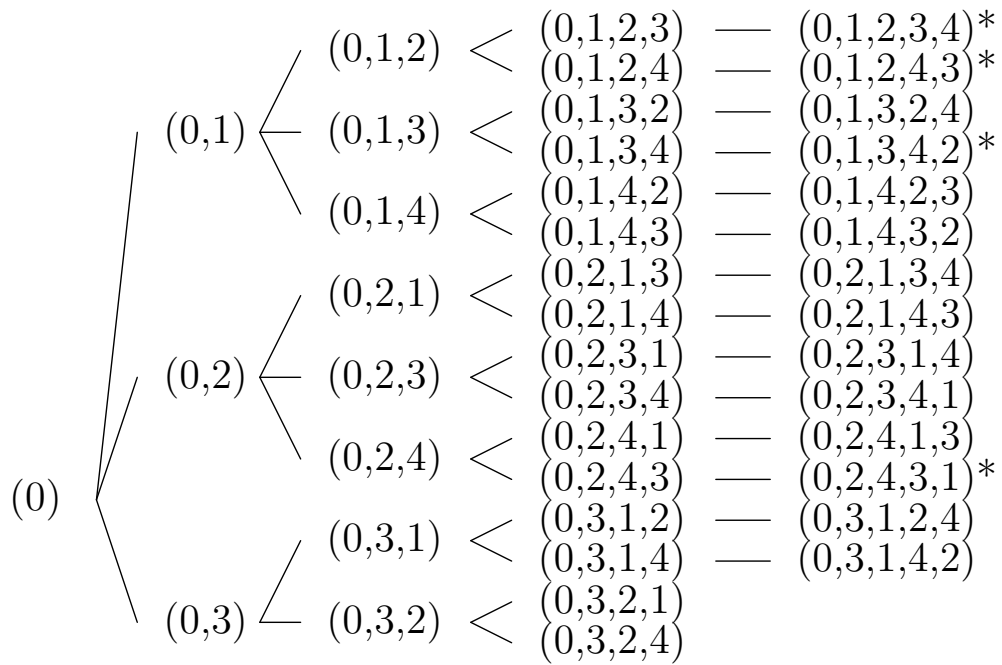
bound = 302.31

length = 308.09



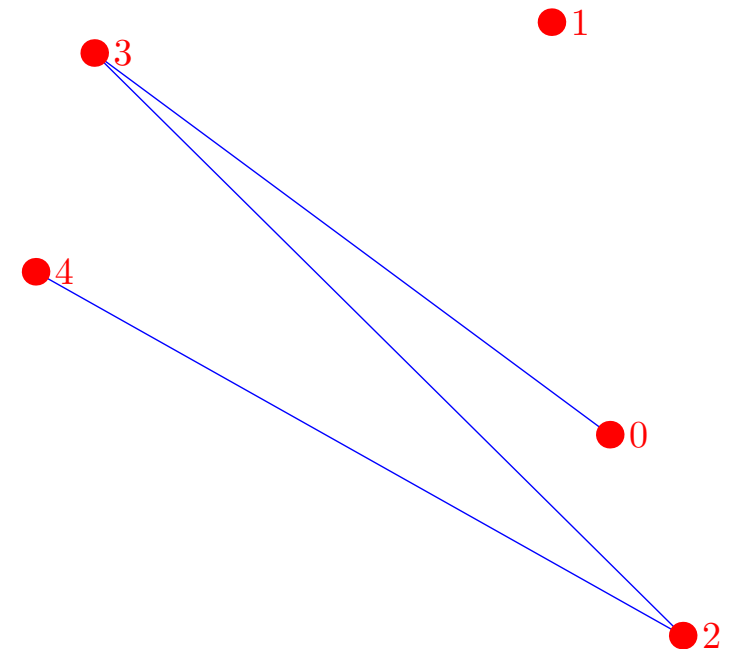
(0,3,2,1)

Branch and Bound in Action



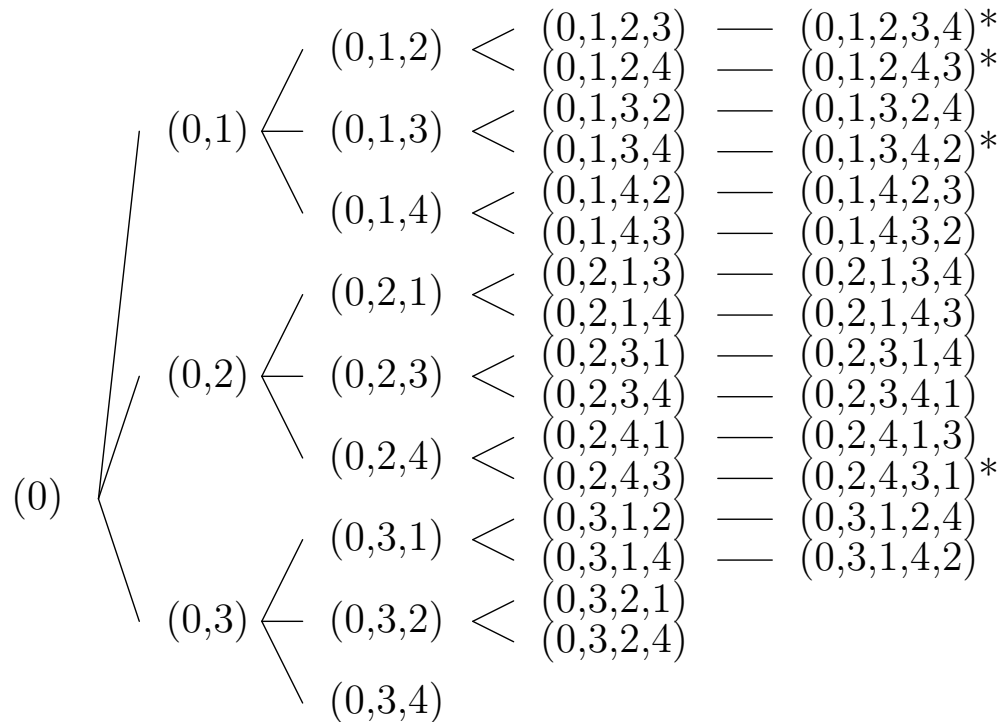
bound = 302.31

length = 324.99



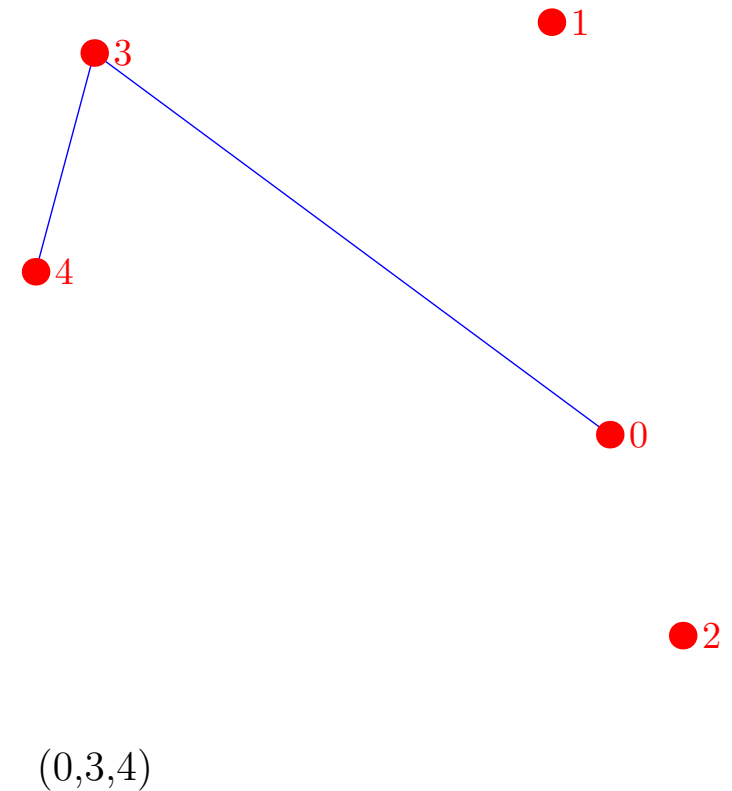
(0,3,2,4)

Branch and Bound in Action

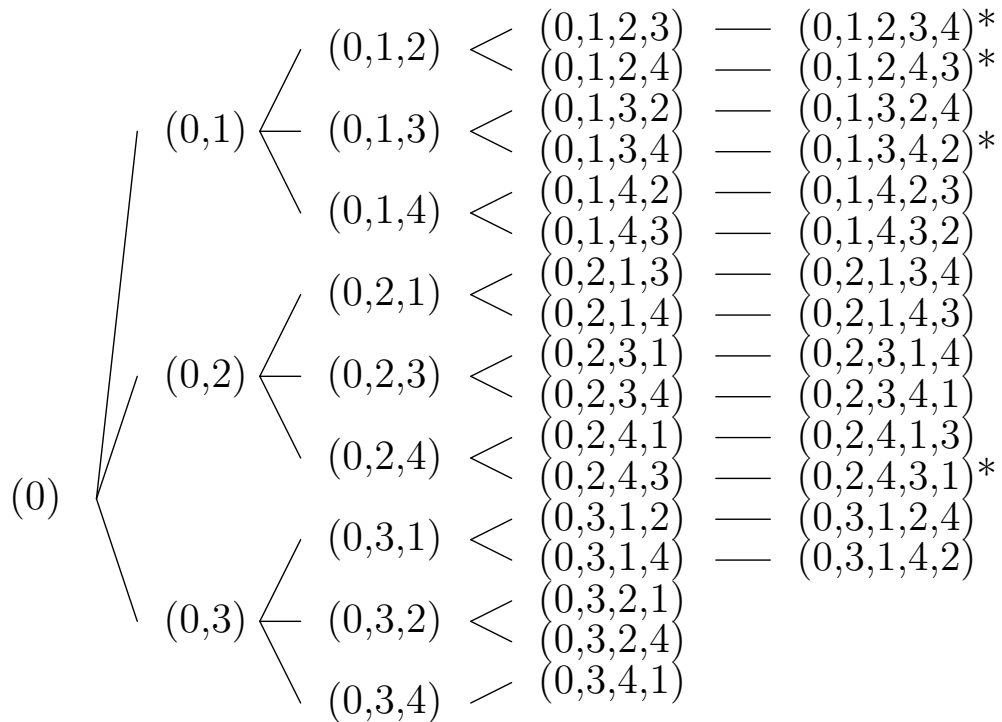


bound = 302.31

length = 127.52

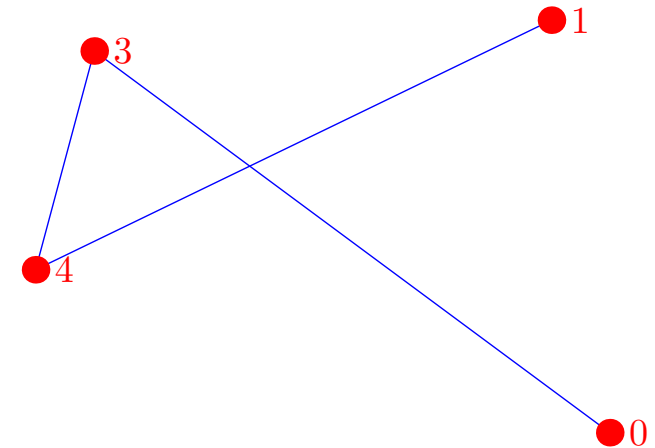


Branch and Bound in Action



bound = 302.31

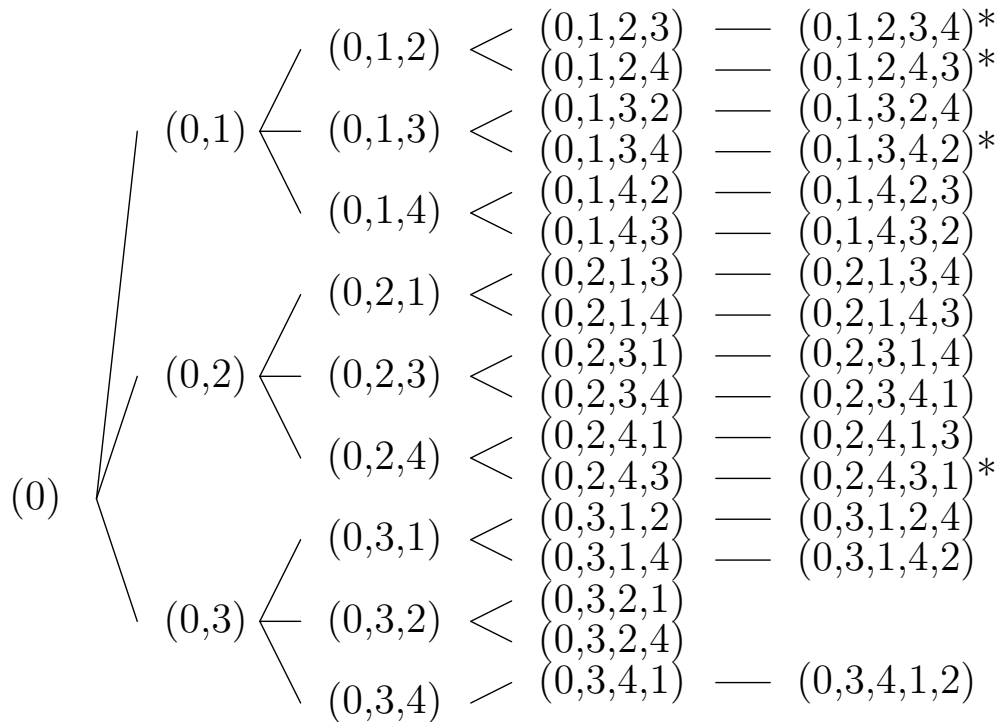
length = 211.72



● 2

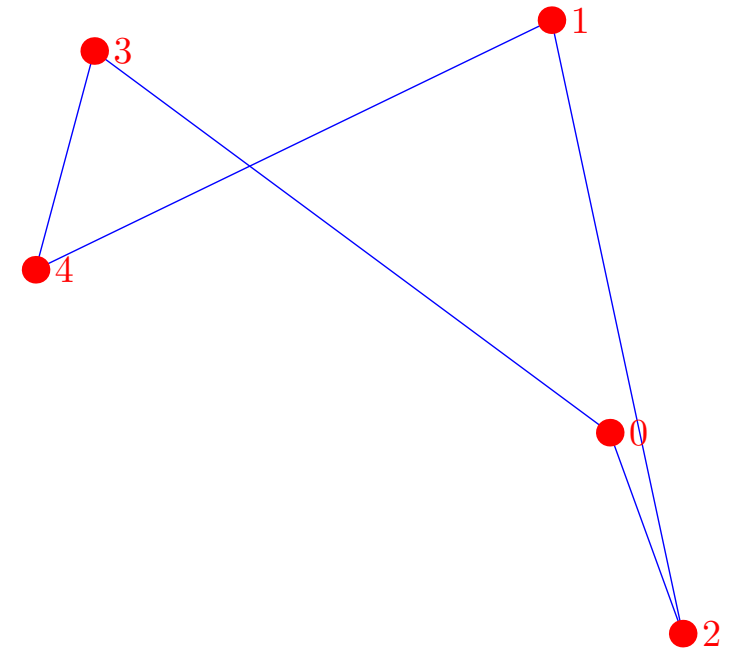
(0,3,4,1)

Branch and Bound in Action



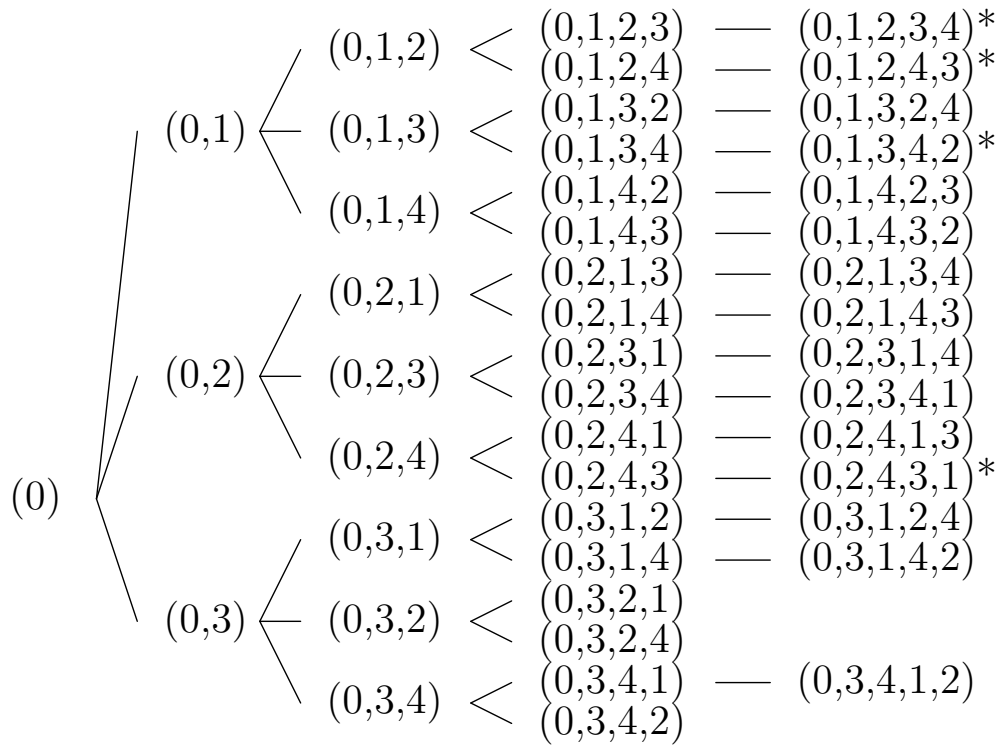
bound = 302.31

length = 335.3



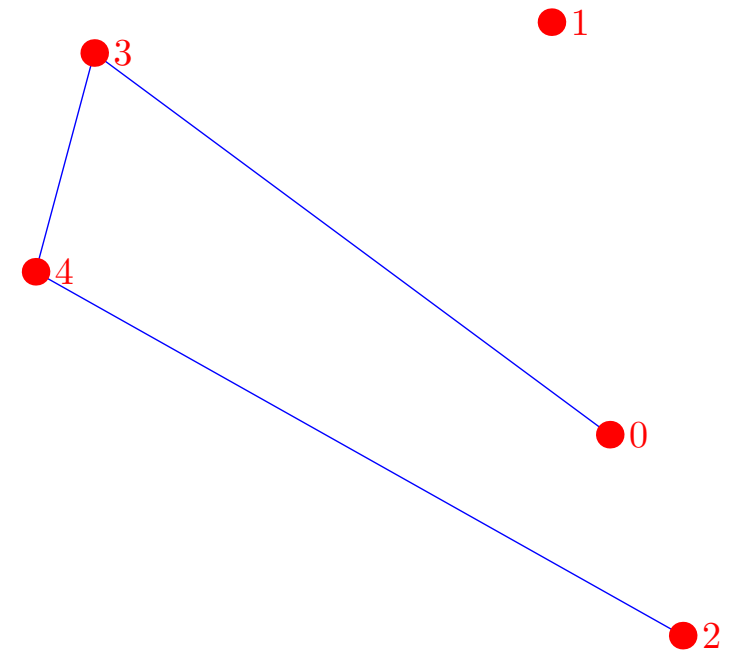
(0,3,4,1,2)

Branch and Bound in Action



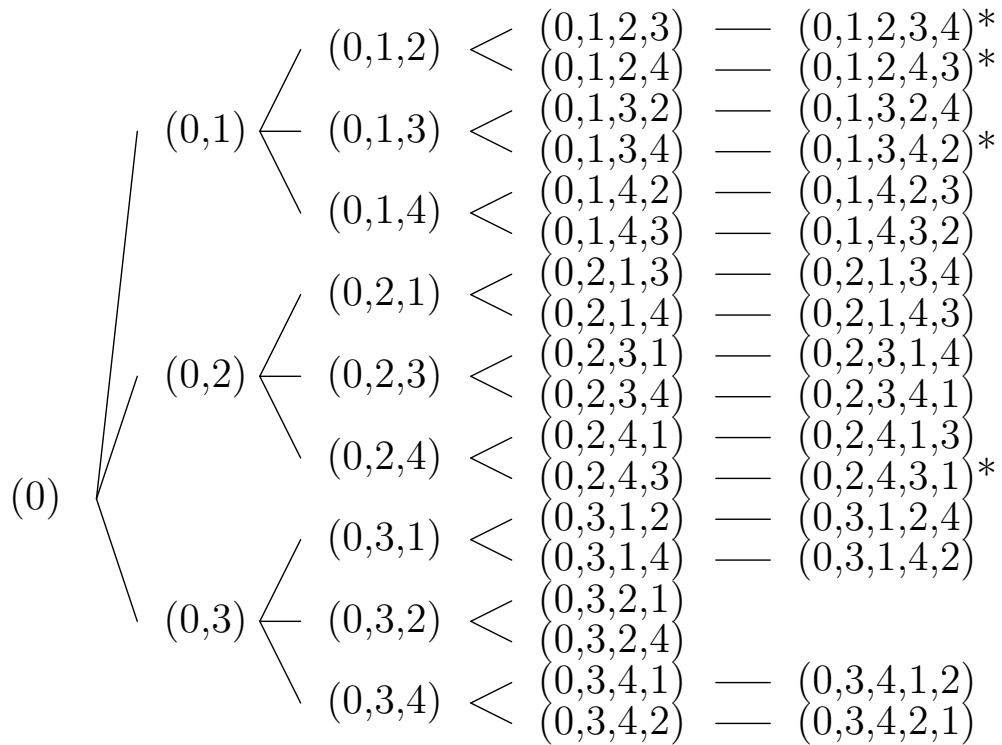
bound = 302.31

length = 236.6



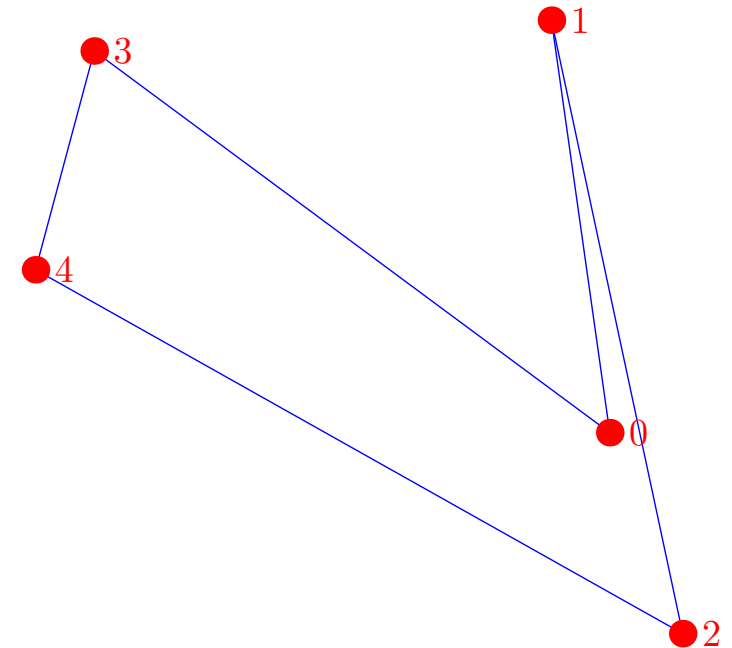
(0,3,4,2)

Branch and Bound in Action



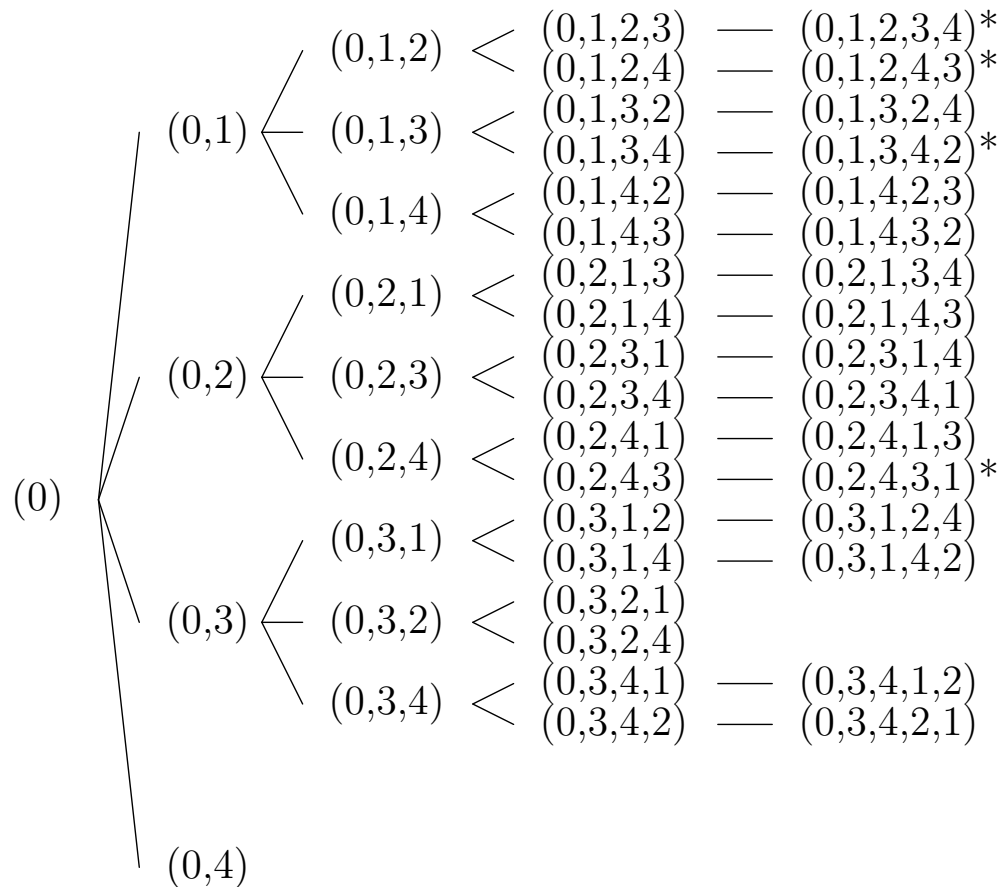
bound = 302.31

length = 389.98



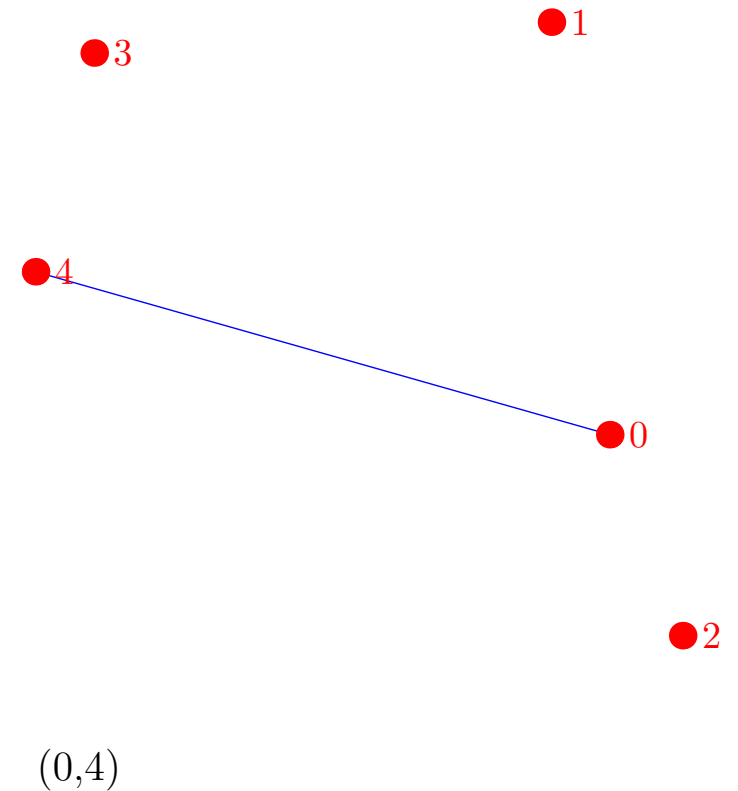
(0,3,4,2,1)

Branch and Bound in Action

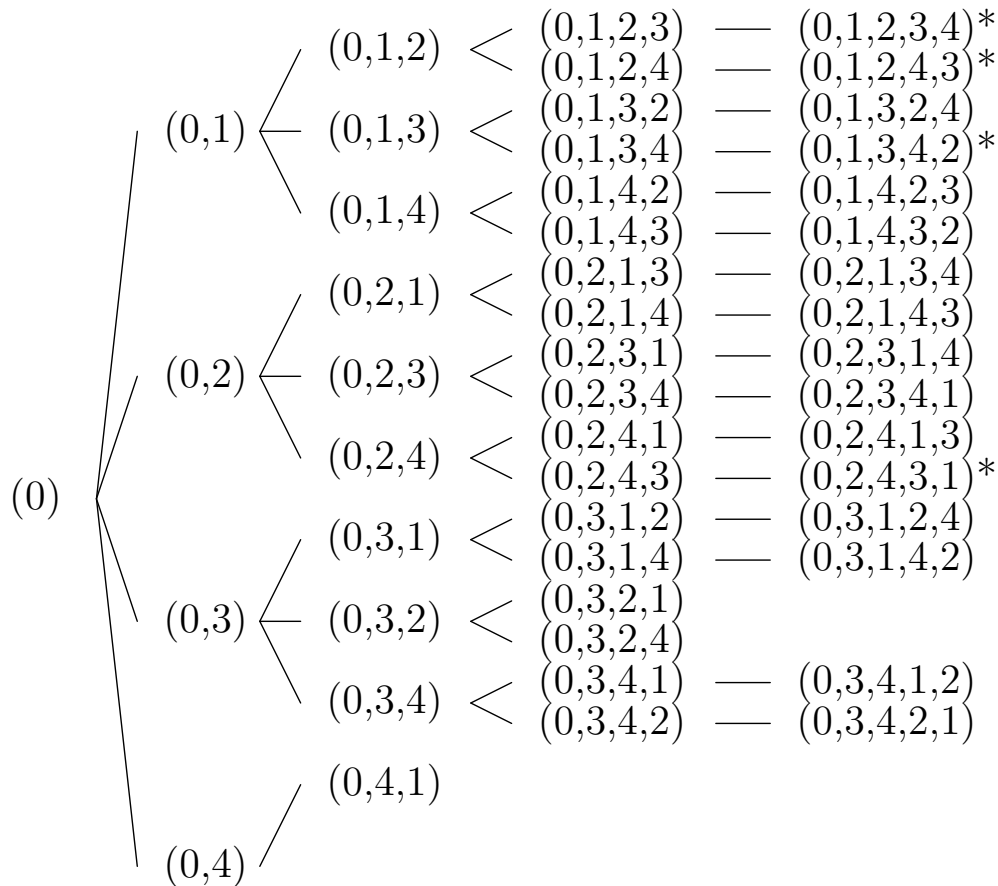


bound = 302.31

length = 87.692

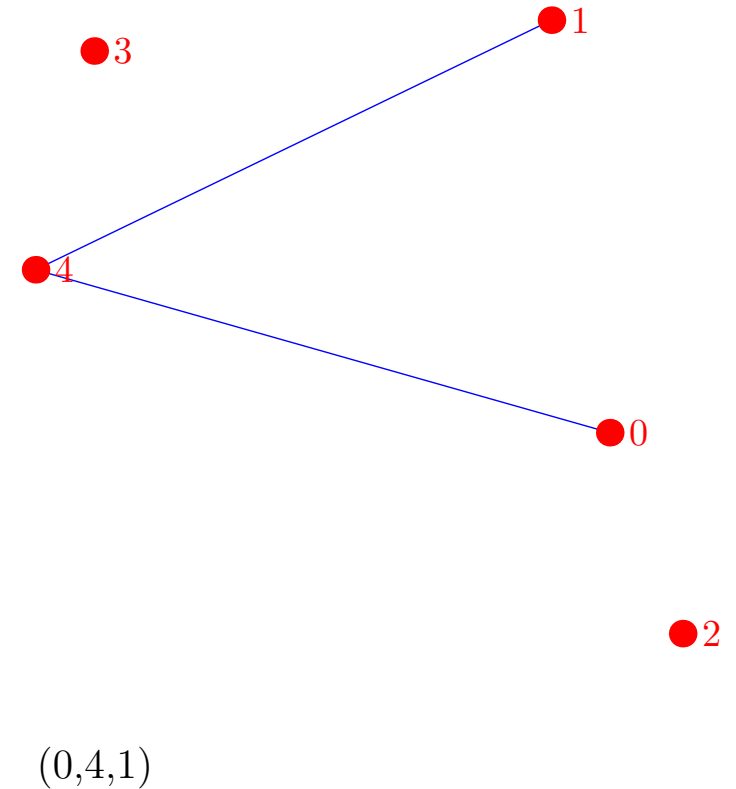


Branch and Bound in Action

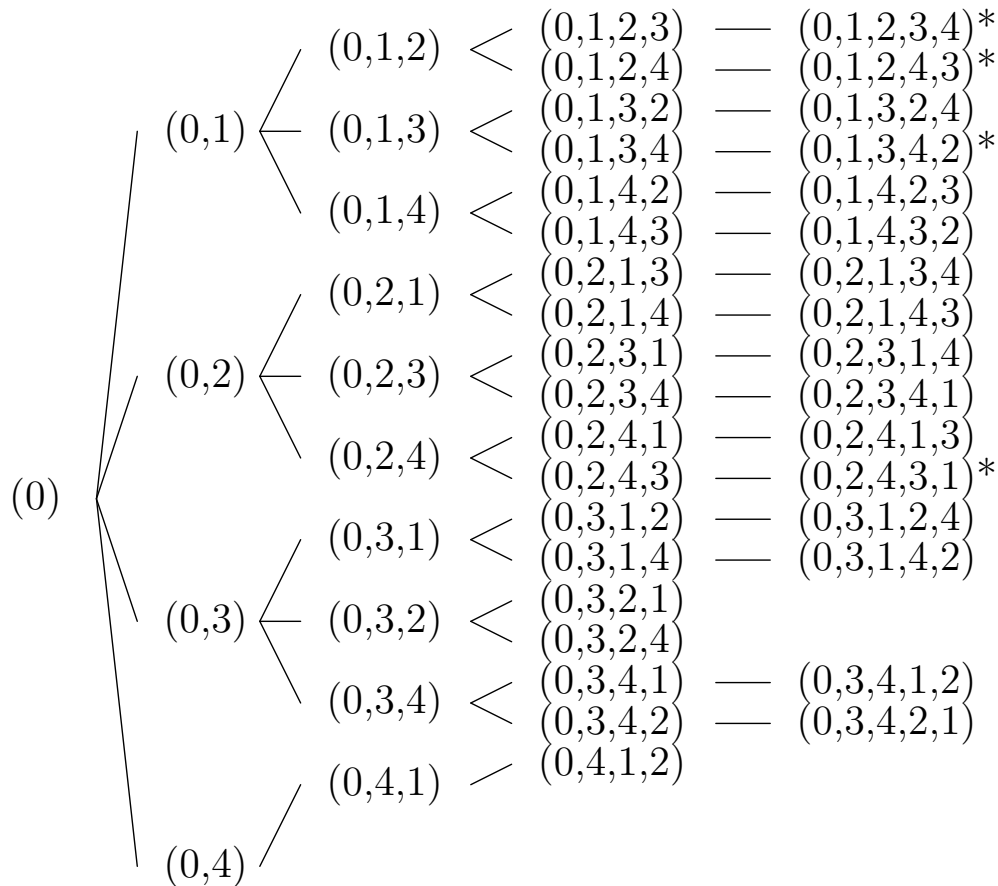


bound = 302.31

length = 171.9

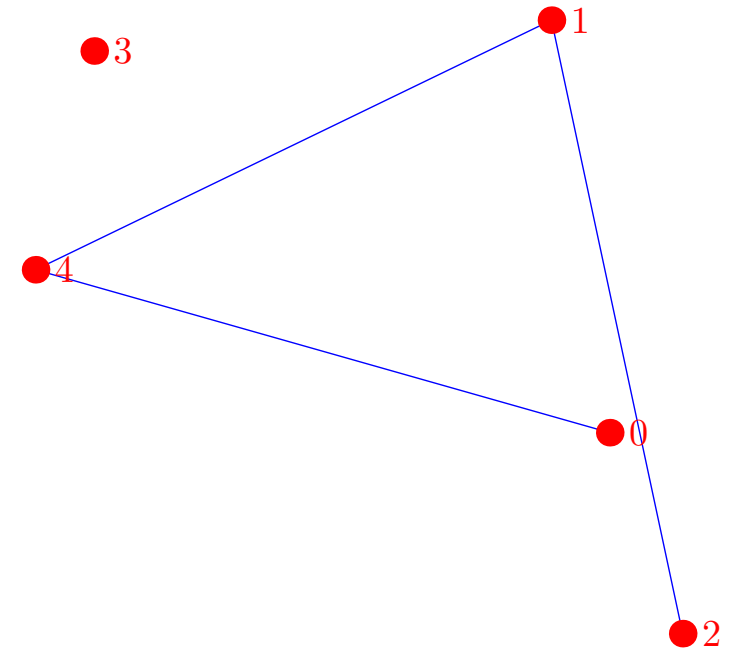


Branch and Bound in Action



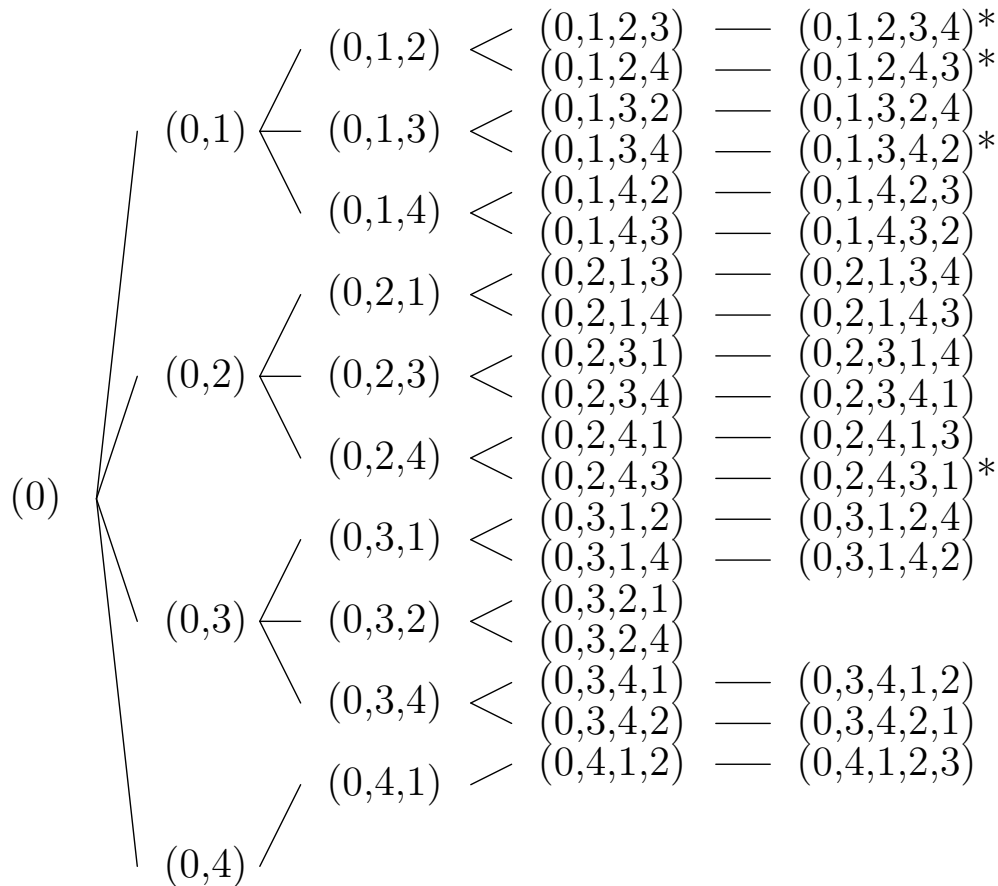
bound = 302.31

length = 264.07



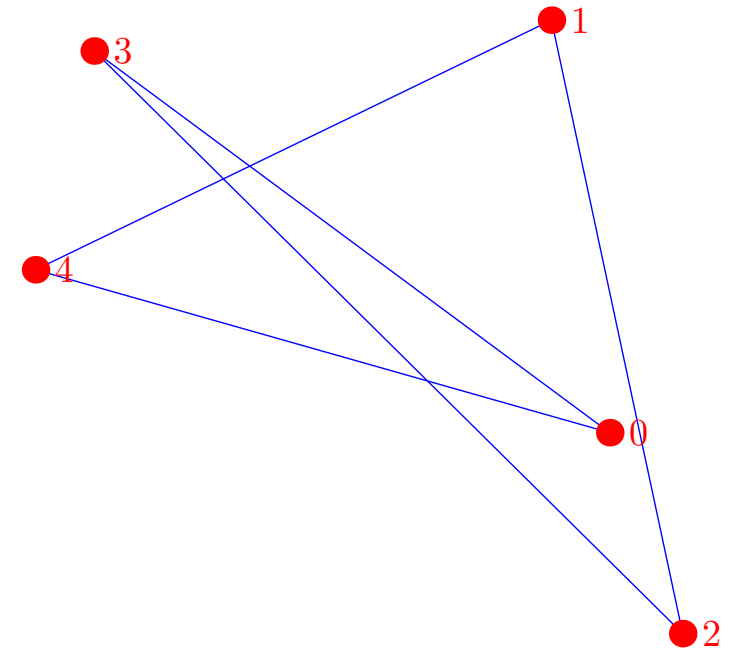
(0,4,1,2)

Branch and Bound in Action



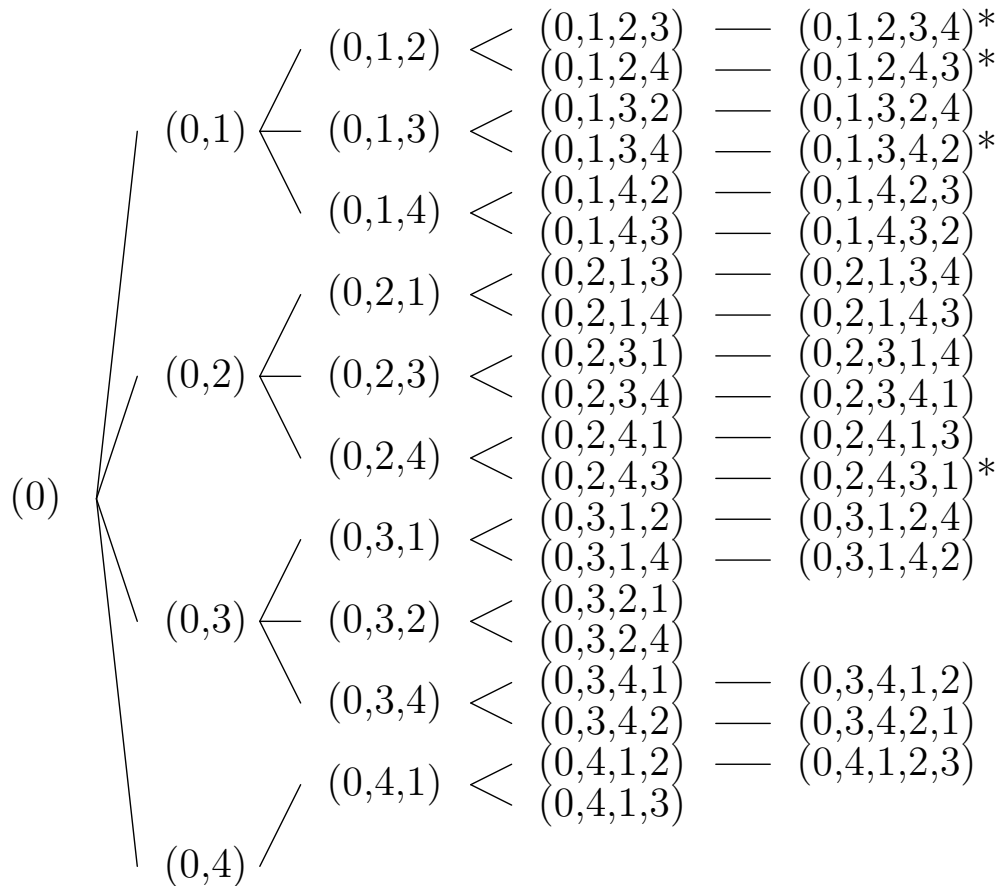
bound = 302.31

length = 479.98



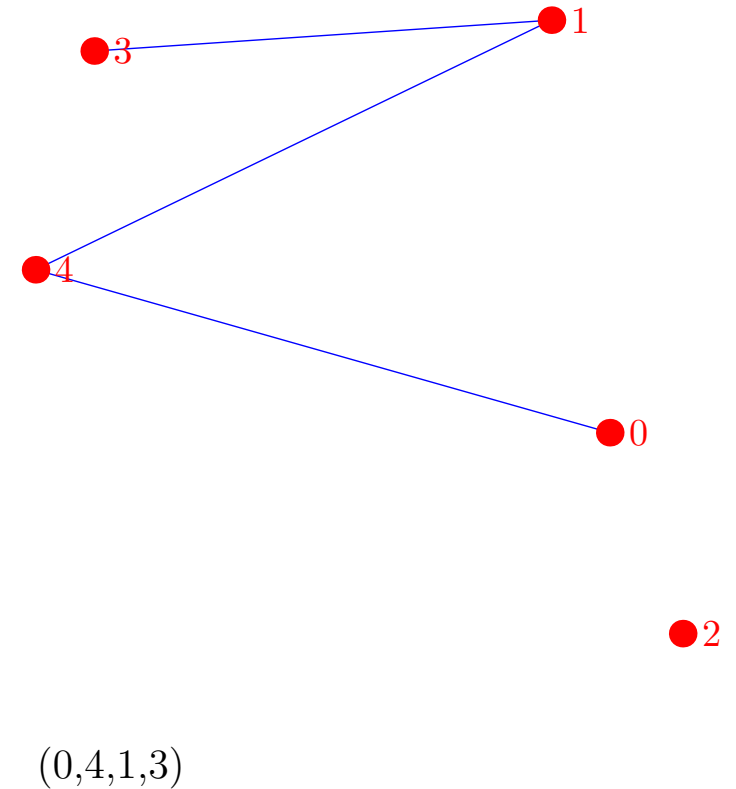
(0,4,1,2,3)

Branch and Bound in Action

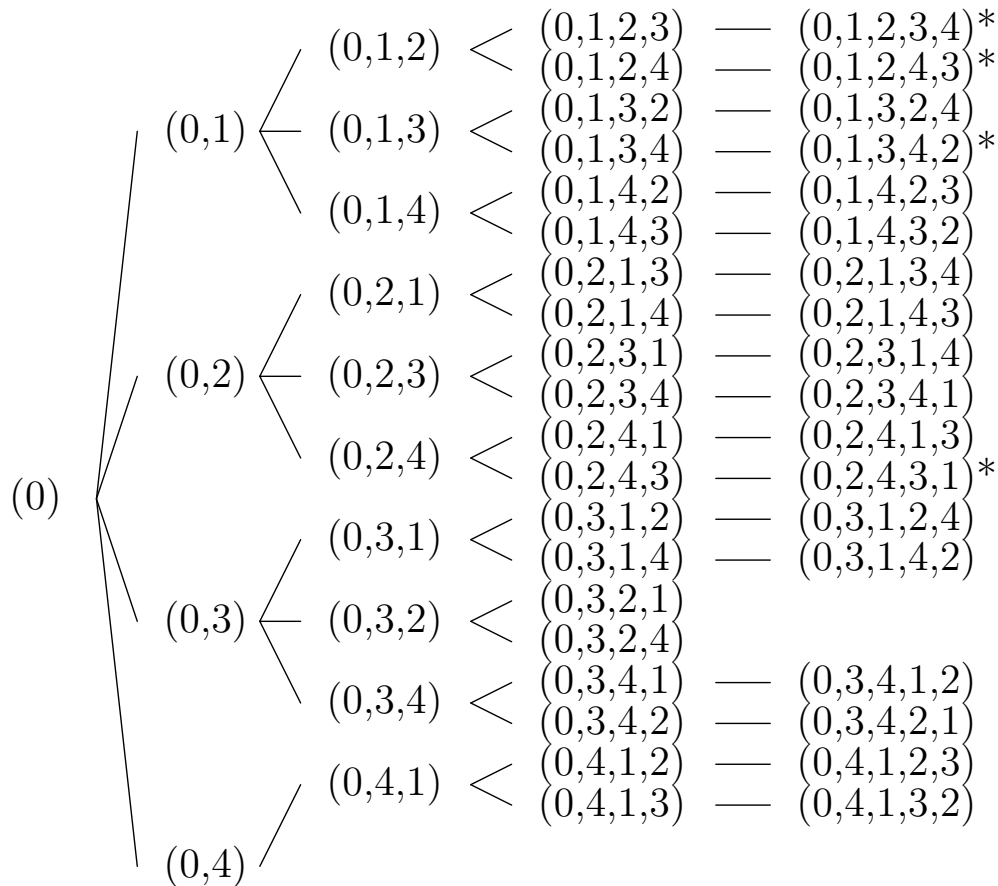


bound = 302.31

length = 239.23

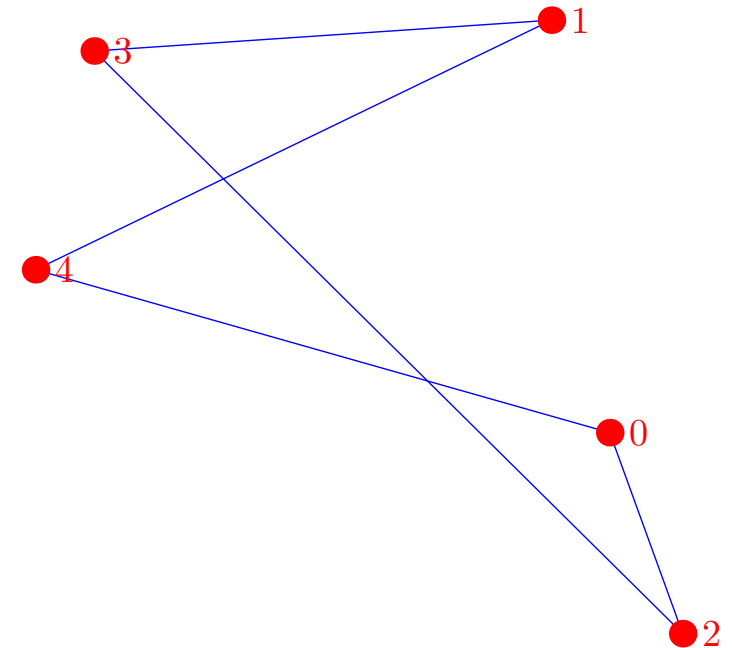


Branch and Bound in Action



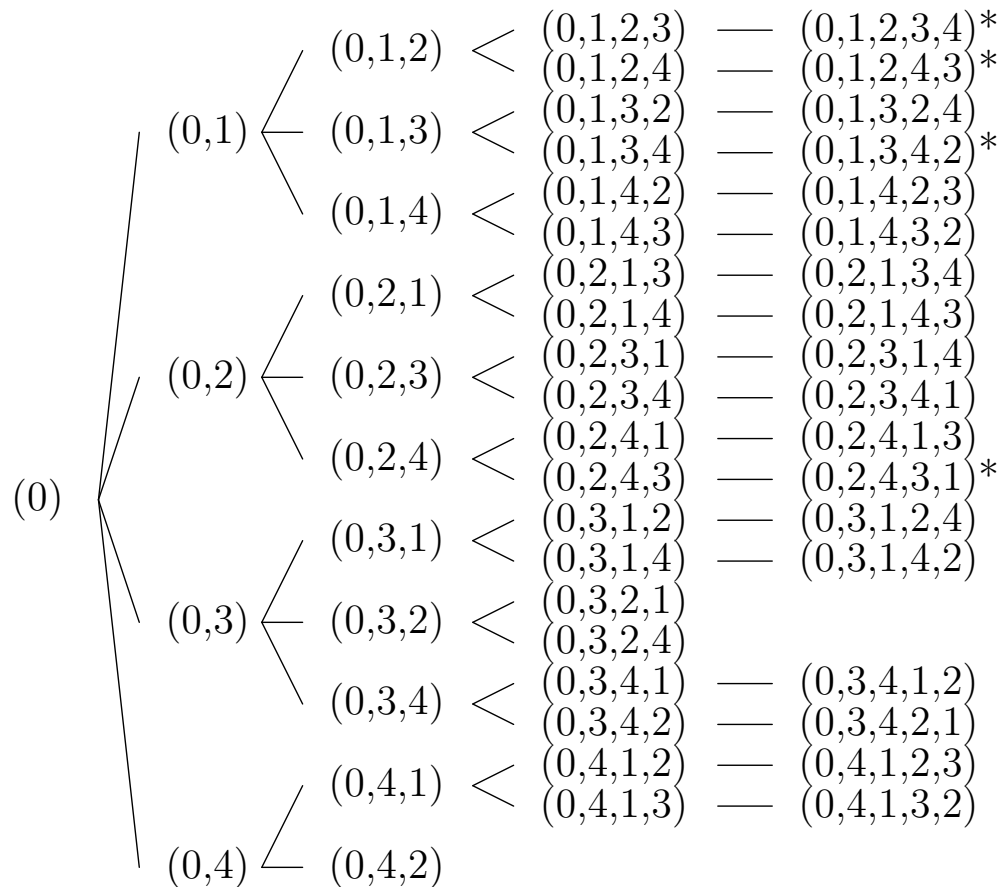
bound = 302.31

length = 392.31



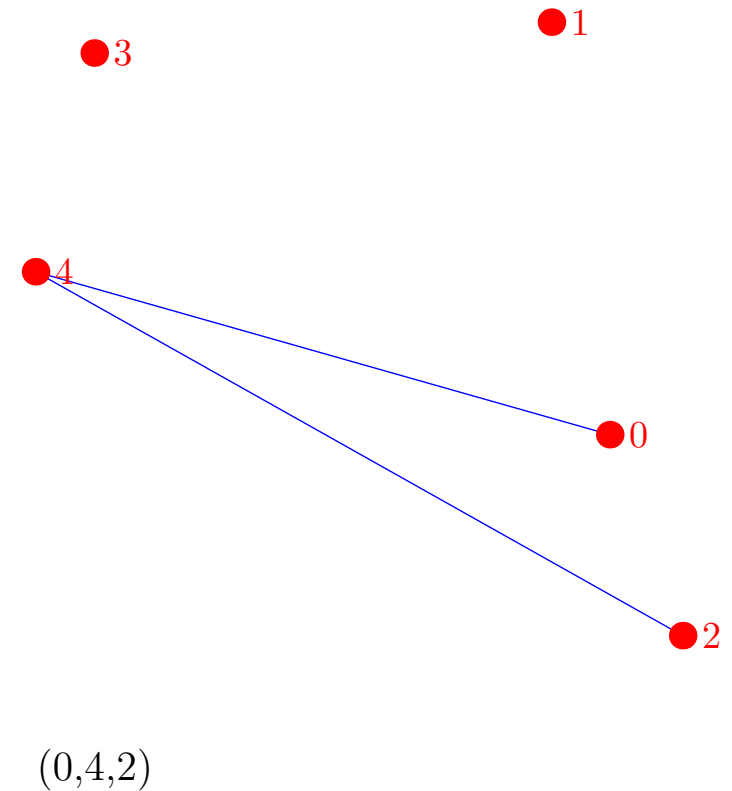
(0,4,1,3,2)

Branch and Bound in Action

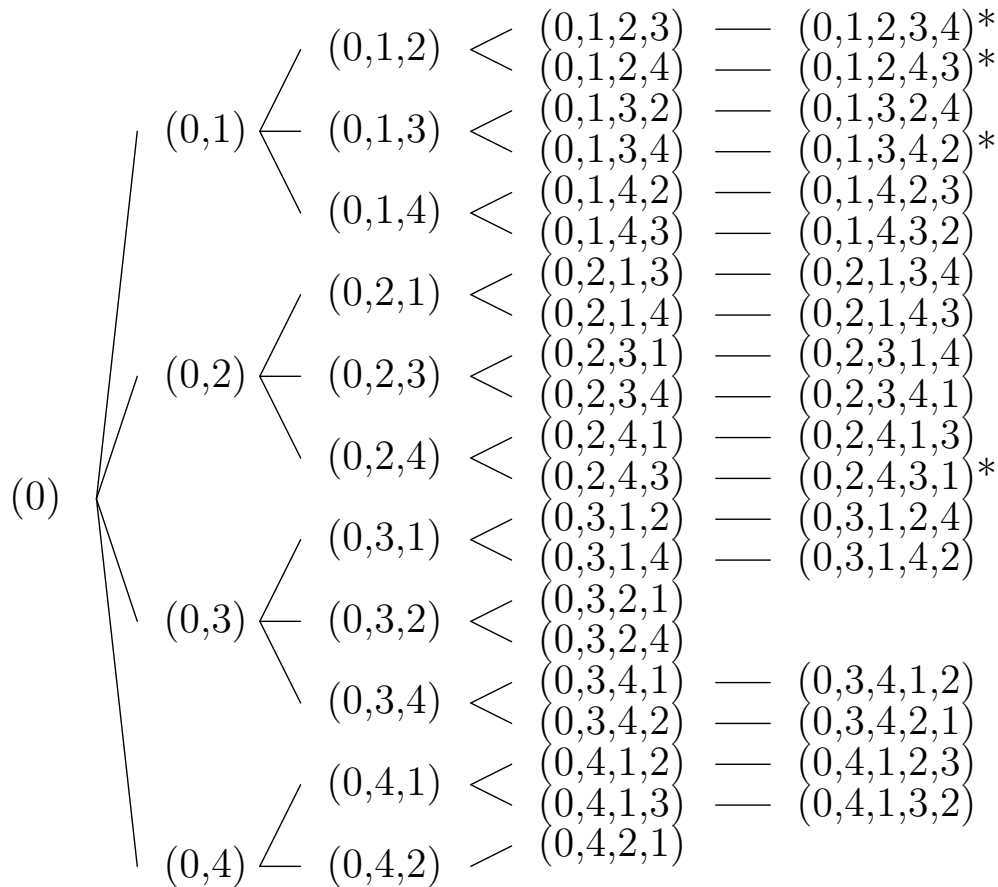


bound = 302.31

length = 196.77

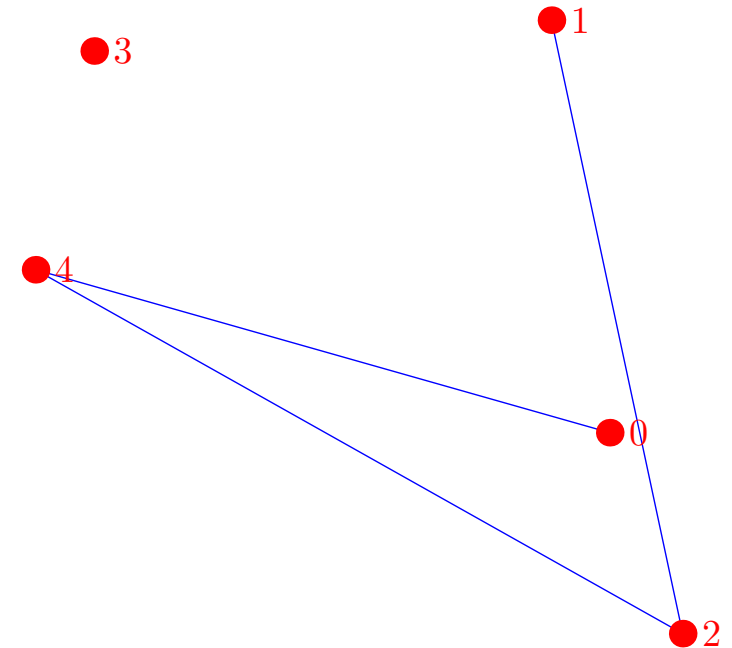


Branch and Bound in Action



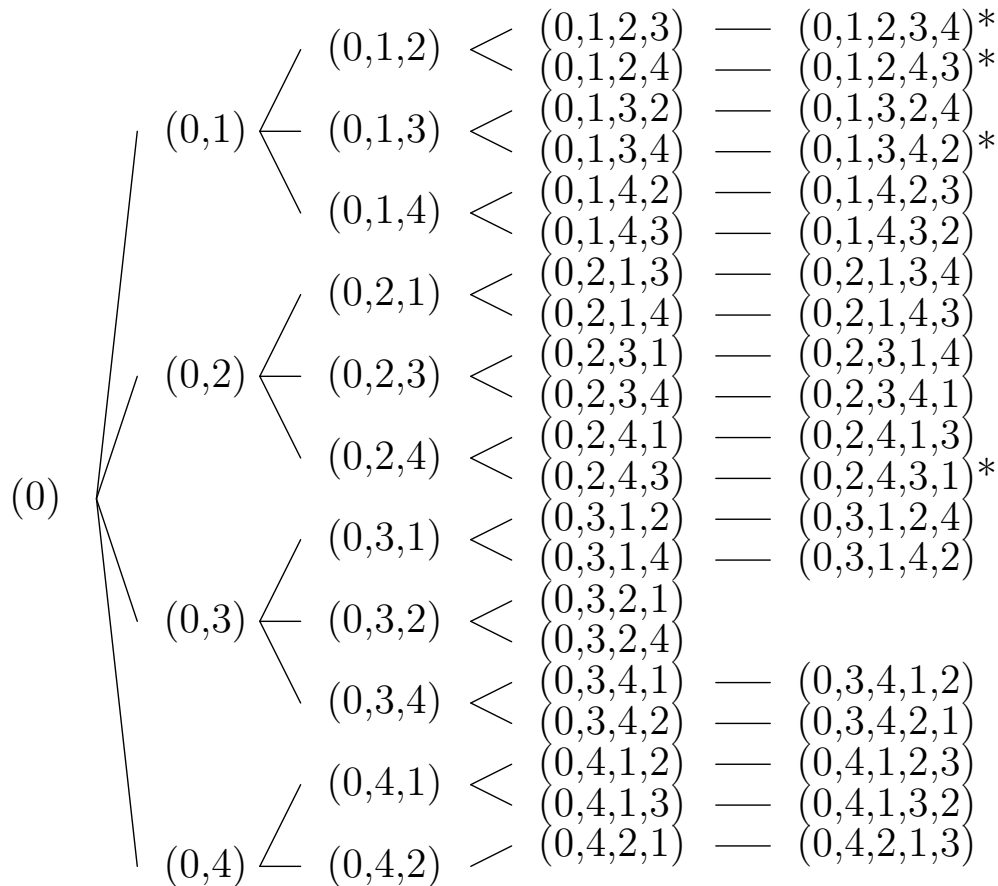
bound = 302.31

length = 288.95



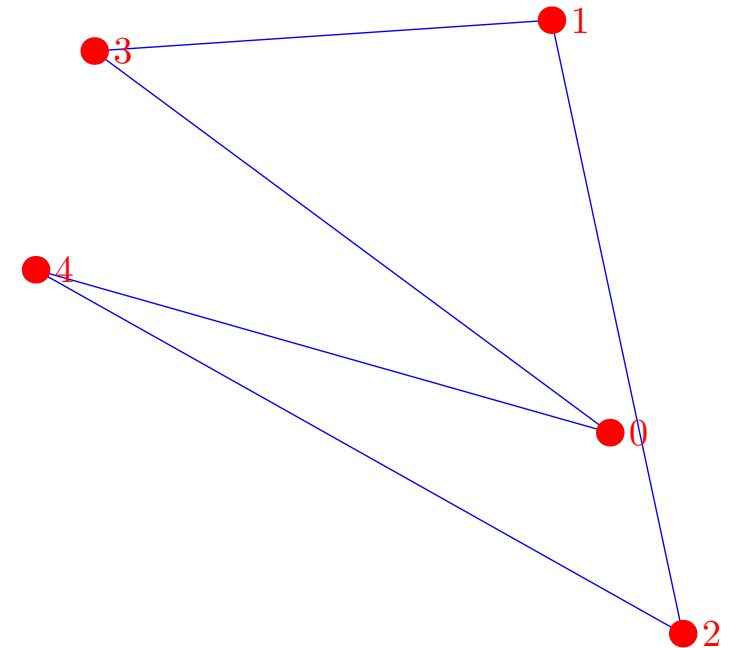
(0,4,2,1)

Branch and Bound in Action



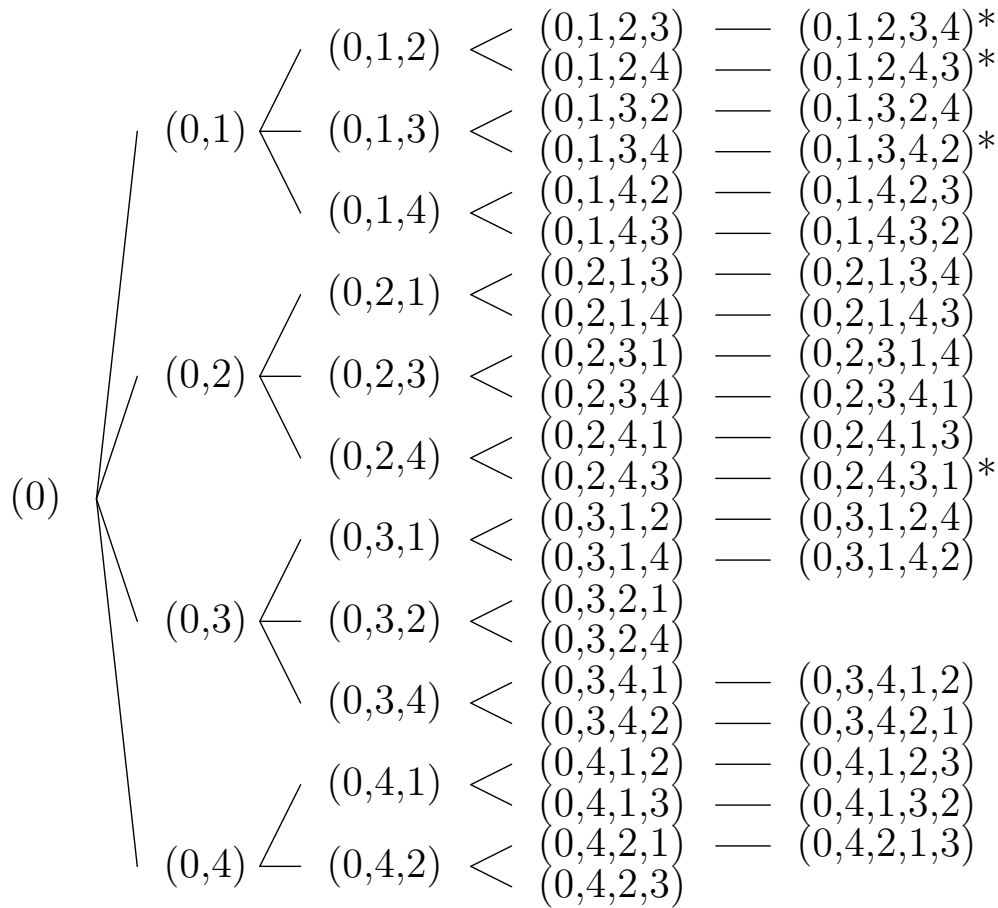
bound = 302.31

length = 450.52



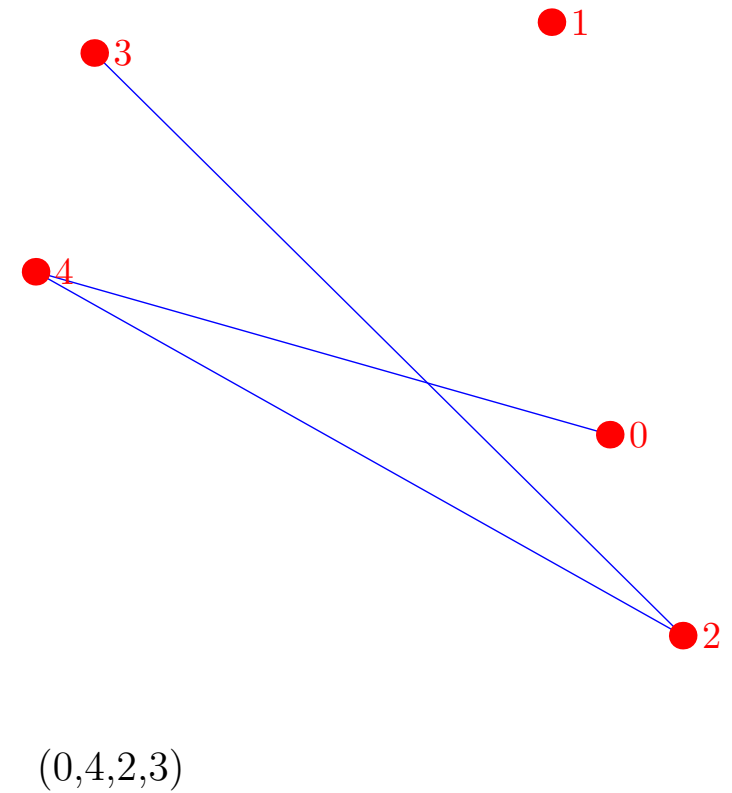
(0,4,2,1,3)

Branch and Bound in Action

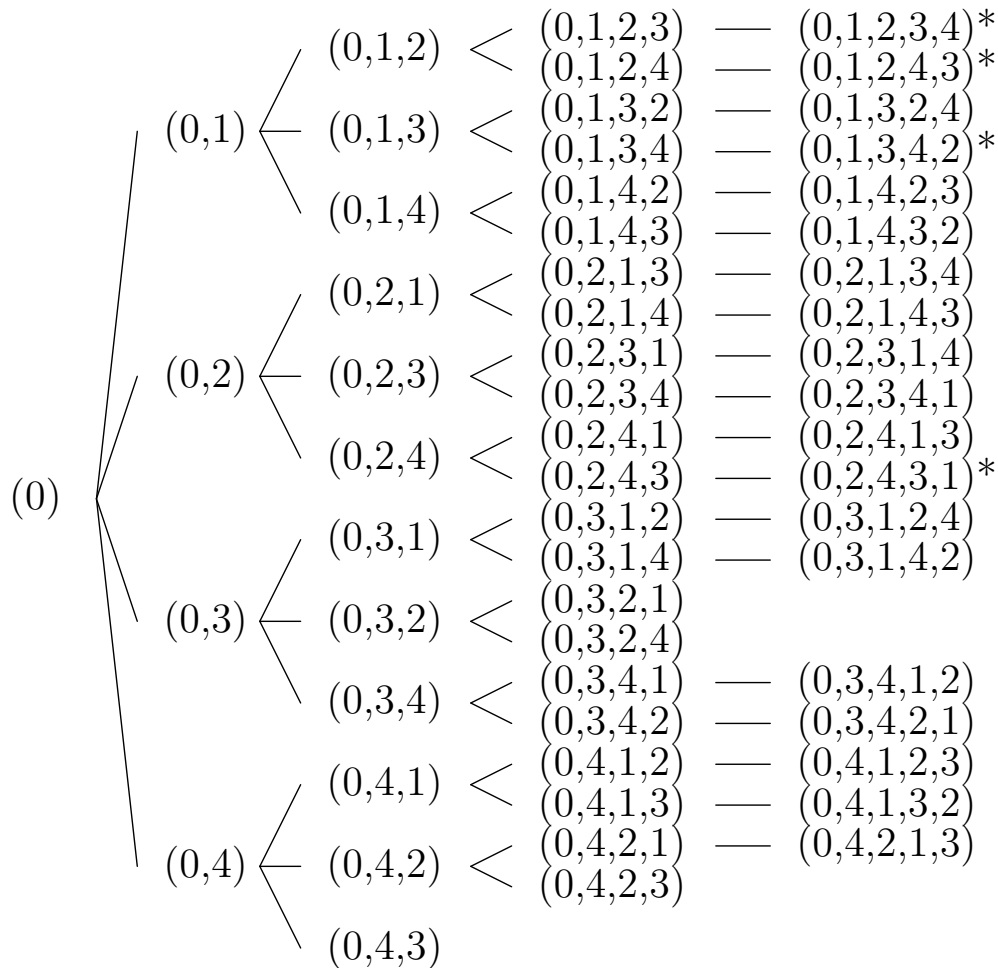


bound = 302.31

length = 318.44

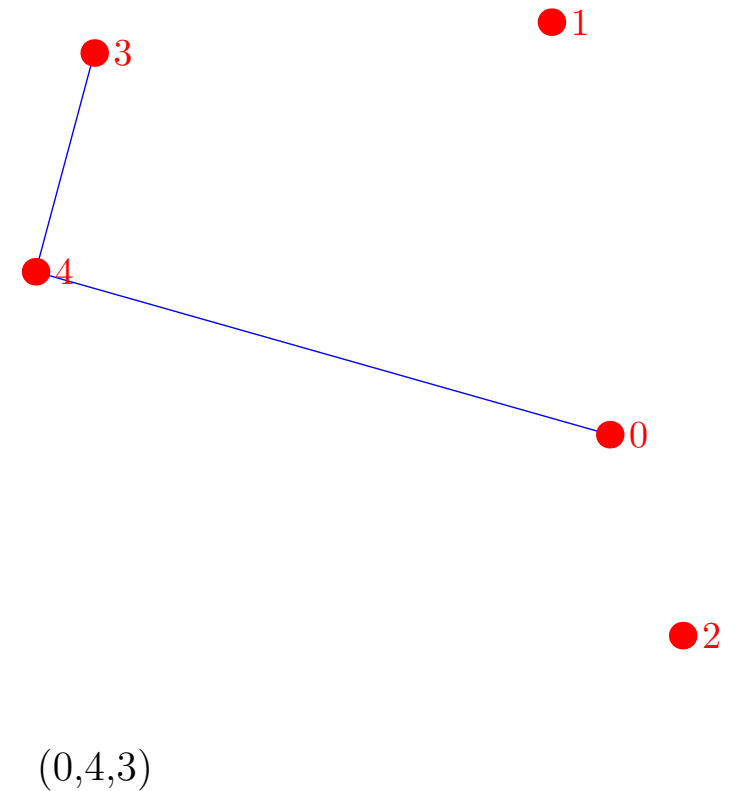


Branch and Bound in Action

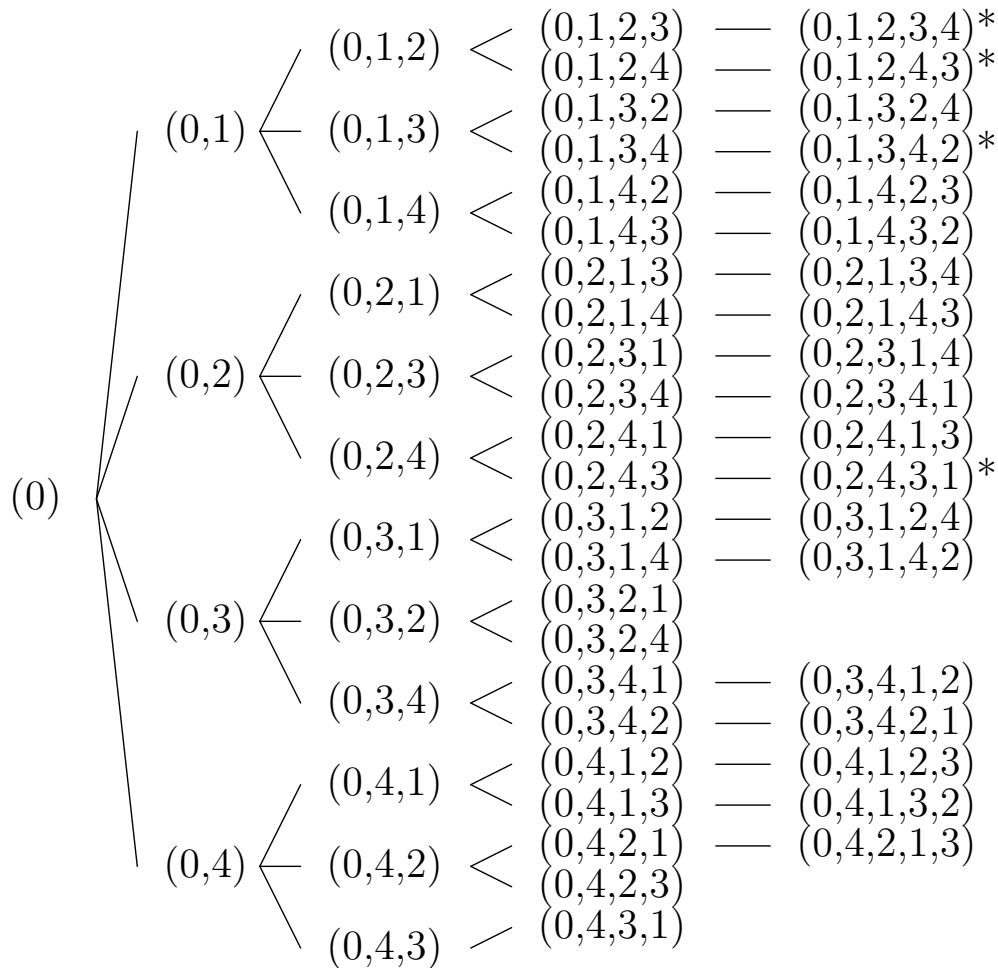


bound = 302.31

length = 120.97

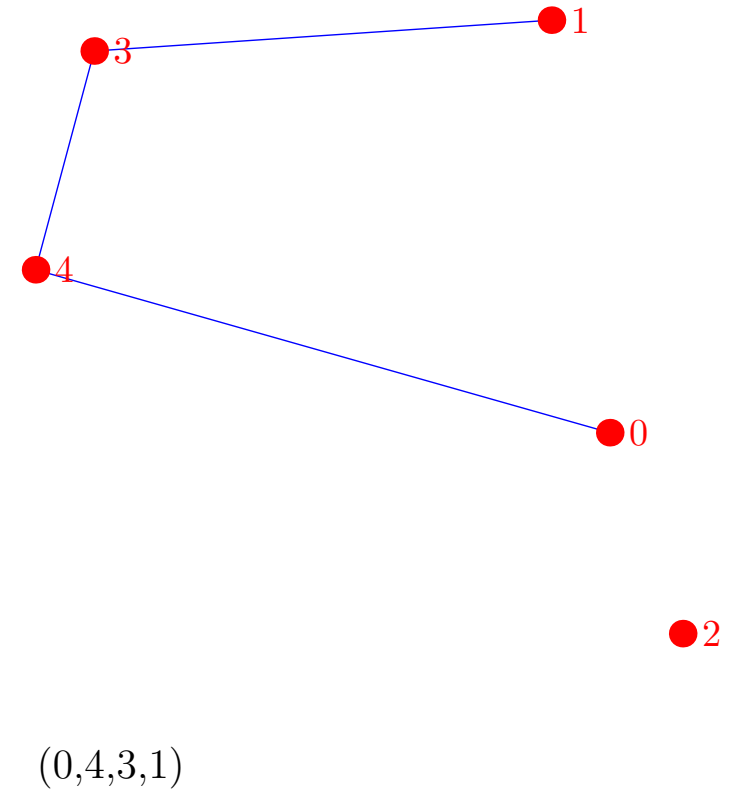


Branch and Bound in Action

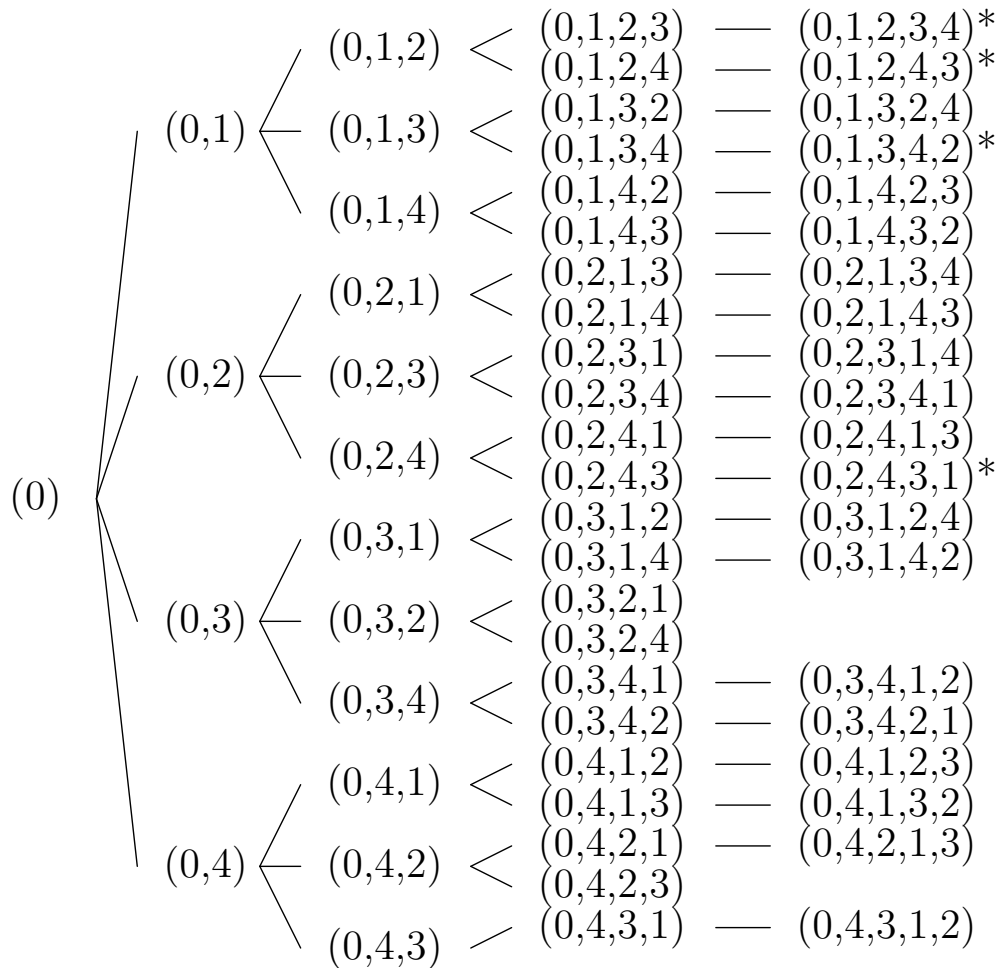


bound = 302.31

length = 188.3

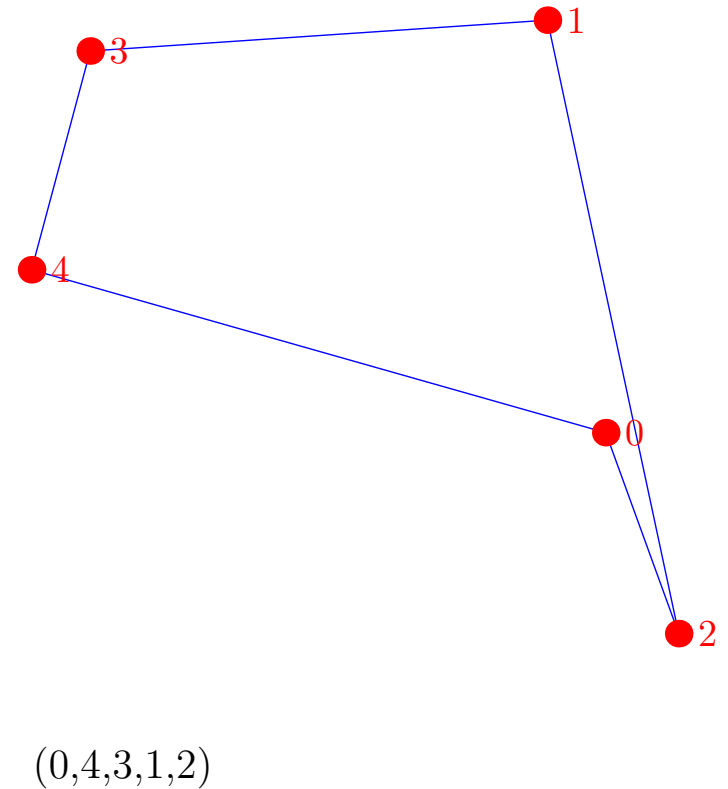


Branch and Bound in Action

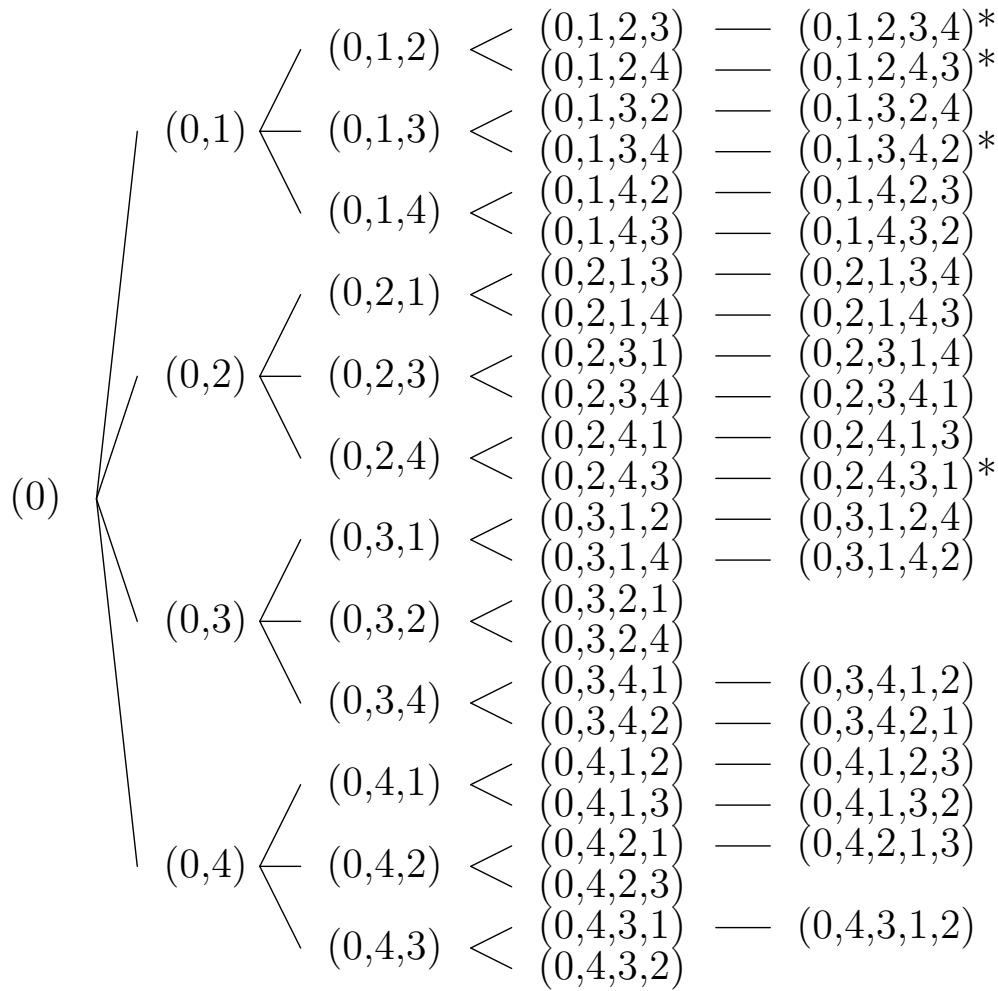


bound = 302.31

length = 311.88

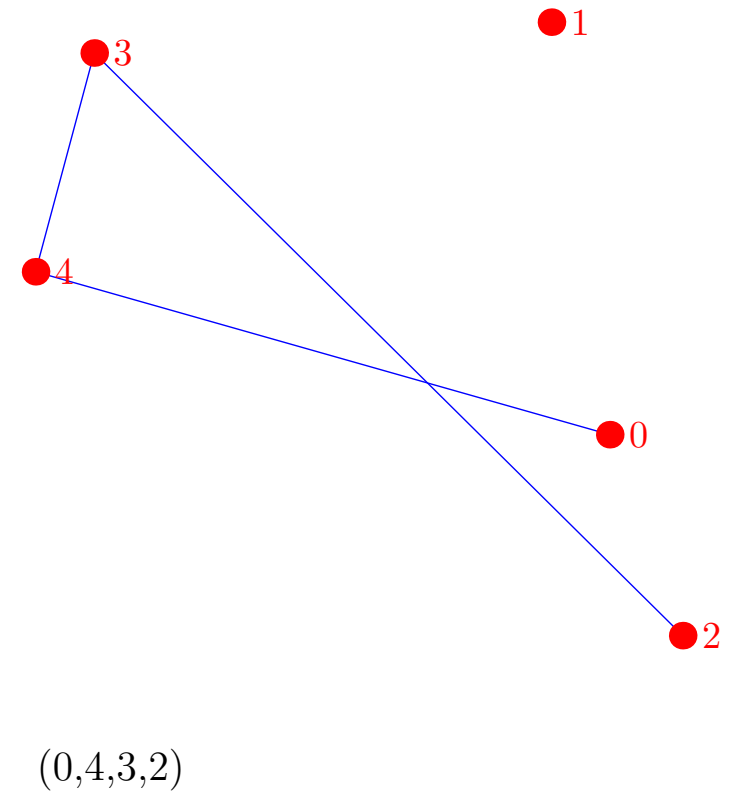


Branch and Bound in Action

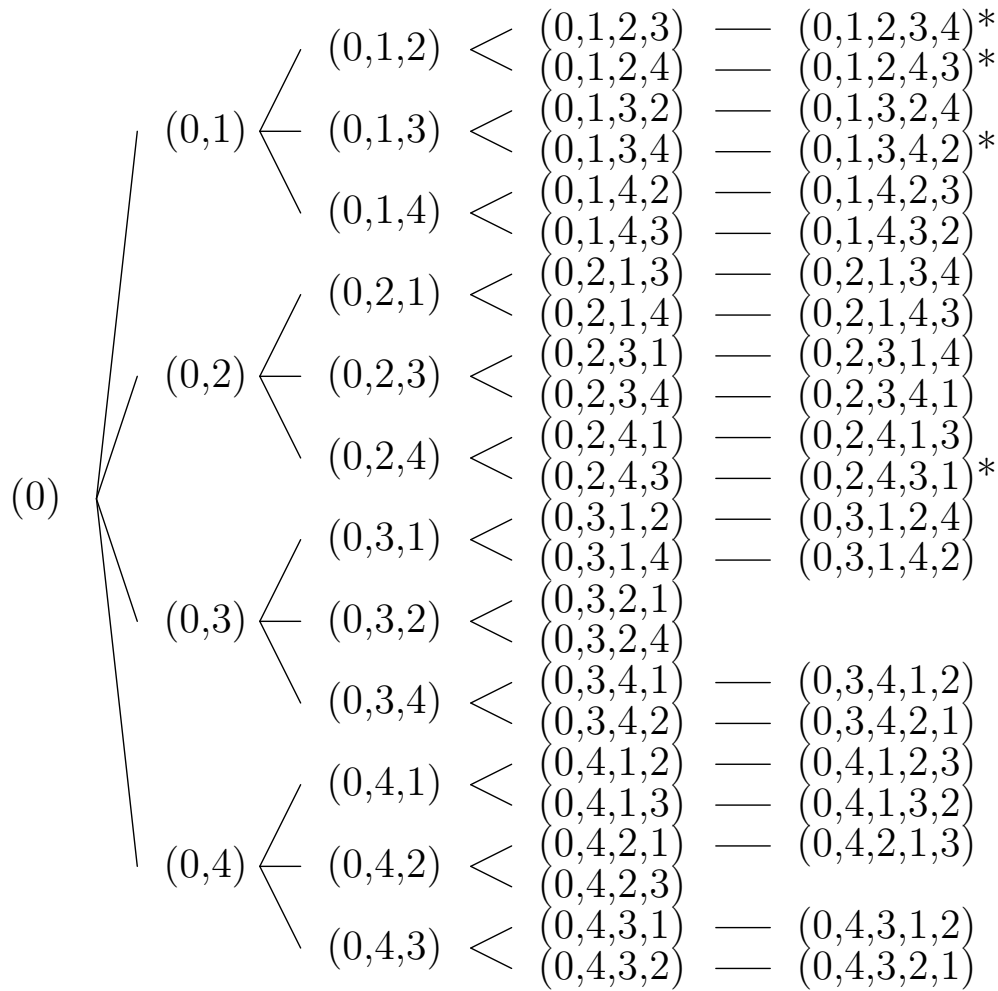


bound = 302.31

length = 242.64

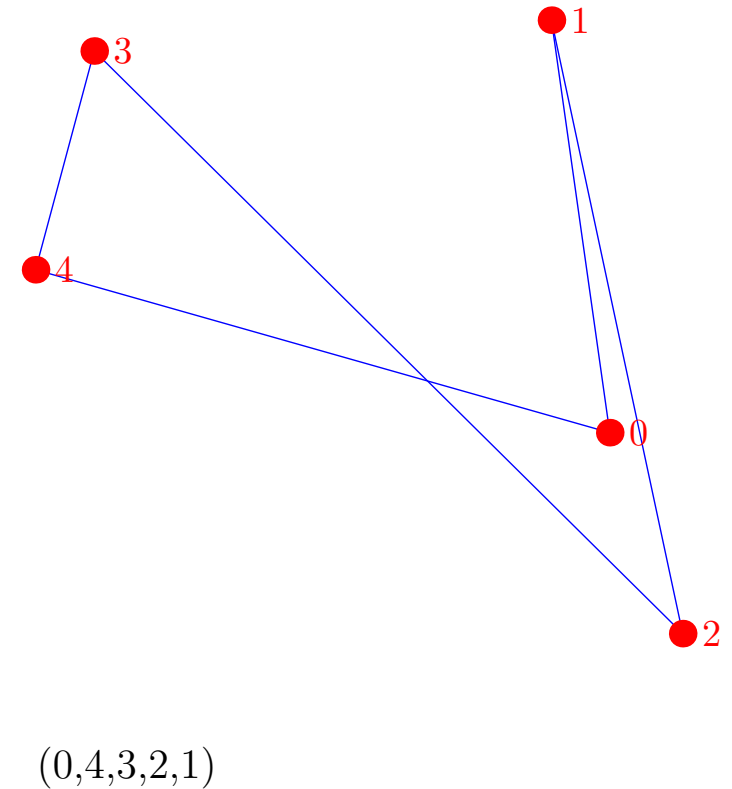


Branch and Bound in Action



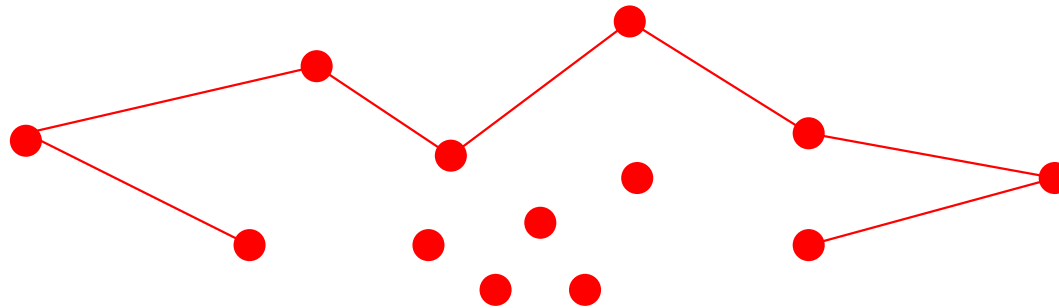
bound = 302.31

length = 396.02



Bound on Partial Solution

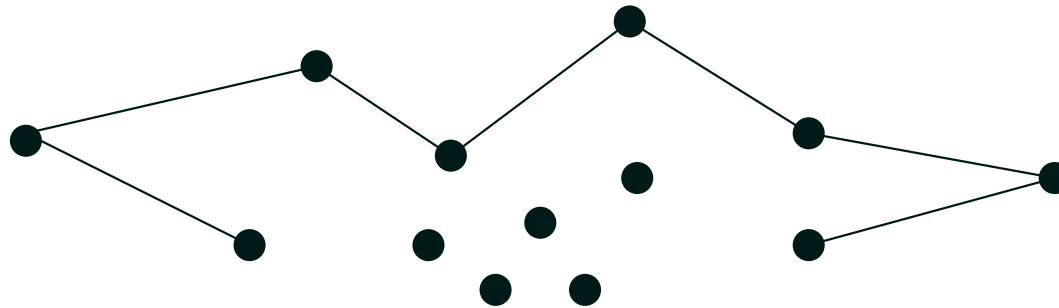
- We know that the partial solution has to include all the remaining cities



- We can use this to obtain a lower bound on the partial solution
- We know the remaining tour will go through each of the unvisited cities and the two edge cities
- In fact the remaining part of the tour is a spanning tree of these vertices (it connects all the vertices once and has no cycles)
- But we know a lower bound for this

Bound on Partial Solution

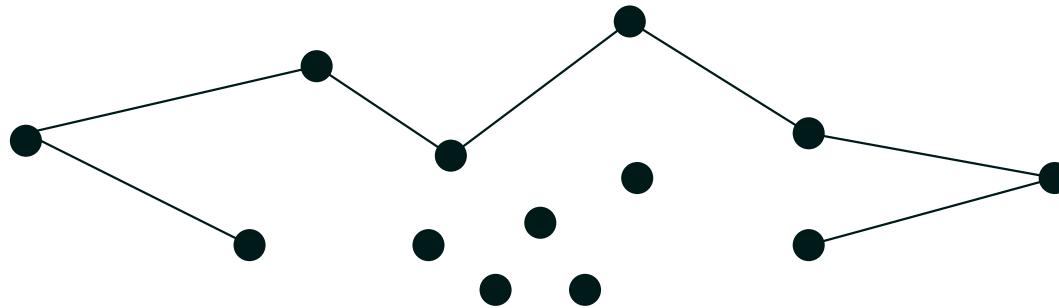
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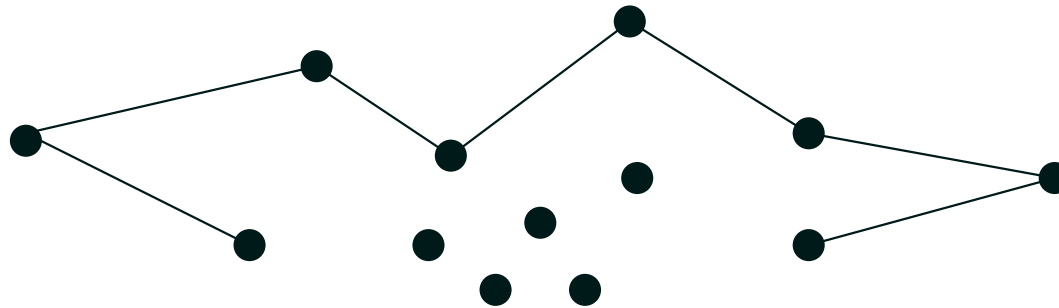
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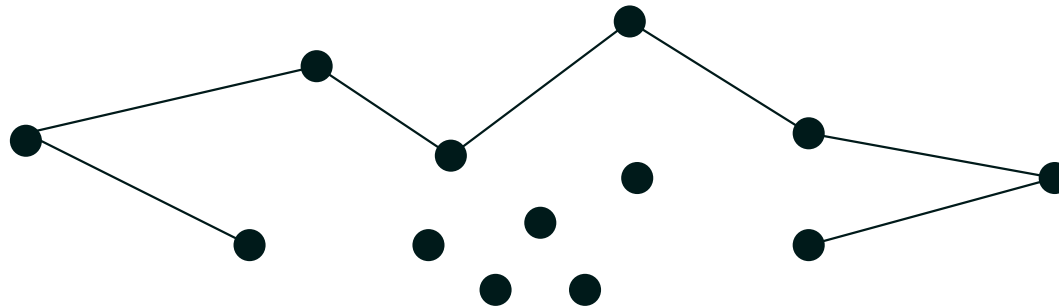
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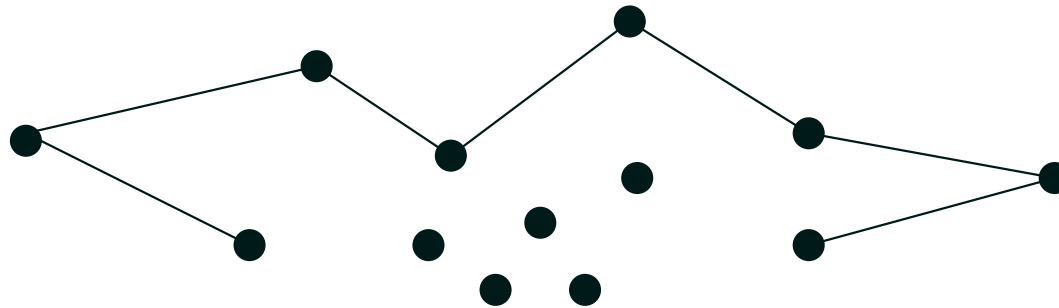
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Bound on Partial Solution

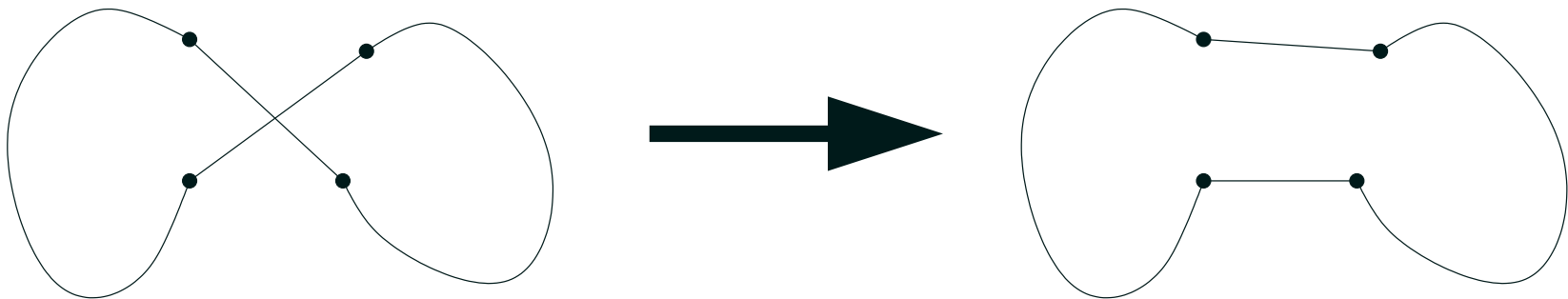
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- We can use this to obtain a lower bound on the partial solution
- We know the remaining tour will go through each of the unvisited cities and the two edge cities
- In fact the remaining part of the tour is a spanning tree of these vertices (it connects all the vertices once and has no cycles)
- But we know a lower bound for this—the minimum spanning tree

Other Cuts

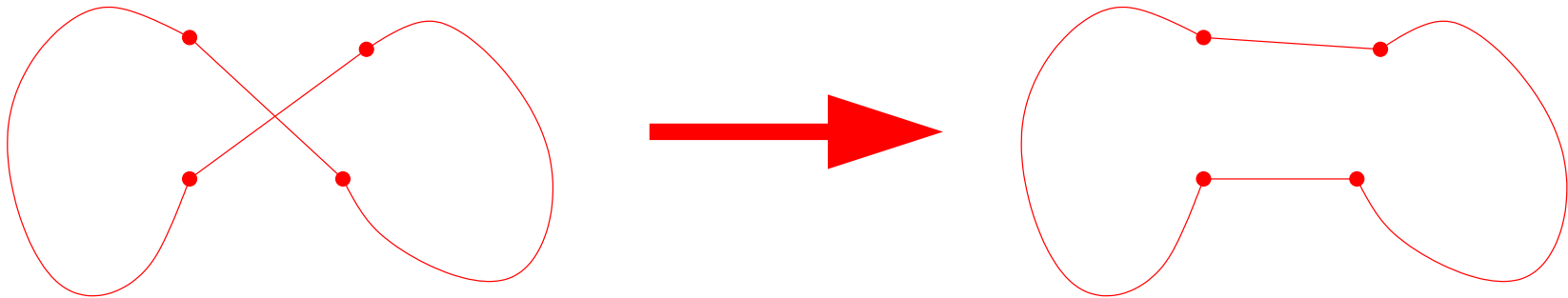
- For 2-D Euclidean TSPs edges should never cross



- In fact we can check that we cannot perform a 2-opt move
- We can also halve the search by considering only one direction—for example, by insisting we visit city 1 before city 2

Other Cuts

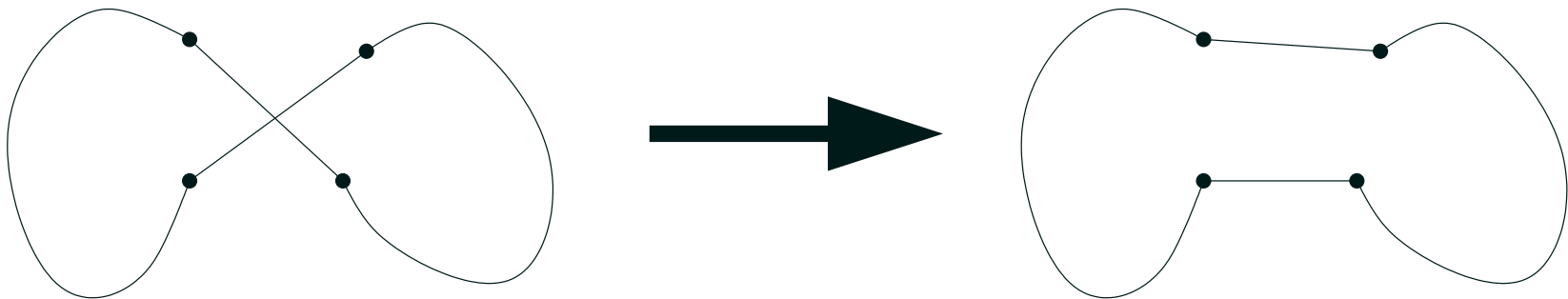
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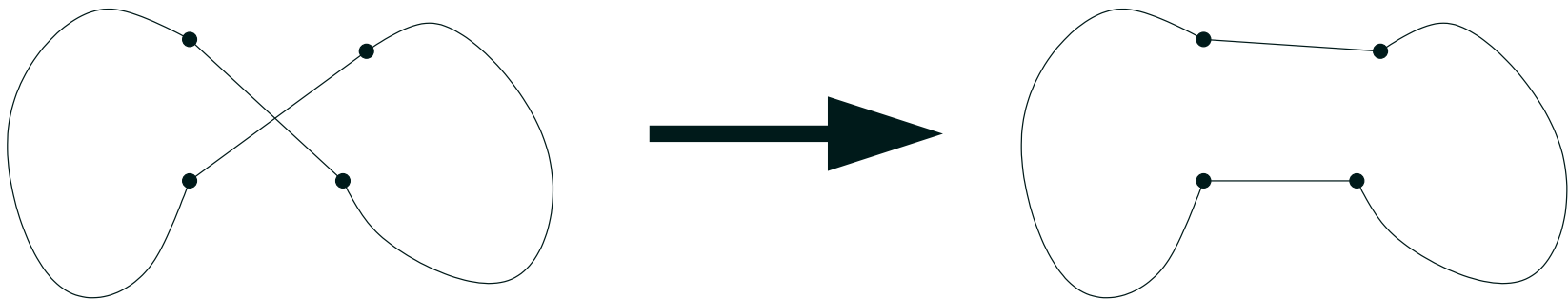
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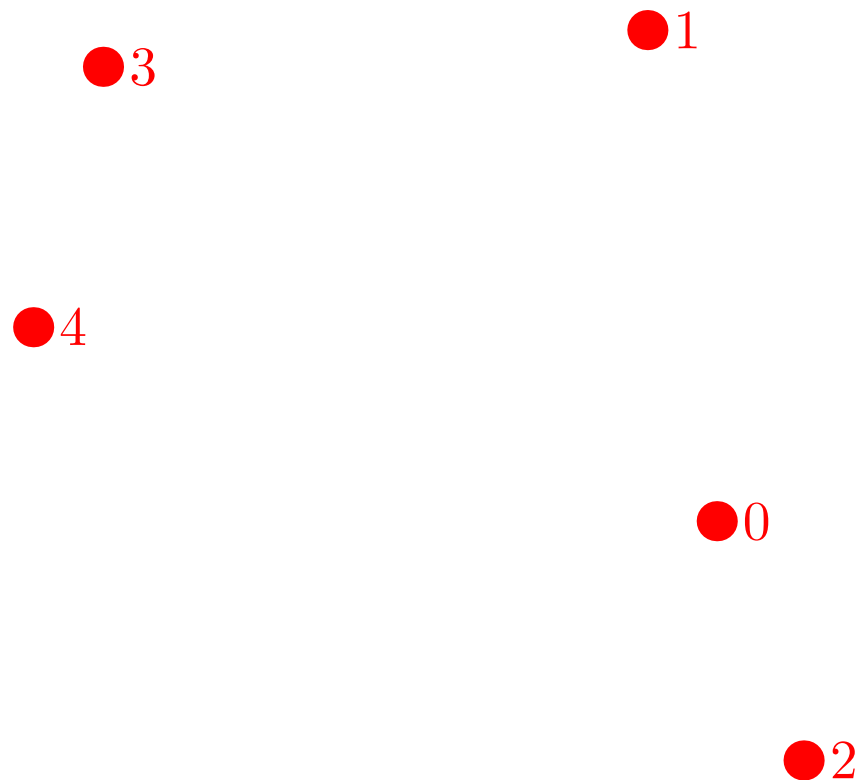
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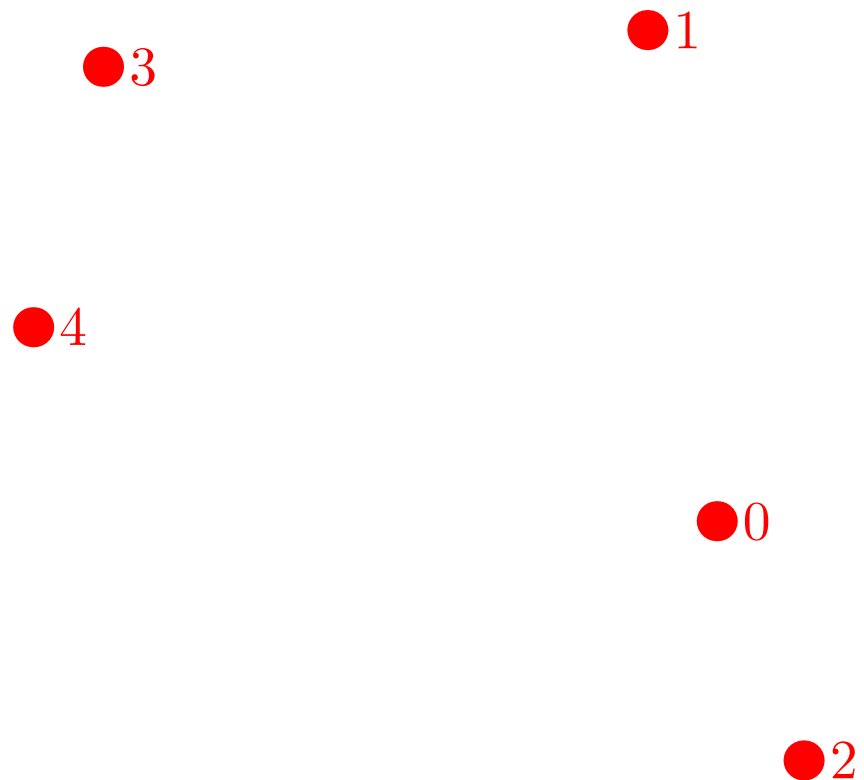
Good Starting Bound

- It helps to start with a good bound
- We can use an *incomplete heuristic algorithm* to find a good solution which will act as a starting bound
- One very simple heuristic is a greedy algorithm



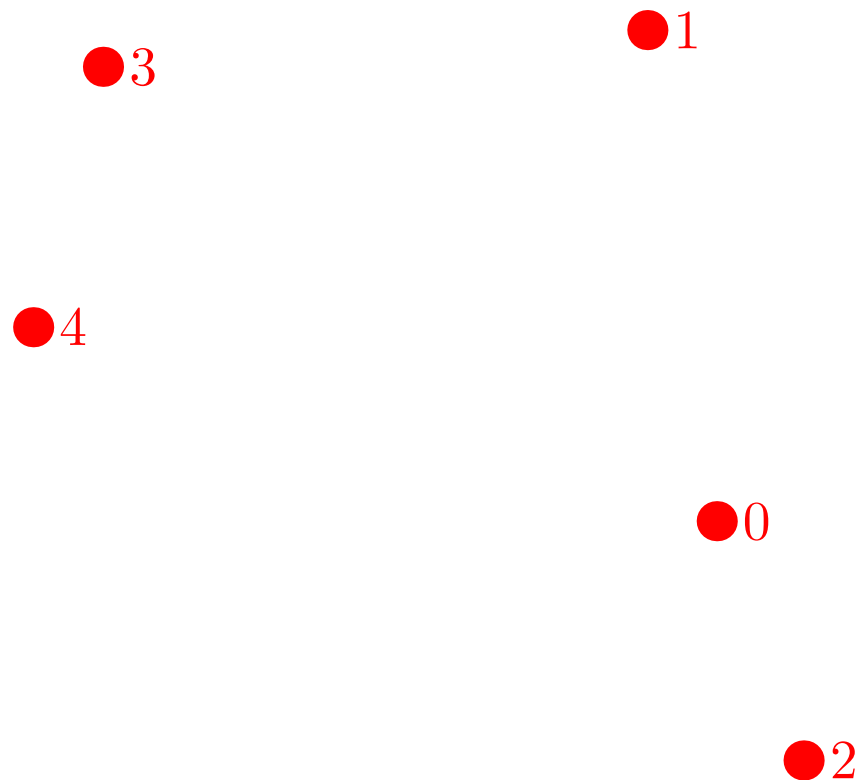
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●3

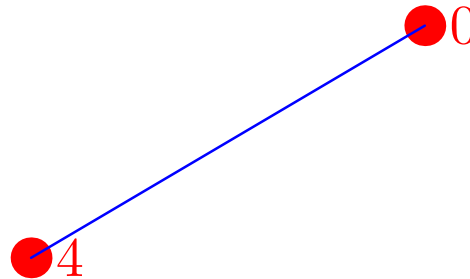
length = 68.558

●2

●1

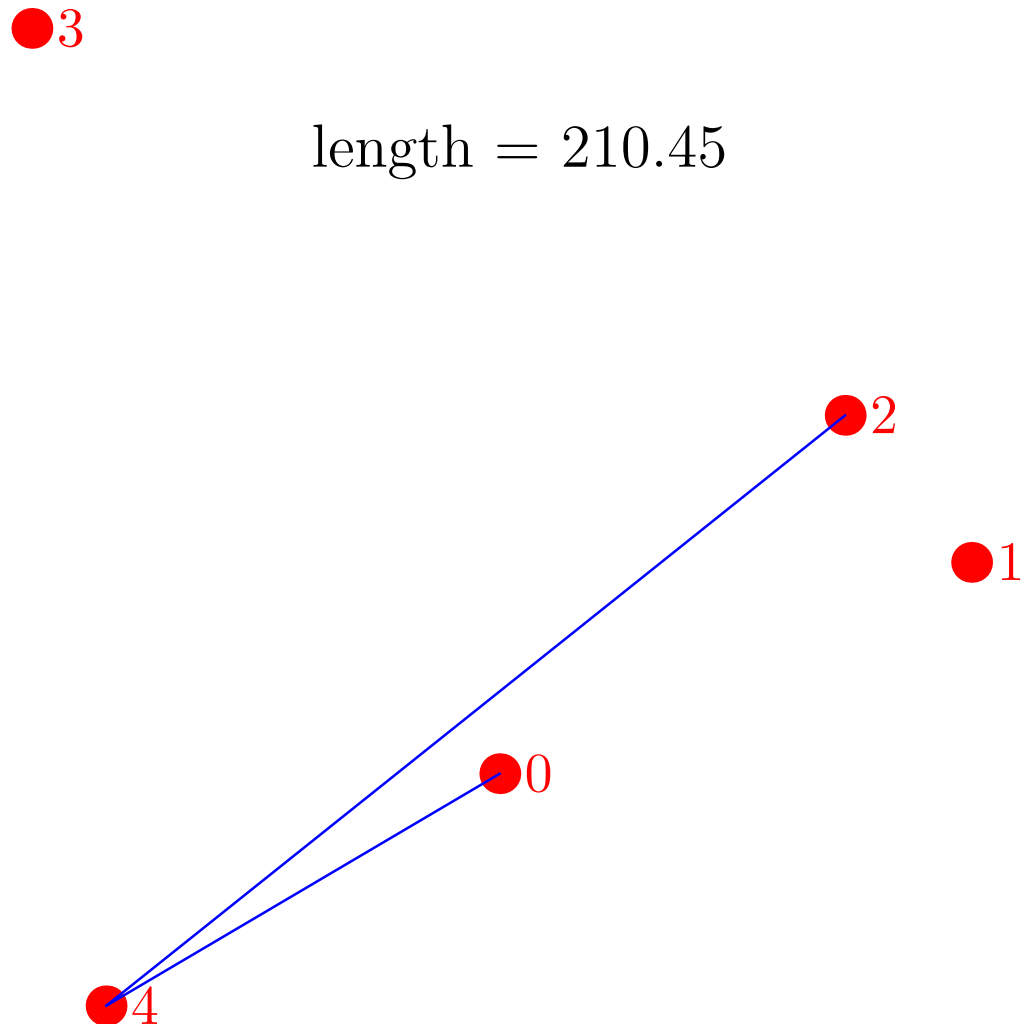
●0

●4



Good Starting Bound

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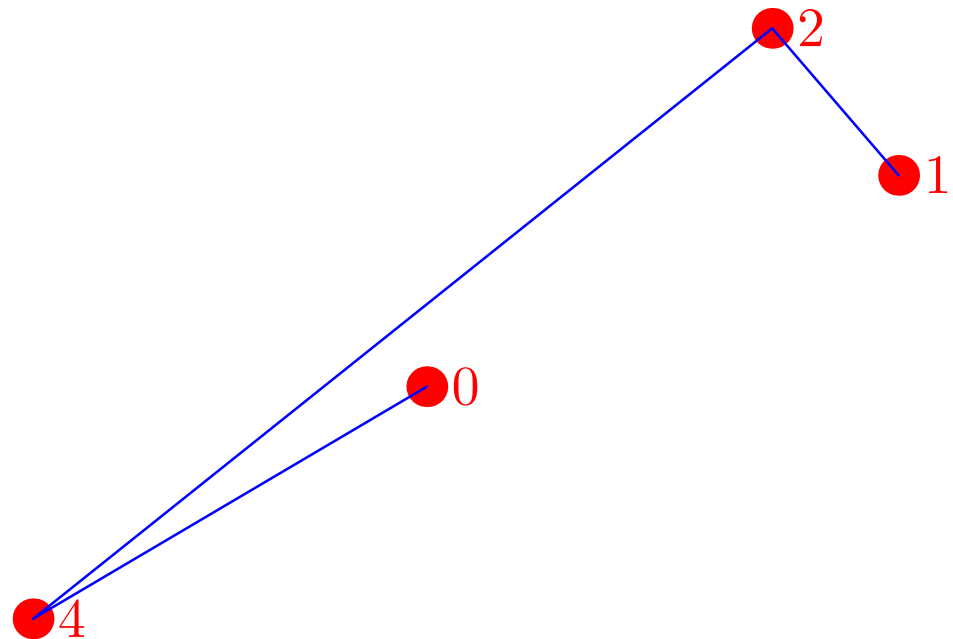


Good Starting Bound

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- One very simple heuristic is a greedy algorithm

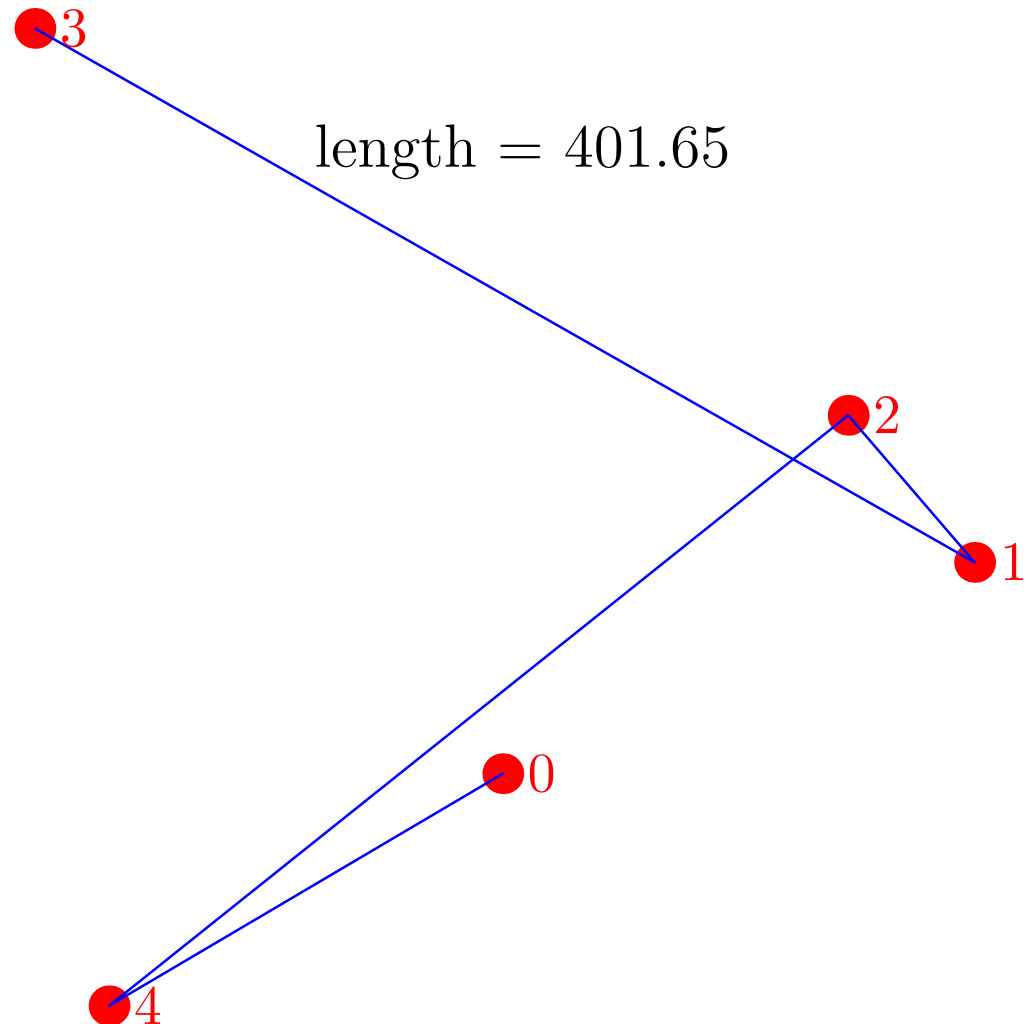
●3

length = 239.54



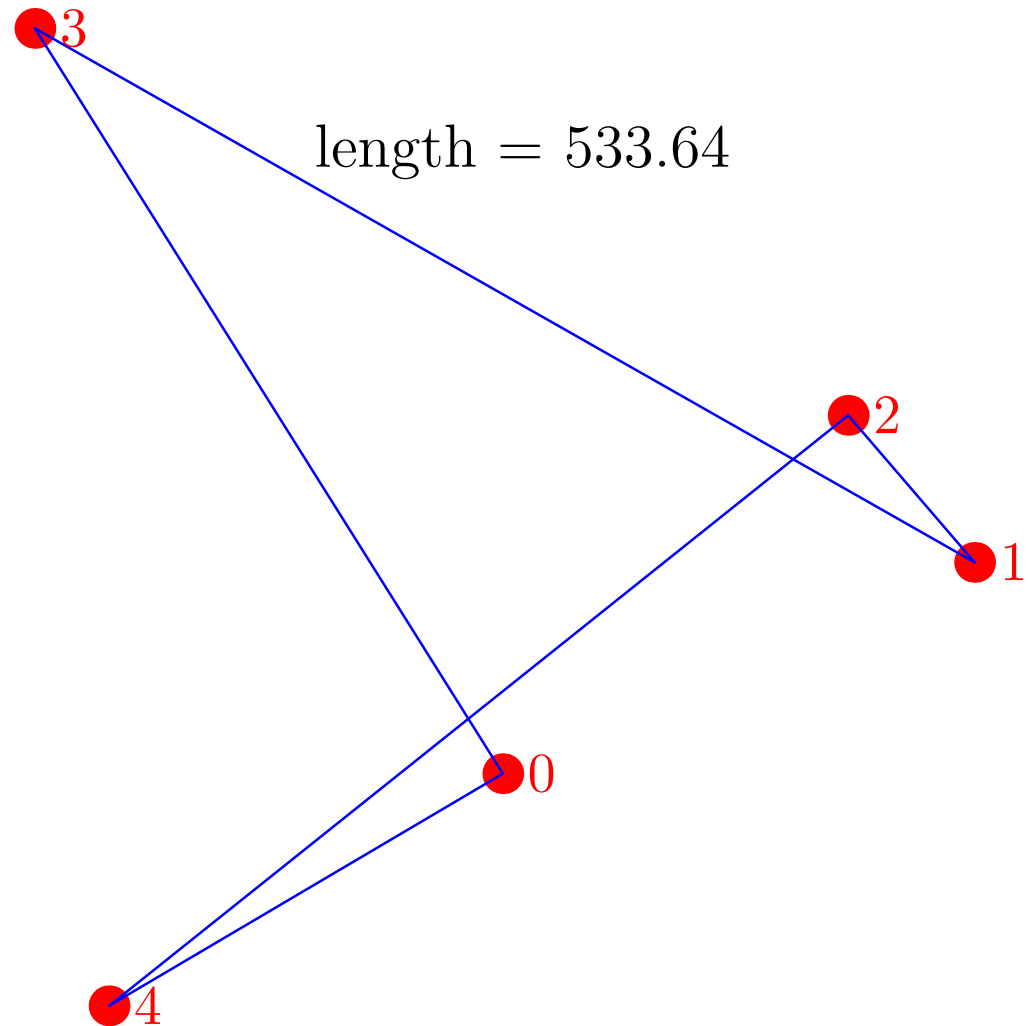
Good Starting Bound

- It helps to start with a good bound
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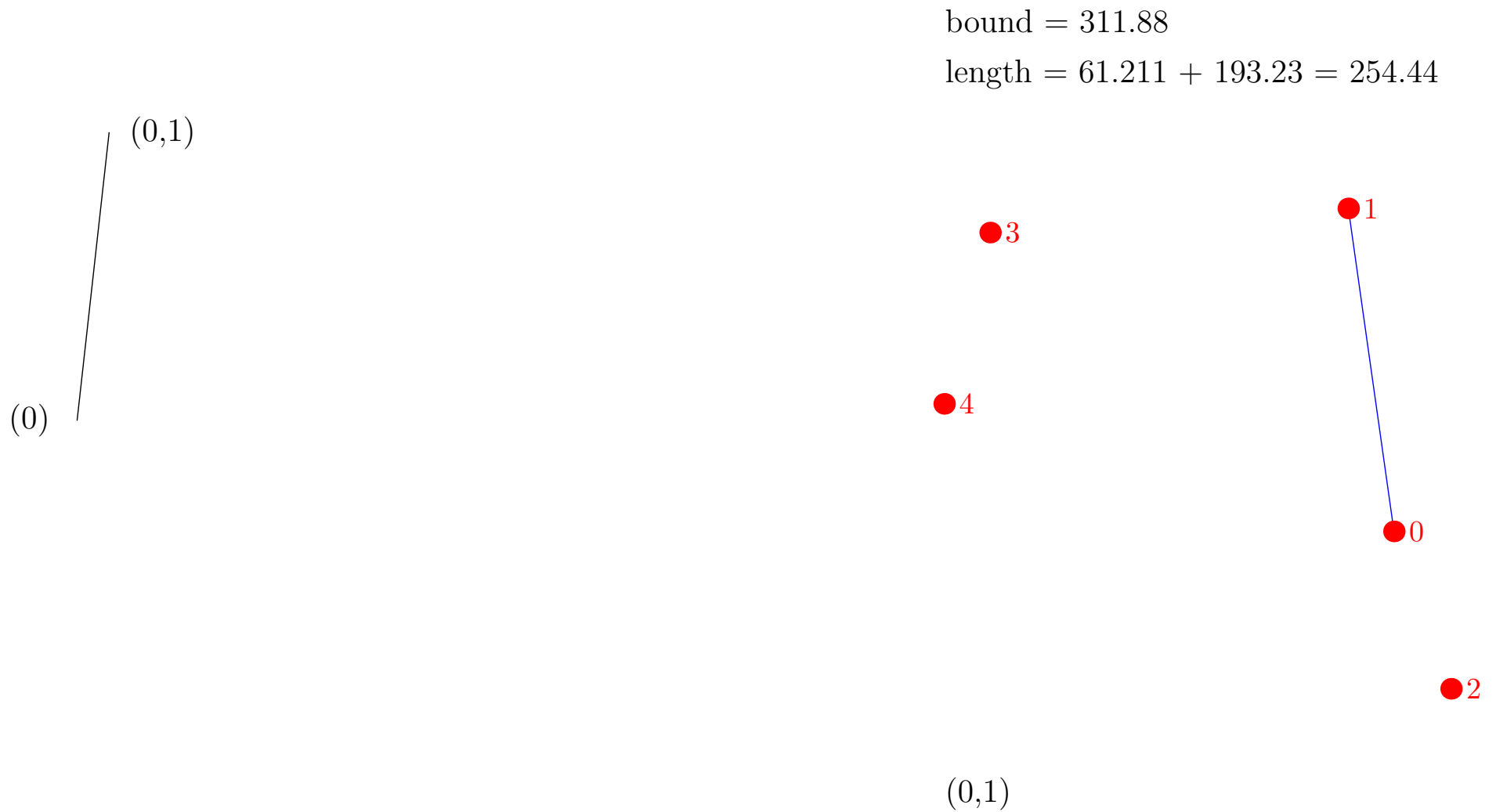


Good Starting Bound

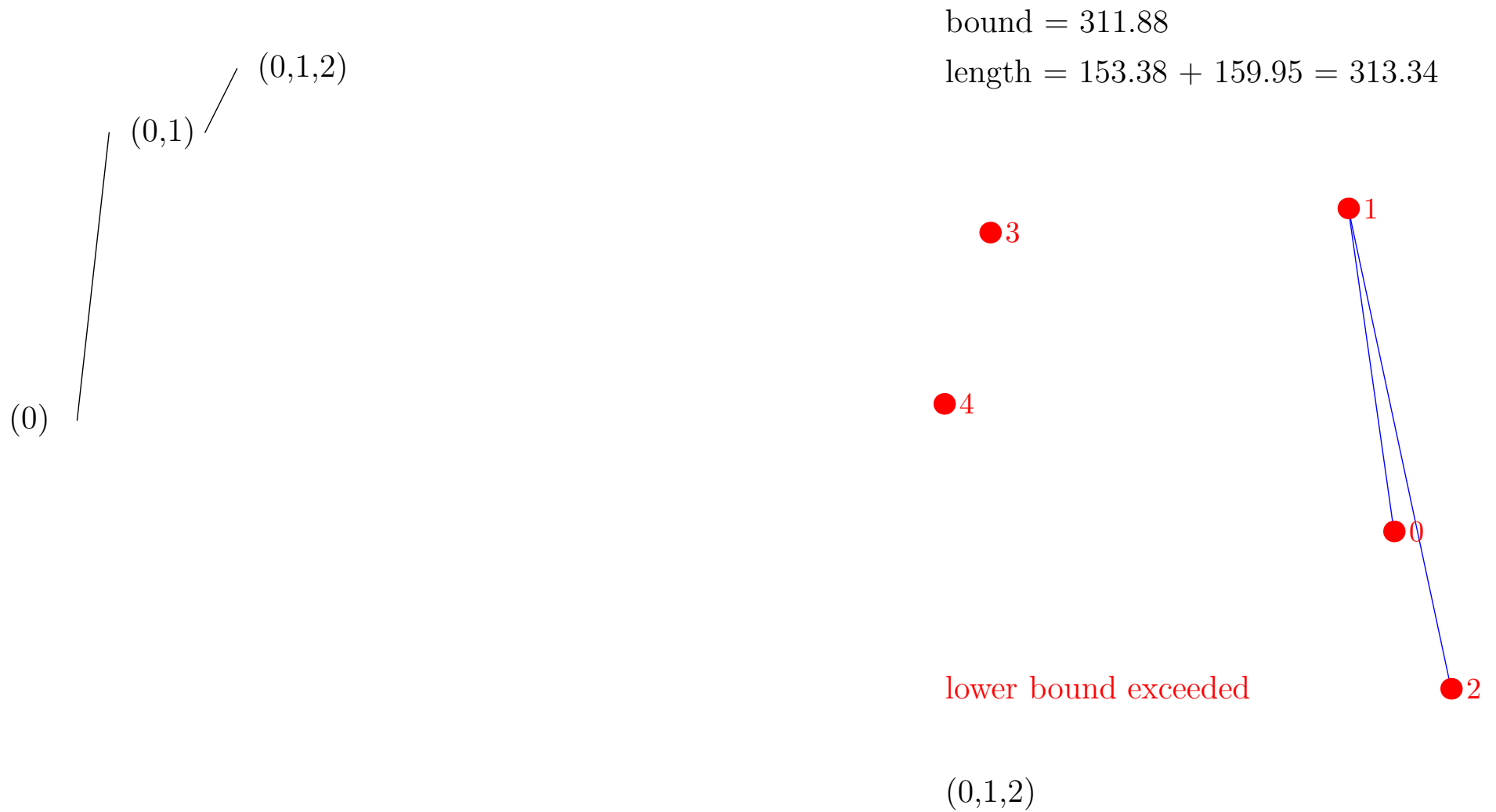
- It helps to start with a good bound
- We can use an *incomplete heuristic algorithm* to find a good solution which will act as a starting bound
- One very simple heuristic is a greedy algorithm



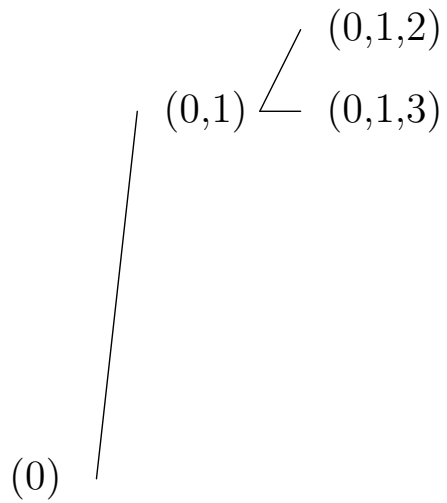
Branch and Bound after Pruning



Branch and Bound after Pruning

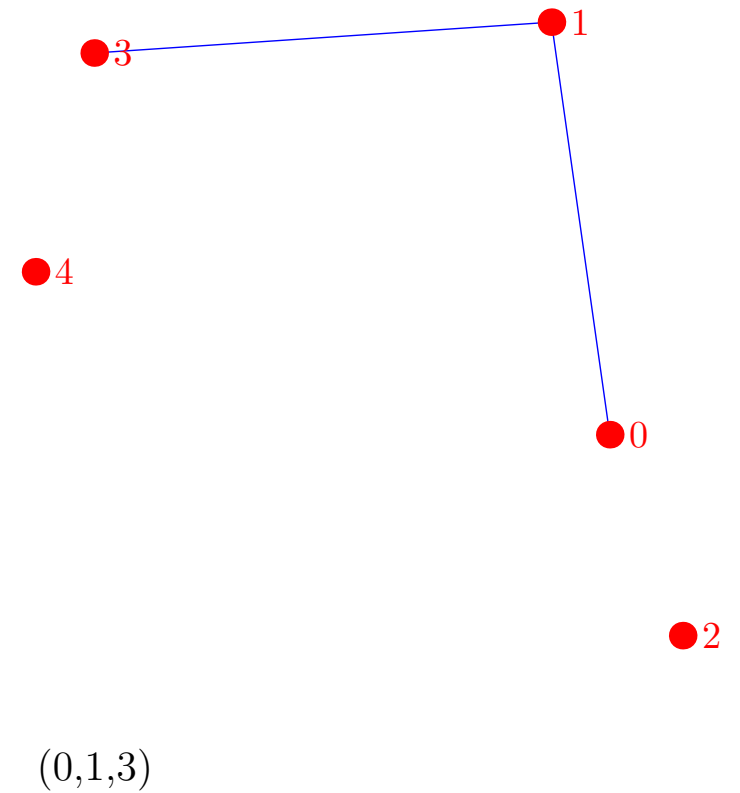


Branch and Bound after Pruning

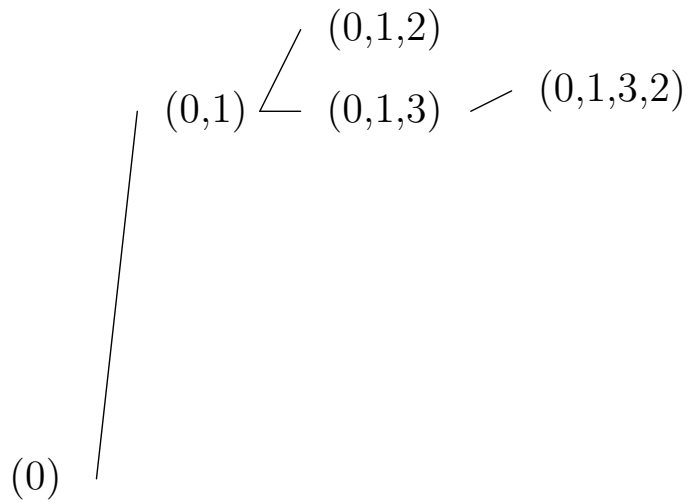


bound = 311.88

length = $128.54 + 159.95 = 288.5$

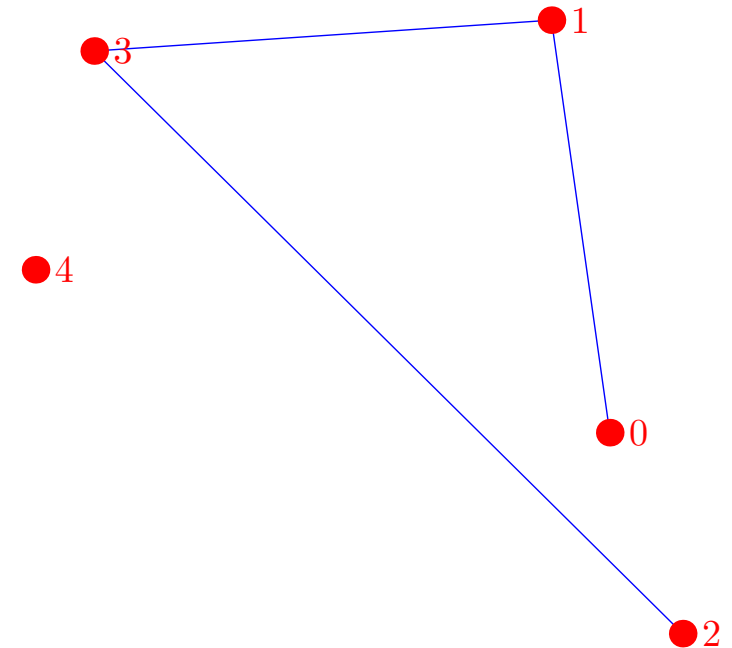


Branch and Bound after Pruning



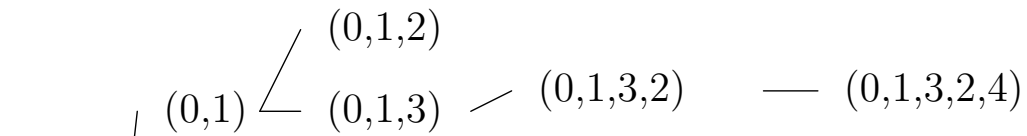
bound = 311.88

length = $250.21 + 31.41 = 281.63$



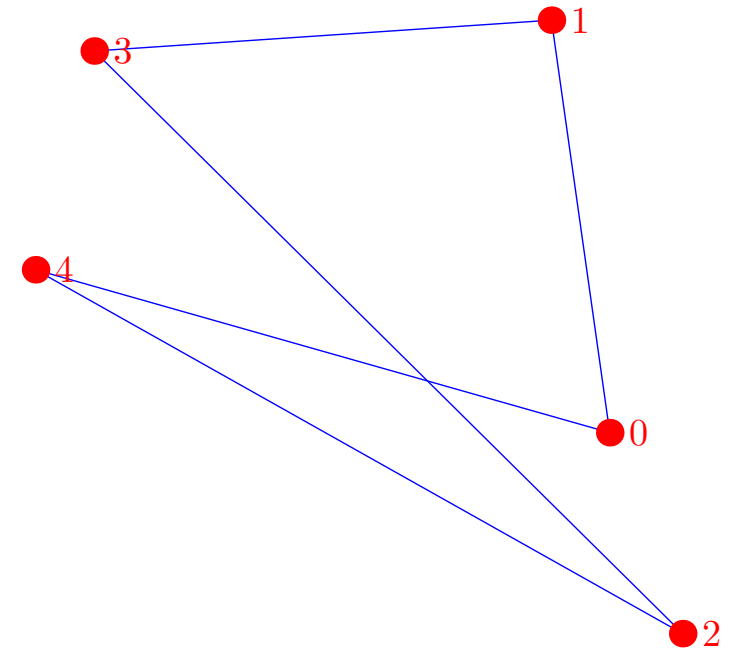
$(0,1,3,2)$

Branch and Bound after Pruning



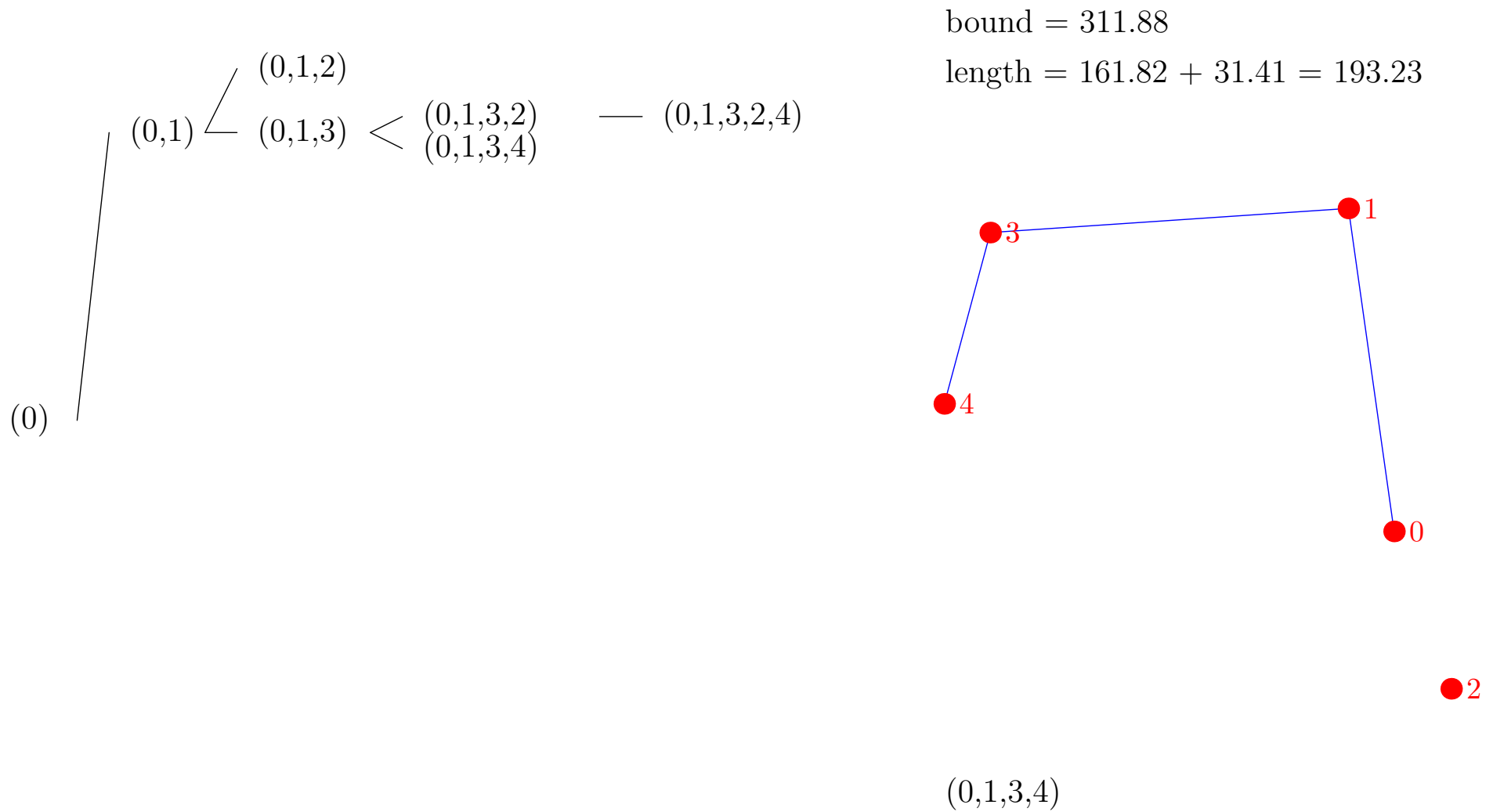
bound = 311.88

length = 446.99 + 0 = 446.99

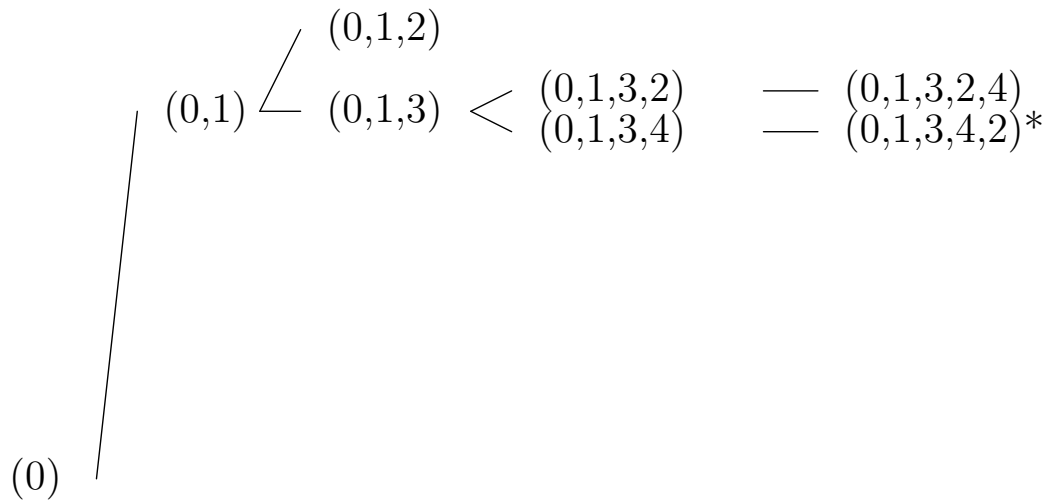


(0,1,3,2,4)

Branch and Bound after Pruning

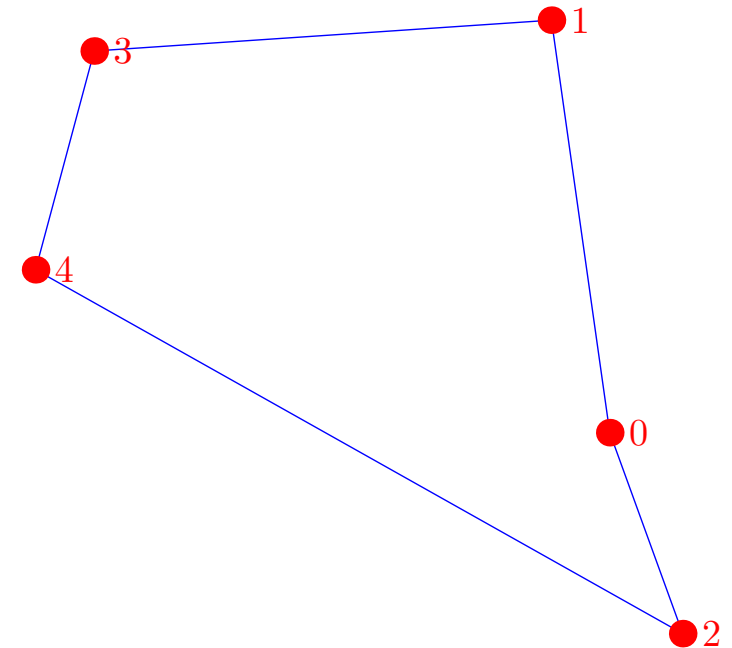


Branch and Bound after Pruning



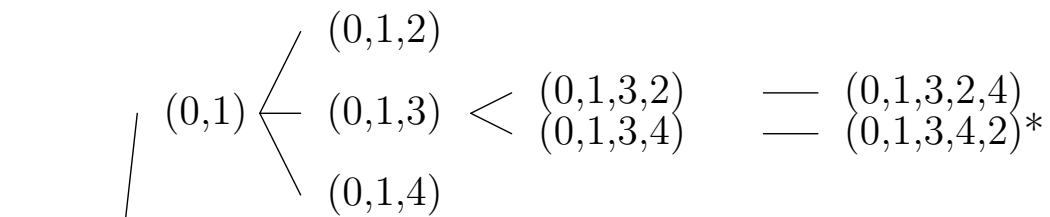
bound = 302.31

length = 302.31 + 0 = 302.31



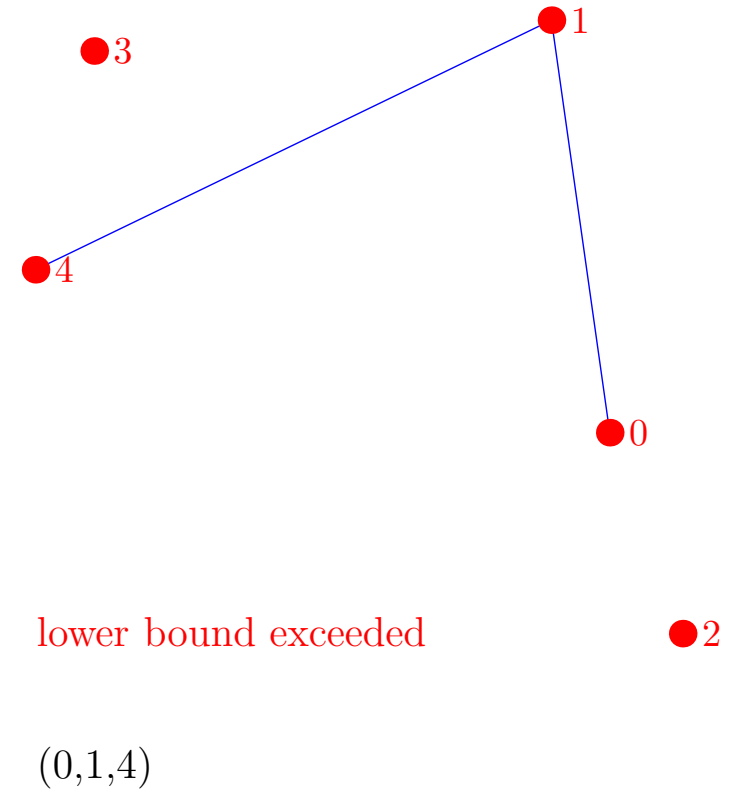
$(0,1,3,4,2)$

Branch and Bound after Pruning

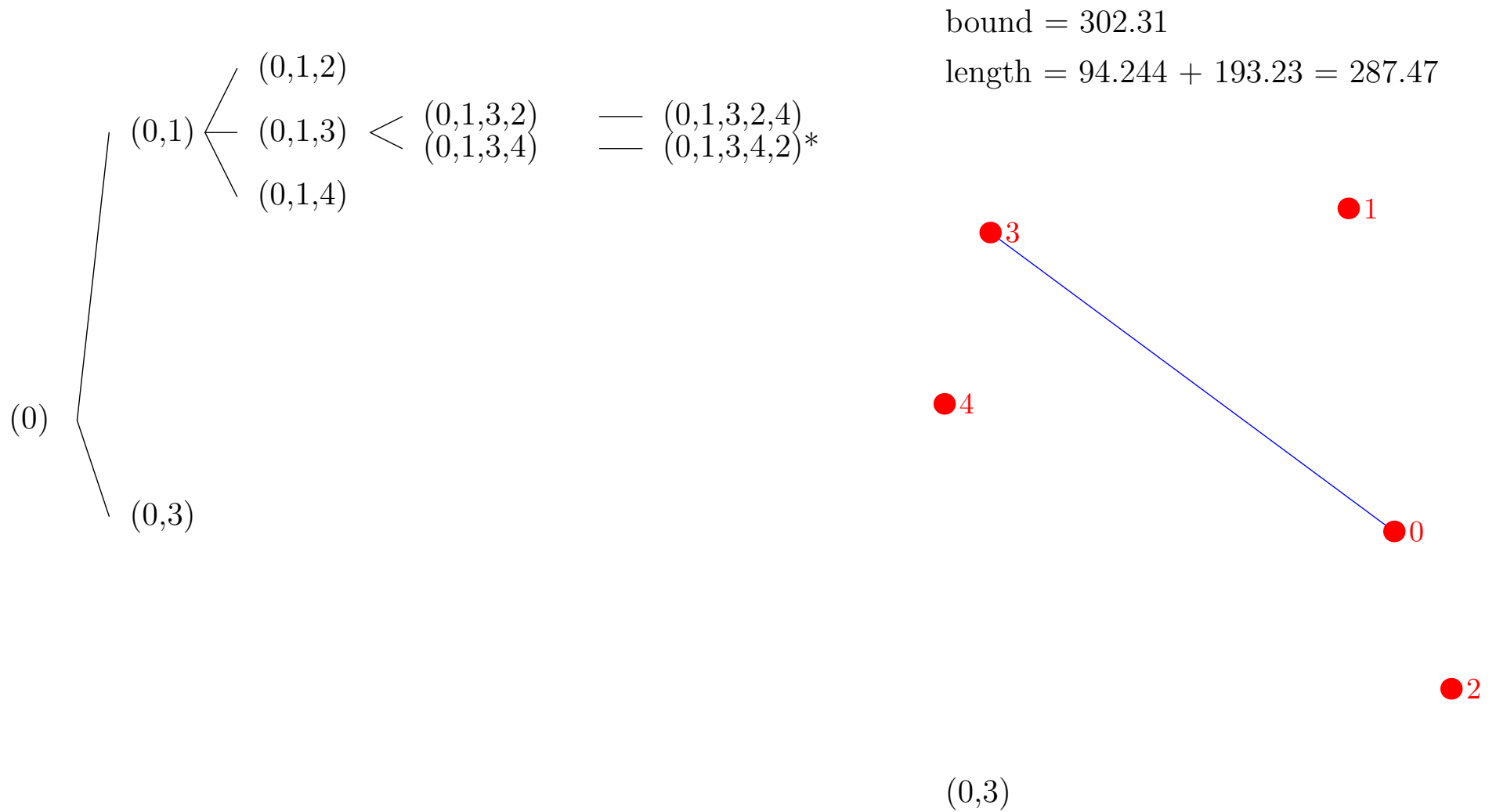


bound = 302.31

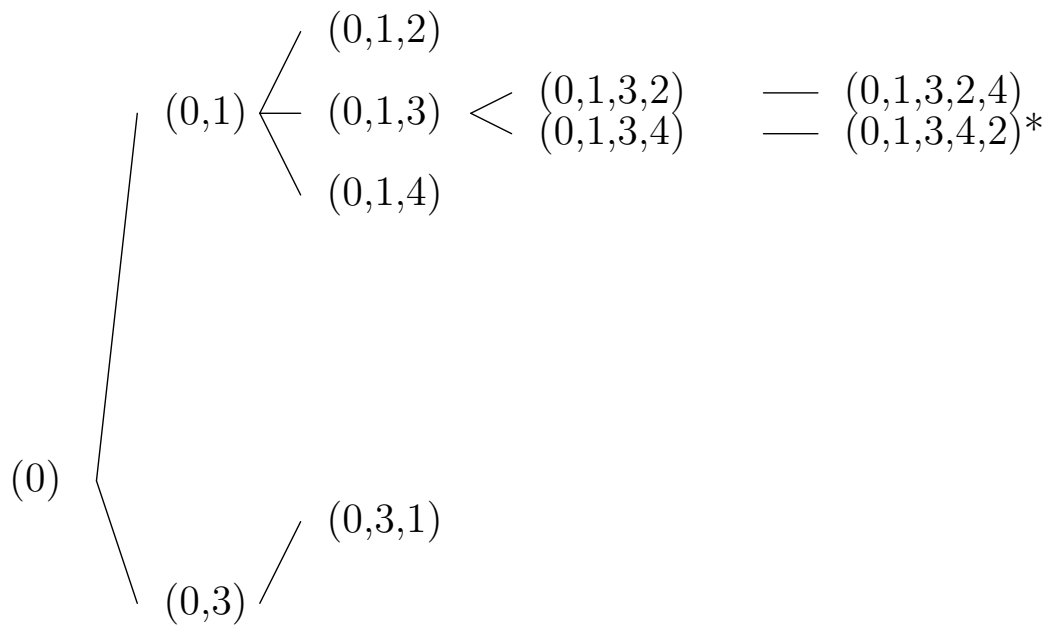
length = 145.41 + 159.95 = 305.37



Branch and Bound after Pruning

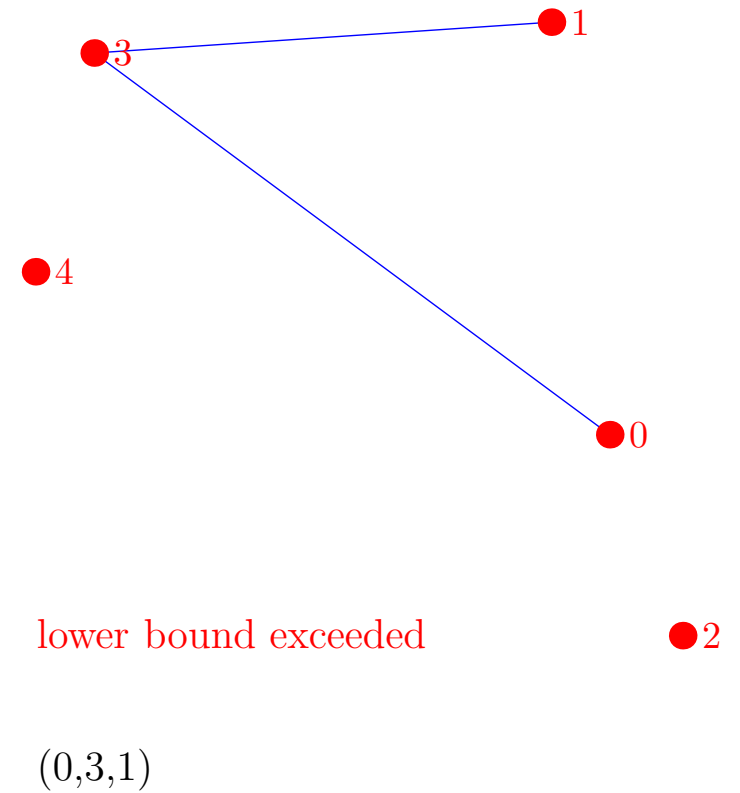


Branch and Bound after Pruning

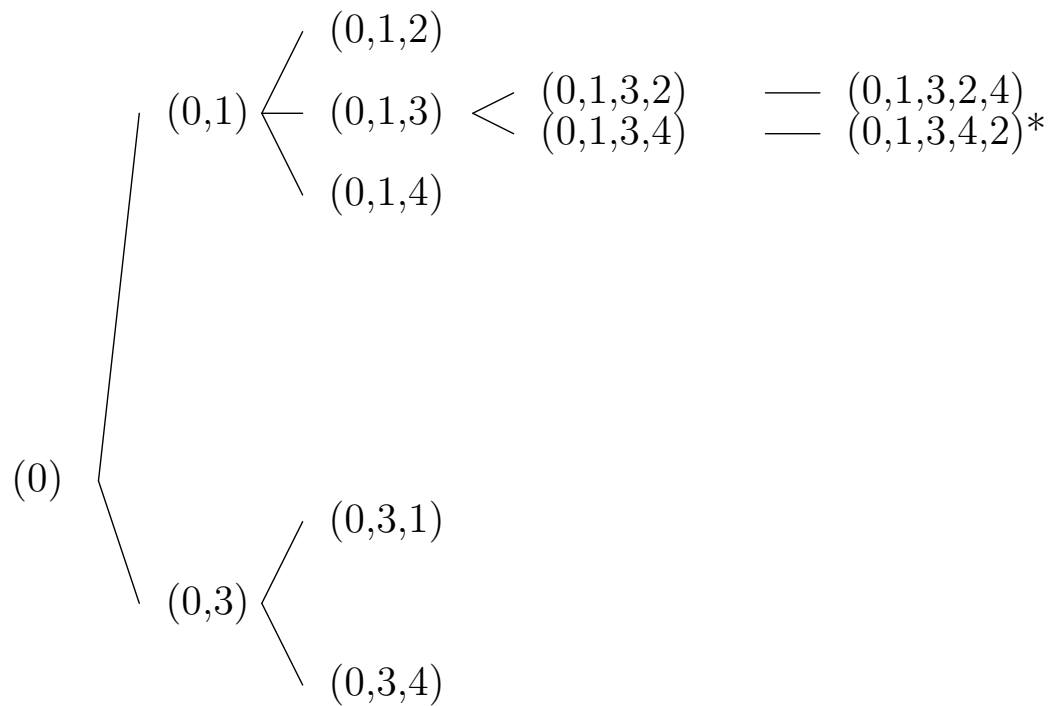


bound = 302.31

length = 161.58 + 159.95 = 321.53

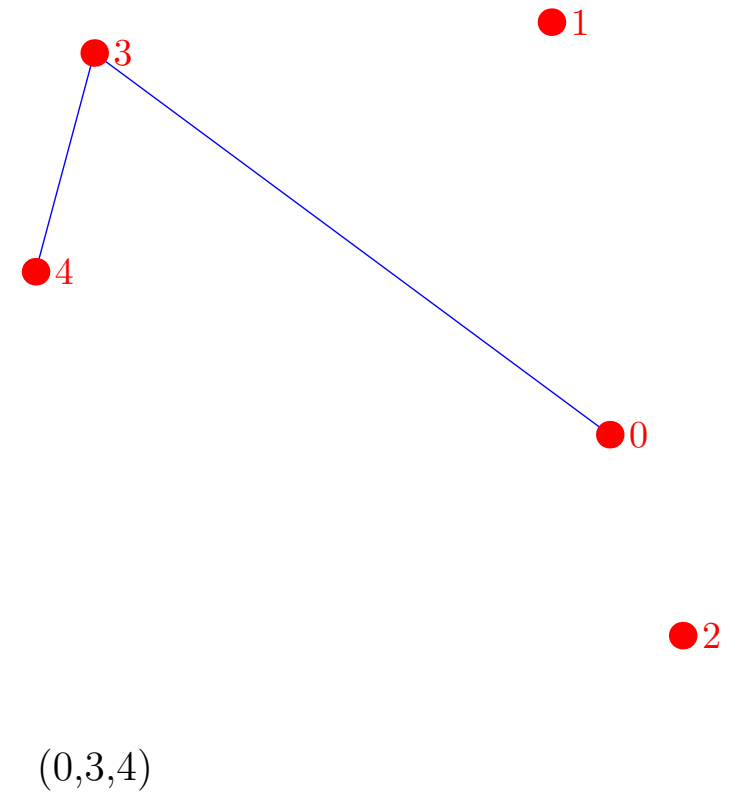


Branch and Bound after Pruning

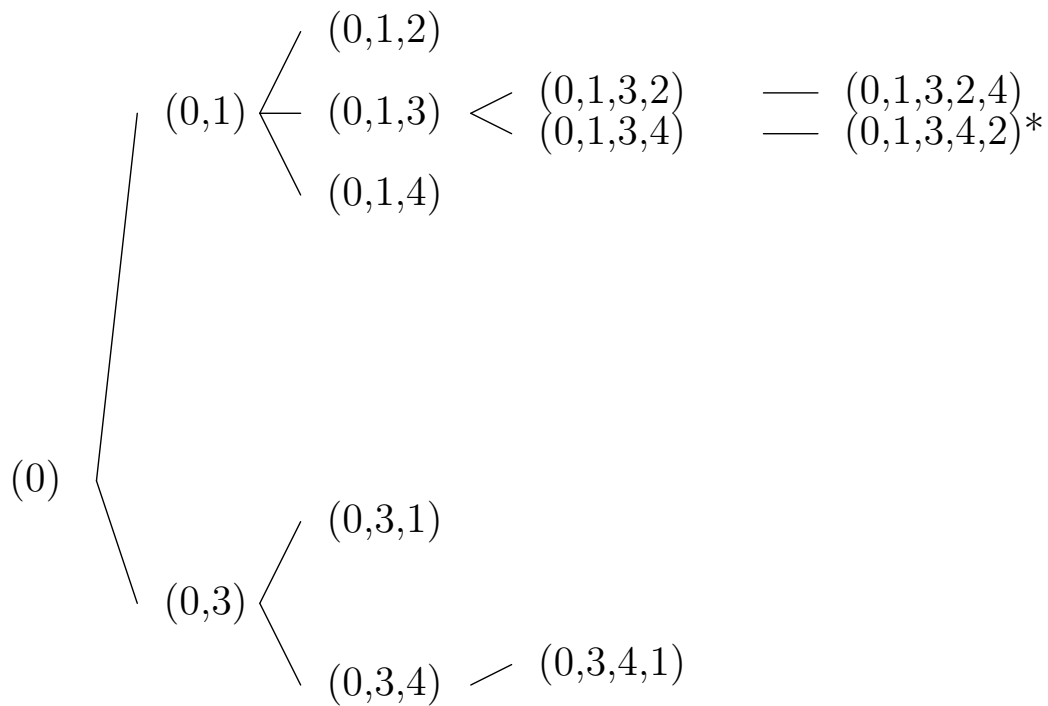


bound = 302.31

length = 127.52 + 159.95 = 287.47

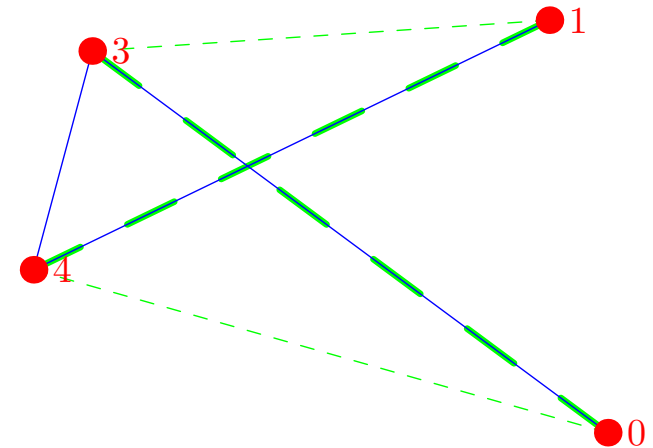


Branch and Bound after Pruning



bound = 302.31

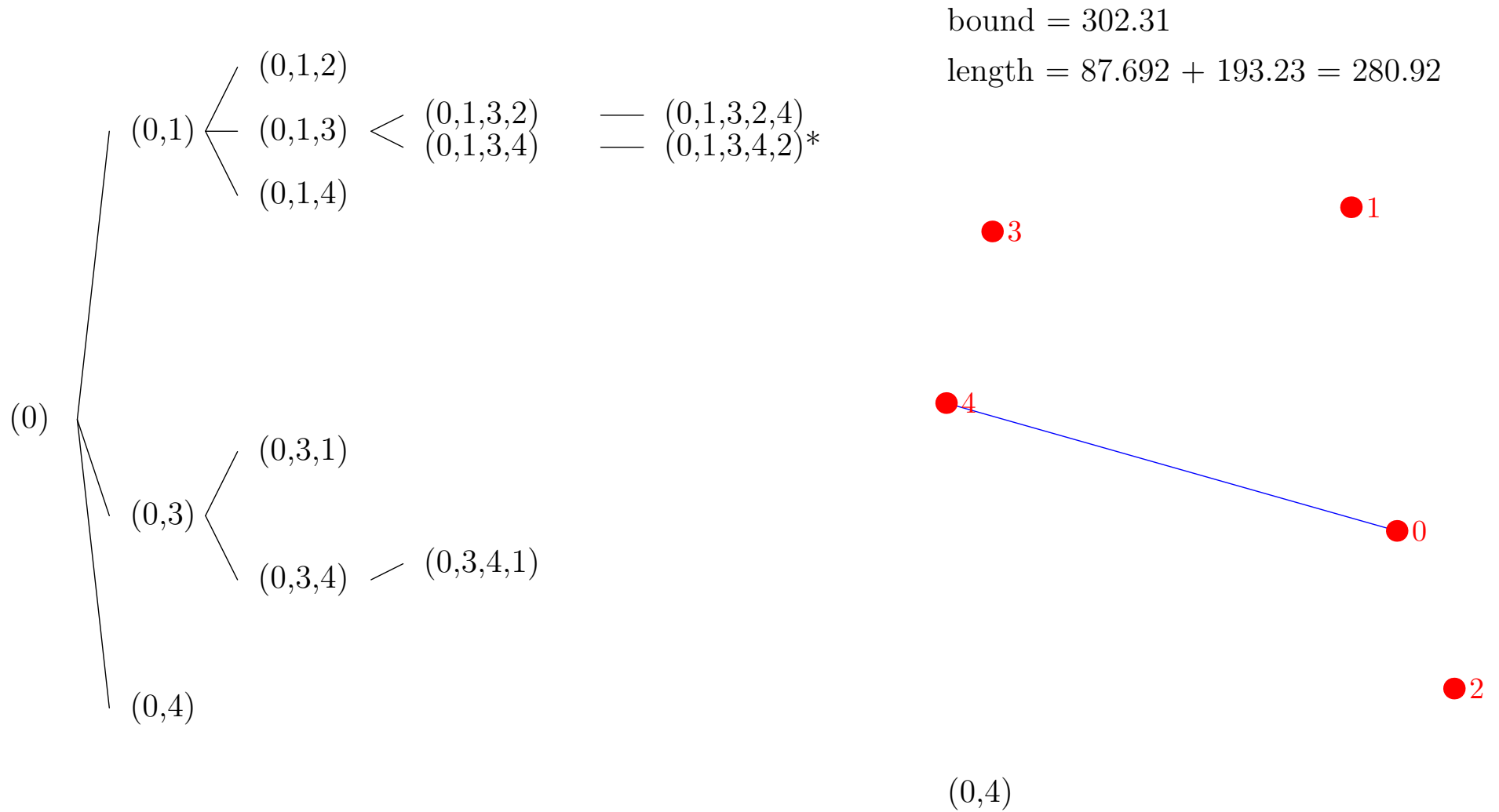
length = 211.72 + 61.211 = 272.93



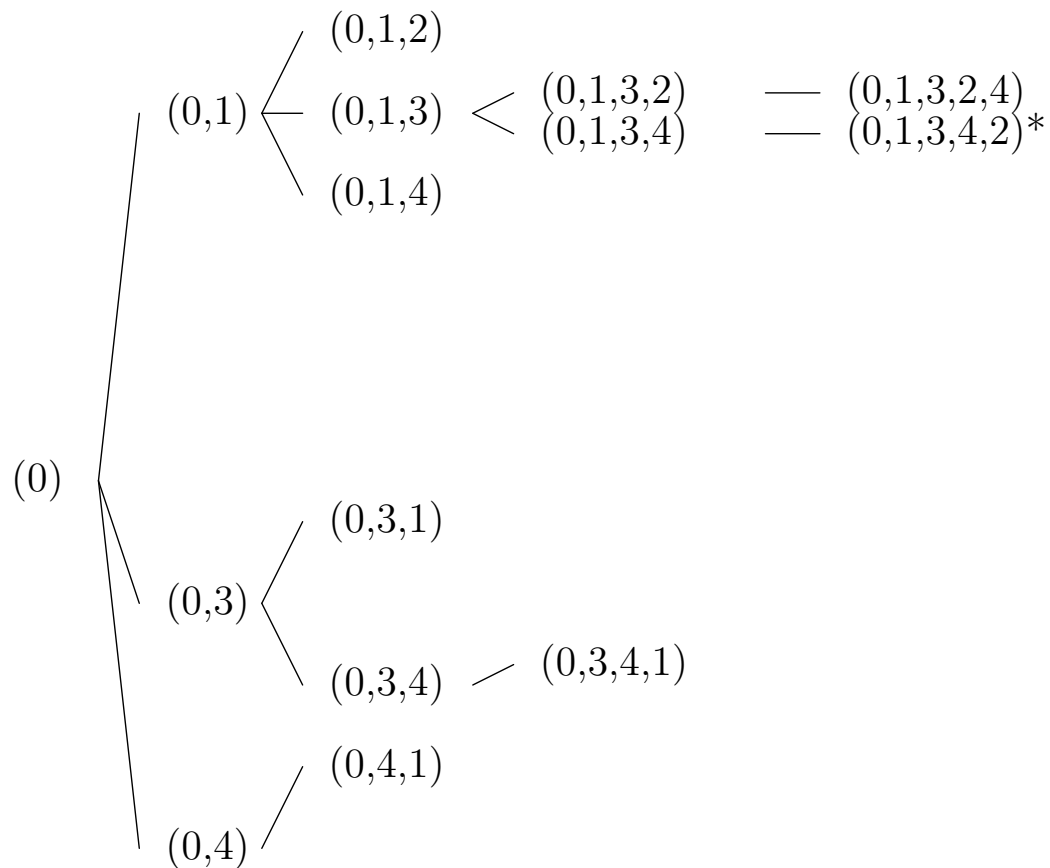
Not 2-opt

$(0,3,4,1)$

Branch and Bound after Pruning

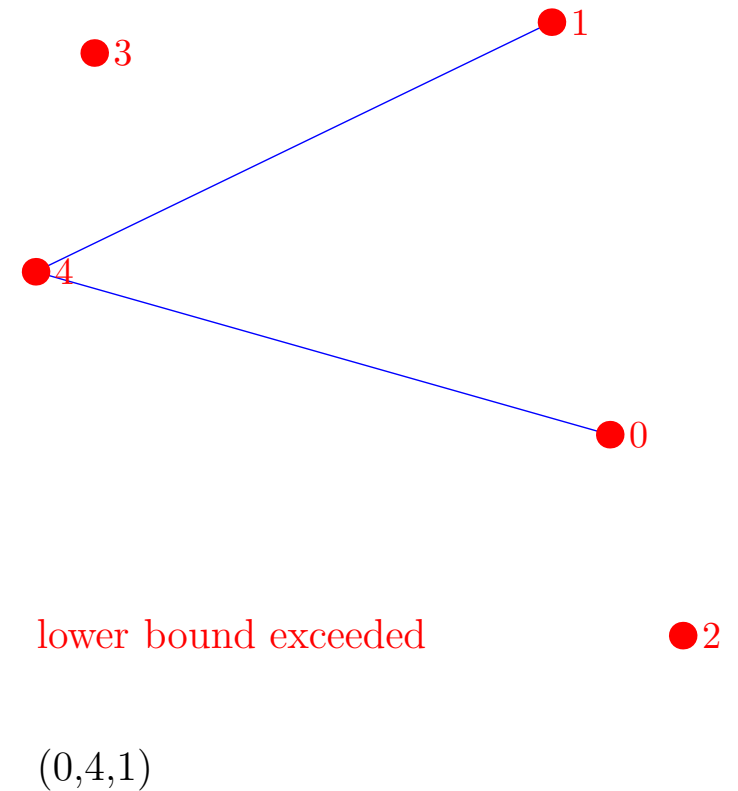


Branch and Bound after Pruning

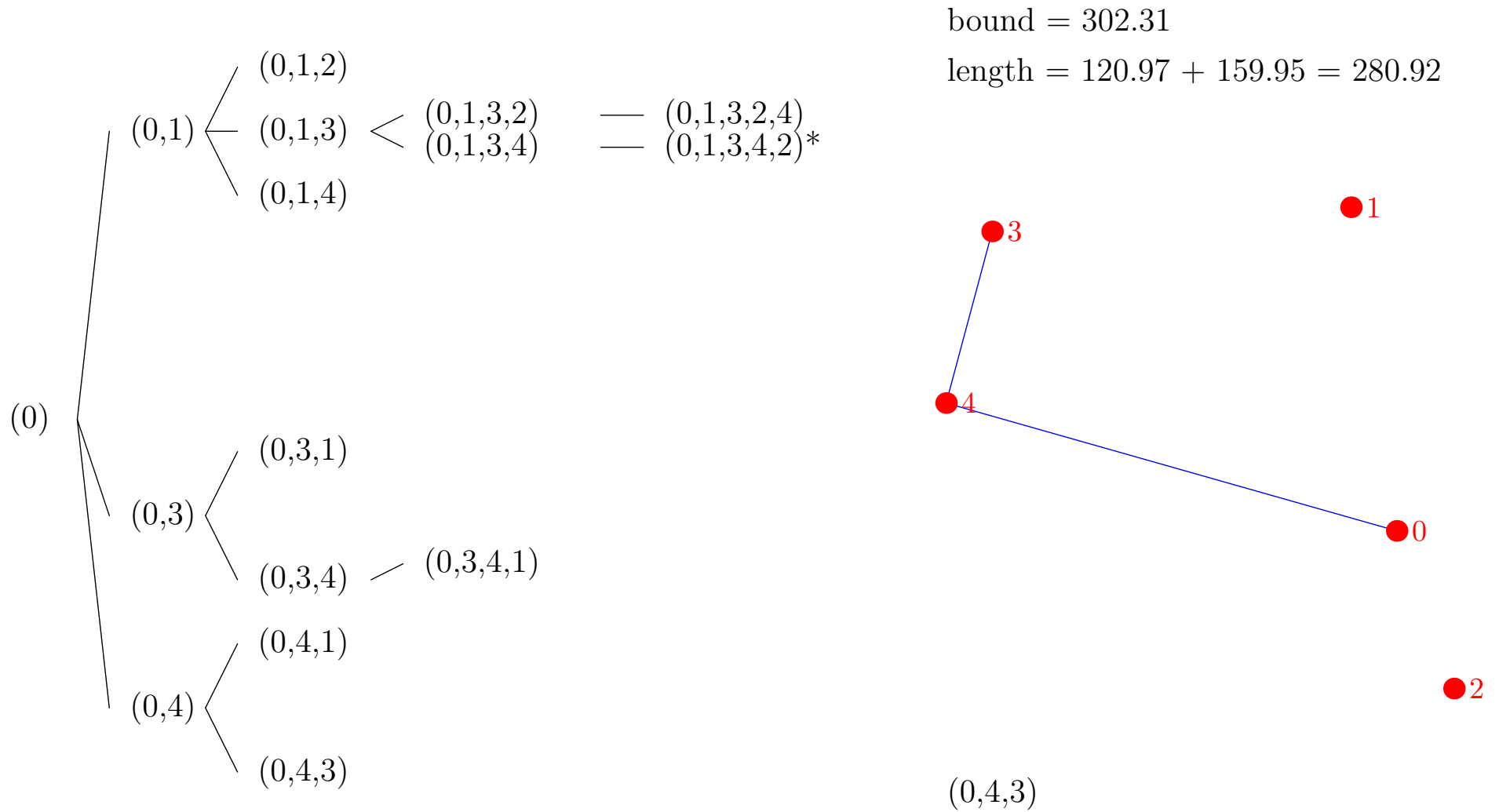


bound = 302.31

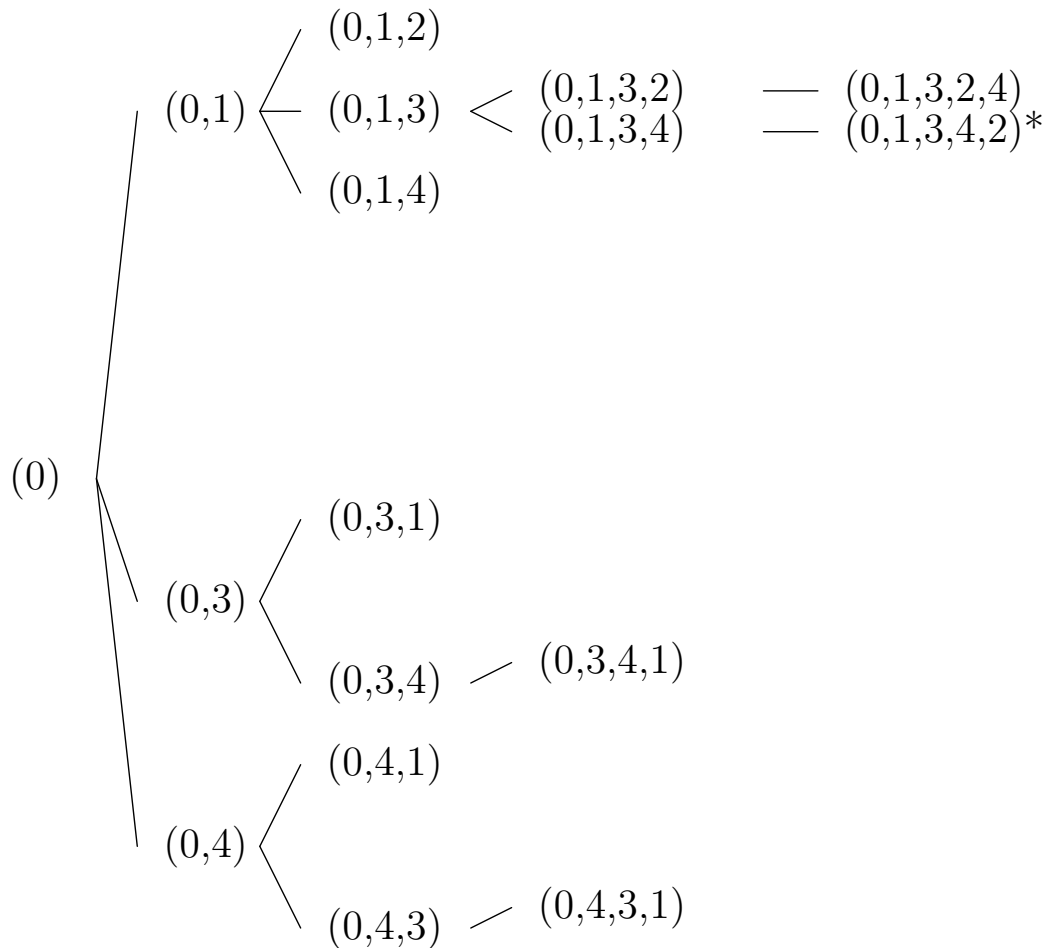
length = 171.9 + 159.95 = 331.85



Branch and Bound after Pruning

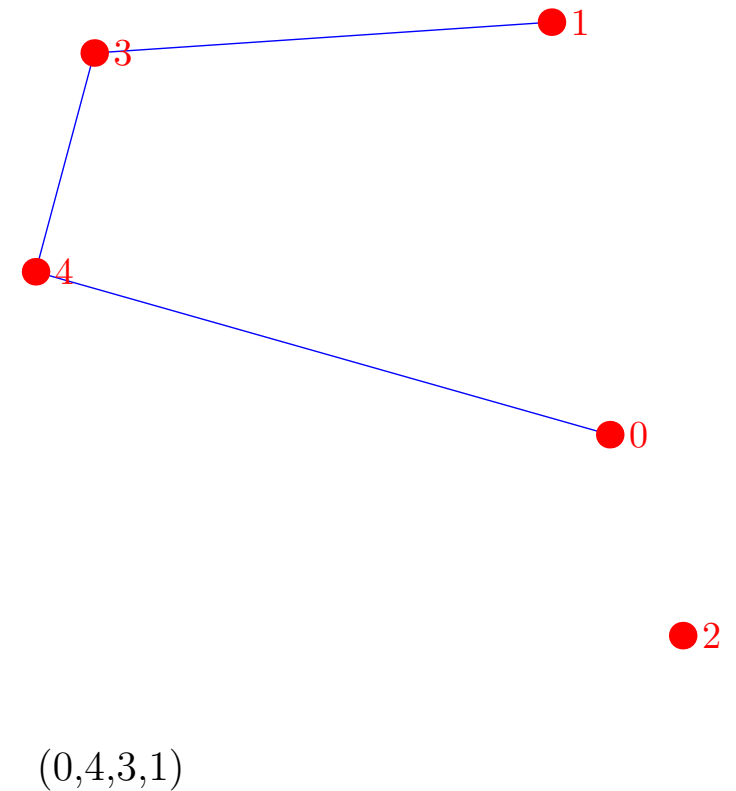


Branch and Bound after Pruning

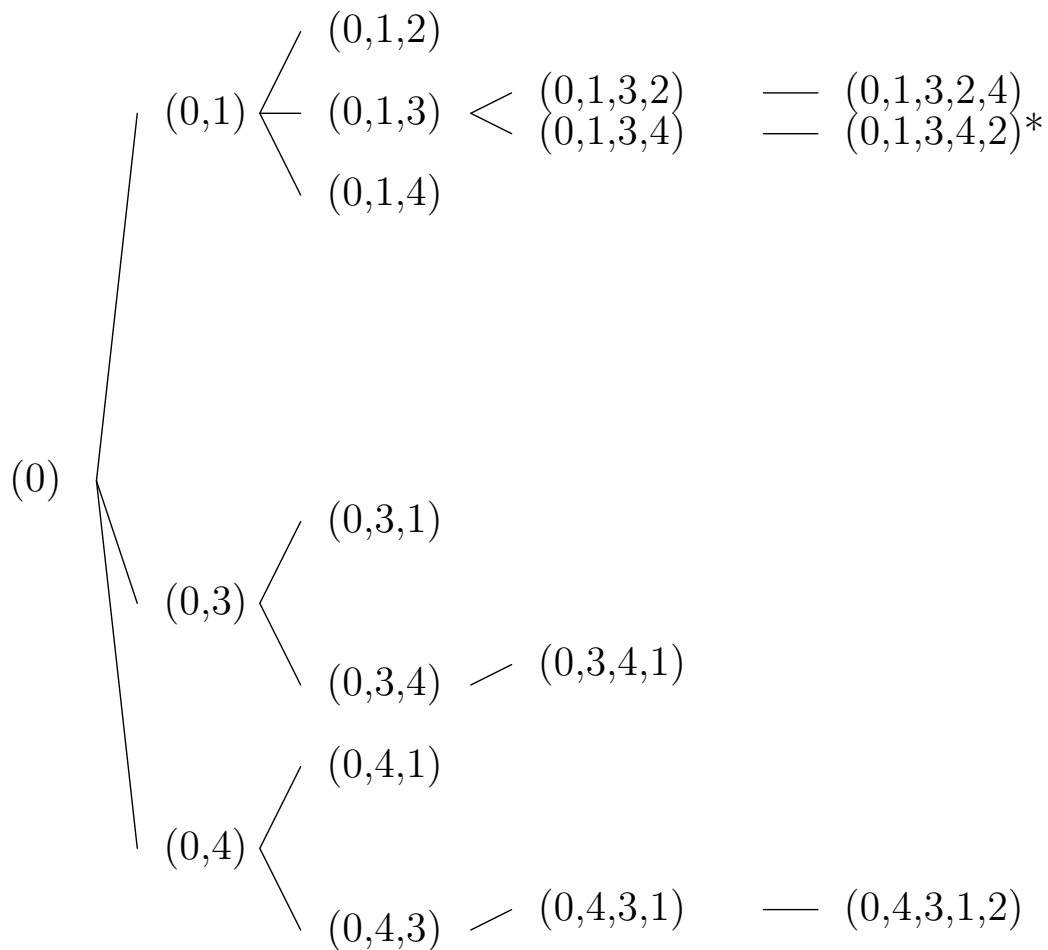


bound = 302.31

length = 188.3 + 61.211 = 249.51

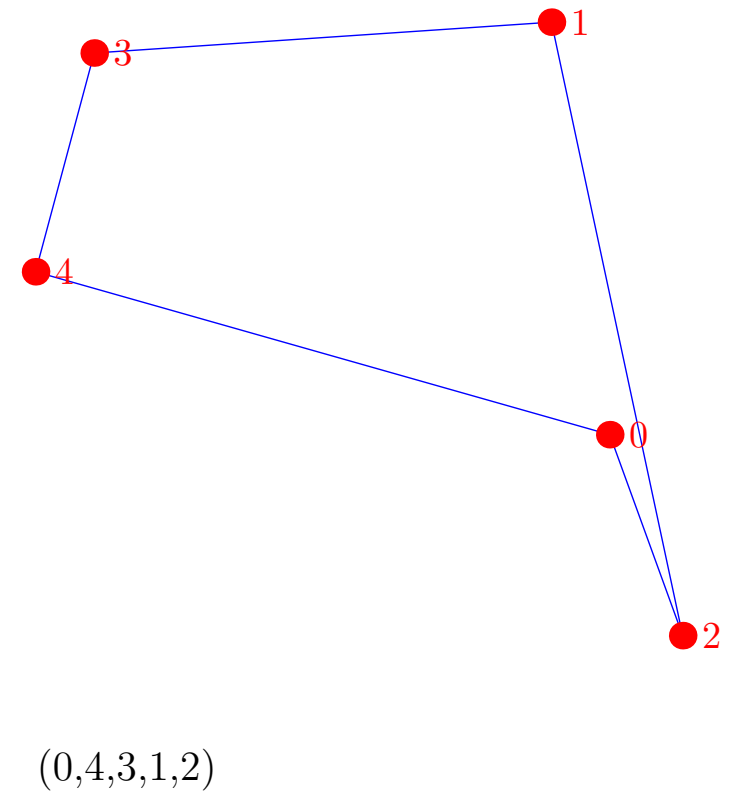


Branch and Bound after Pruning



bound = 302.31

length = 311.88 + 0 = 311.88



Applications of Branch and Bound

- Branch and bound works for many optimisation problems
- It's drawback is that you often end up still searching an exponentially large search space even though it might be massively faster than exhaustive enumeration
- To make it work well requires considerable work
- This is not an instantaneous algorithm, you may be waiting hours before you find a solution
- For really large problems branch and bound might be too slow

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- For really large problems branch and bound might be too slow

Applications of Branch and Bound

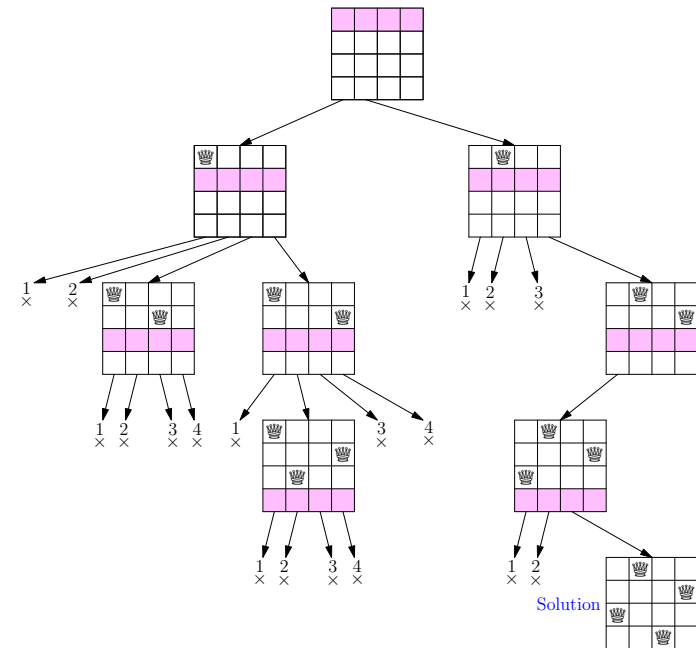
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Outline

1. Search Trees
2. Backtracking
3. Branch and Bound
4. **Search in AI**



Other Search Strategies

- Search is a big topic in AI
- The algorithms used depends on the information available
- A classic search scenario is when there is “heuristic” information which provides a hint as to where an optimal solution lies
- Algorithms such as A^* exist which will find the best route given an (admissible) heuristic as efficiently as possible
- You should learn about this next year in AI

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- Search is also used to find the best action to take in planning problems and game playing (e.g. computer chess)
- Again it is useful to think in terms of a search tree
- Searching all paths on the search tree is usually infeasible
- Look for ways of pruning the search tree to focus on good moves
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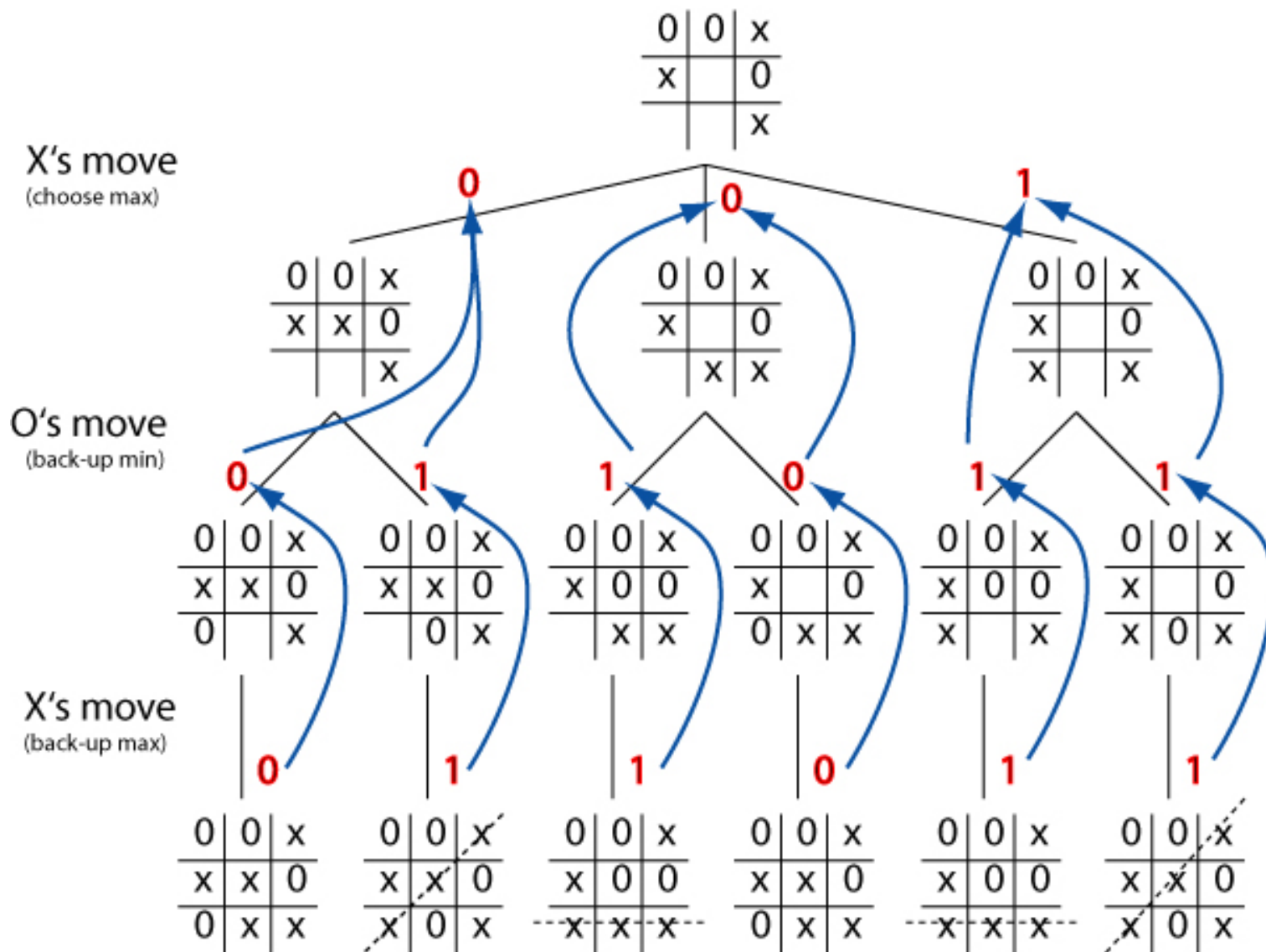
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Minimax with Alpha-Beta Pruning



Lessons

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- It is helpful to consider the search space as a tree whose branch corresponds to possible actions
- Backtracking is useful in search trees with constraints
- For optimisation problems branch and bound uses backtracking and costs of partial solutions as constraints
- Widely applicable, but can take too long

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